MAT2125: Introduction to real analysis

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Chapter 1

The real numbers

The field of real numbers, $\mathbb{R}$, is a fundamental starting point for analysis. It is rich with structure: we will begin with field operations $(+, -, \times, \div)$ which are applied to finitely many elements at a time[1], and then order relations $(<, >, \leq, \geq)$; then we will see the operations of least upper bound and greatest lower bound, which are applied to (typically infinite) sets of real numbers.

We will describe the real numbers by giving axioms; the real numbers $\mathbb{R}$ is the unique structure satisfying these axioms. Proving that the object $\mathbb{R}$ exists is a non-trivial feat, which will not be covered in this course. Rather, we will take $\mathbb{R}$ and its axioms as given, and prove many other great things using them.

1.1 Field axioms

Definition 1.1.1. *A* field *is a set F equipped with two operations,* addition $+ : F \times F \to F$, *and* multiplication, $\cdot : F \times F \to F$, *which satisfy the following properties:*

(F1) For all $a, b \in F$, we have $a + b = b + a$; (*$+$ is* commutative*)*

(F2) For all $a, b, c \in F$, we have $(a + b) + c = a + (b + c)$; (*$+$ is* associative*)*

(F3) There is an element $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$; (*$+$ has an* identity*)*

(F4) For any $a \in F$, there is an element $-a \in F$ such that $a + (-a) = (-a) + a = 0$; (*$+$ has* inverses*)*

(F5) For all $a, b \in F$, we have $a \cdot b = b \cdot a$; (*$\cdot$ is* commutative*)*

(F6) For all $a, b, c \in F$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$; (*$\cdot$ is* associative*)*

(F7) There is an element $1 \in F$ such that $1 \neq 0$ and $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$; (*$\cdot$ has an* identity*)*

(F8) For any $a \in F \setminus \{0\}$, there is an element $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$; (*$\cdot$ has* inverses*)*

(F9) For any $a, b, c \in F$, $(a + b) \cdot c = a \cdot c + b \cdot c$. (distributivity*)*

Example 1.1.2. The real numbers $\mathbb{R}$ is a field. $\mathbb{Q}$ and $\mathbb{C}$ are also fields; $\mathbb{N}$ and $\mathbb{Z}$ are not.

[1] Later in the course, we discuss infinite sums, but we need to develop limits before this is possible.

All of the valid operations in arithmetic are encoded by the field axioms. In this course, you are allowed to use intuition about the consequences of these field axioms (e.g., "if $ab = ac$ and $a \neq 0$ then $b = c$" – this is true, and can be proven directly from the axioms, but a proof from the axioms isn't expected). We will also use familiar notation from arithmetic, such as $a + b + c$ for $(a + b) + c$, $a - b$ for $a + (-b)$, ab for $a \cdot b$, and $\frac{a}{b}$ for ab^{-1}.

Exercise 1.1.1. [TBB08, Exercise 1.3.5] Define $\mathbb{Z}/5\mathbb{Z}$ to be the set $\{0, 1, 2, 3, 4\}$ with $+$ and $\cdot$ defined by the following tables.

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\cdot$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Prove that $\mathbb{Z}/5\mathbb{Z}$ is a field.

Exercise 1.1.2. [TBB08, Exercise 1.3.6] Define $\mathbb{Z}/6\mathbb{Z}$ to be the set $\{0, 1, 2, 3, 4, 5\}$ with $+$ and $\cdot$ defined by the following tables.

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$\cdot$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

Prove that $\mathbb{Z}/6\mathbb{Z}$ isn't a field. Determine precisely which axioms (F1)–(F9) don't hold in $\mathbb{Z}/6\mathbb{Z}$.

1.2 Order structure

Definition 1.2.1. *An* ordered field *is a field F equipped with a binary relation $<$ satisfying the following properties*

(O1) If $a < b$ and $b < c$ then $a < c$; *($<$ is* transitive*)*

(O2) For all $a, b \in F$, exactly one of the following hold: $a = b$ or $a < b$ or $b < a$; *($<$ is a* total order*)*

(O3) For all $a, b, c \in F$, if $a < b$ then $a + c < b + c$; *($<$ is* compatible *with $+$)*

(O4) For all $a, b, c \in F$, if $a < b$ and $0 < c$ then $a \cdot c < b \cdot c$; *($<$ is* compatible *with $\cdot$)*

Example 1.2.2. The real numbers $\mathbb{R}$ is an ordered field. So is $\mathbb{Q}$, but $\mathbb{C}$ is not (if we try to use the operation $a + bi < c + di$ if $a = c$ and $b < d$, then (O2) fails; see Exercise 1.2.3 to see that no other relation works).

We can define $\leq, >, \geq$ in terms of $<$:

$$\begin{aligned} &a \leq b \text{ means } a < b \text{ or } a = b, \\ &a > b \text{ means } b < a, \text{ and} \\ &a \geq b \text{ means } a > b \text{ or } a = b. \end{aligned}$$

Exercise 1.2.1. Prove that if F is an ordered field then $1 > 0$ in F.
[*Hint. If not, then* $0 > 1$*; in this case, prove that* $-1 > 0$ *and use this to prove* $1 > 0$ *after all.*]

Exercise 1.2.2. Prove that if F is an ordered field then for all $a, b, c \in F$, if $a < b$ and $c < 0$ then $a \cdot c > b \cdot c$.

Exercise 1.2.3. Prove that for any binary relation $<$ on $\mathbb{C}$, we do not get an ordered field.
[*Hint. By contradiction. Consider two cases, depending on whether* $i > 0$ *or* $i < 0$.]

Exercise 1.2.4. Prove that for any binary relation on $\mathbb{Z}/5$ (from Exercise 1.1.1), we do not get an ordered field. More generally, if F is a field in which $1 + \cdots + 1 = 0$ (some finite, nonempty sum), then for any binary relation on F, we do not get an ordered field.

Exercise 1.2.5. Let F be an ordered field and let $a, b \in F$. Show that if $a < b$ then there exists $c \in F$ such that

$$a < c < b.$$

[*Hint. Use the previous exercise to justify that one can average* a *and* b.]

1.3 Bounded sets, infima, and suprema

Definition 1.3.1. *Let* F *be an ordered field, let* $S \subseteq F$ *be a subset, and let* $a \in F$*. Then* a *is an* upper bound *for* S *if for any* $x \in S$*,*

$$x \leq a.$$

Likewise, a *is a* lower bound *for* S *if for any* $x \in S$*,*

$$a \leq x.$$

S *is* bounded above *if there exists an upper bound for* S*.* S *is* bounded below *if there exists a lower bound for* S*.* S *is* bounded set *if it is bounded above and bounded below.*

Example 1.3.2. Consider $S := [0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$. An upper bound for S is 1. Another upper bound for S is 2. A lower bound for S is 0. Another lower bound for S is -100. S is bounded.

Example 1.3.3. Consider $\mathbb{N}_{\geq 1} = \{1, 2, 3, \dots\}$. This set is not bounded above. (However, we don't have enough information about $\mathbb{R}$ to prove this yet! It will be Theorem 1.3.13.) It is bounded below, for example by 0.

Example 1.3.4. Consider $T := \{1/n : n \in \mathbb{N}_{\geq 1}\}$. Then T is bounded below by 0 and bounded above by 1.

Sometimes a set S is bounded below by an element a in that set: for example $[0, 1]$ is bounded below by 0, which is in $[0, 1]$. In such a case, we say that a is the *minimum* of S, and write $a = \min S$. Likewise, if a set S contains an upper bound b, then b is the *maximum* of S, and write $a = \max S$.

However, ***some bounded sets don't contain their bounds***. For example, $T := \{1/n : n \in \mathbb{N}_{\geq 1}\}$ doesn't contain any element which is a lower bound for T (Exercise 1.3.1).

Remember: when you write $a = \min S$ ***or*** $a = \max S$***, this implies that*** $a \in S$***.***

Definition 1.3.5. *Let* F *be an ordered field, let* $S \subseteq F$ *be a nonempty subset, and let* $a \in F$*.* a *is a* least upper bound*(or* supremum*) for* S *if:*

(i) it is an upper bound for S*, and*

(ii) for any upper bound b *of* S*, we have* $b \geq a$*.*

When a is a least upper bound for S, we write

$$a = \sup S.$$

Similarly, a is a greatest lower bound *(or* infimum*) for S, written $a = \inf S$, if*

(i) it is a lower bound for S, and

(ii) for any lower bound b of S, we have $b \leq a$.

By writing $a = \sup S$ or $a = \inf S$, it suggests that the supremum and infimum of a set (if they exist) are unique; indeed, this is the case (Exercise 1.3.3).

We also have the following conventions:

- If S is not bounded above, we write $\sup S = \infty$.
- If S is not bounded below, we write $\inf S = -\infty$.
- We write $\sup \emptyset = -\infty$ and $\inf \emptyset = \infty$.

We emphasize however that $\pm\infty$ ***are not real numbers***, and we are not allowed to manipulate them as if they were.

(Note that every real number is both an upper bound and a lower bound for $\emptyset$ – so although $\emptyset$ has upper and lower bounds, it does not have *least, greatest* ones respectively.)

Example 1.3.6. Consider the set $S := (0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$. Then 1 is the least upper bound for S. Certainly, by definition, we see that 1 is an upper bound. For (ii), suppose for a contradiction that b is another upper bound and $b \leq 1$ doesn't hold. Then since $<$ is a total order, it follows that $b < 1$. By Exercise 1.2.5, we can find $x \in \mathbb{R}$ such that $b < x < 1$. But then $x \in S$ and since $b < x$, this contradicts that b is an upper bound.

Example 1.3.7. If $a = \max S$ then $a = \sup S$. To see this, first a must be an upper bound. For (ii), suppose that b is another upper bound for S. Then since $a \in S$ and $b \geq x$ for all $x \in S$, we have $b \geq a$.

Definition 1.3.8. *An ordered field F is* complete *if for any nonempty set $S \subseteq F$ which is bounded above, there is an element $a \in F$ which is a least upper bound for S. In short, if $S \subseteq F$ is nonempty and bounded above, then* $\sup S$ *exists.*

Example 1.3.9. The real numbers $\mathbb{R}$ is complete. (We will not prove this – but rather take it as a fundamental fact.)

Interesting fact 1.3.10. In fact (although we also won't prove it), $\mathbb{R}$ is the unique ordered field which is complete: in other words, if F is some other ordered field which is complete then there is a bijection $\phi : F \to \mathbb{R}$ which preserves $+, \cdot, \leq$, i.e., for all $a, b \in F$,

$$\phi(a + b) = \phi(a) + \phi(b),$$
$$\phi(a \cdot b) = \phi(a) \cdot \phi(b),$$

and

$$a \leq b \text{ if and only if } \phi(a) \leq \phi(b).$$

Proposition 1.3.11. *For any set $S \subseteq \mathbb{R}$ which is nonempty and bounded below,* $\inf S$ *exists.*

Proof. Exercise 1.3.4 □

Here is an example of how one might use completeness.

Proposition 1.3.12. *There exists a real number $a \in \mathbb{R}$ such that $a^2 = 2$.*

Proof. Set

$$S := \{x \in \mathbb{R} : x \geq 0 \text{ and } x^2 \leq 2\}.$$

Claim. For $b > 0$, b is an upper bound for S if and only if $b^2 \geq 2$.

To see this, first suppose $b^2 \geq 2$. If b isn't an upper bound then there is $x \in S$ such that $x > b$. But then $x^2 > x \cdot b > b^2 \geq 2$, which contradicts that $x \in S$.

Conversely, suppose that b is an upper bound for S and, for a contradiction, that $b^2 < 2$. Clearly $b \geq 1$. Then set $\epsilon := 2 - b^2 > 0$ and $x := b + \frac{\epsilon}{3b}$, and observe that

$$\begin{aligned} x^2 &= b^2 + \frac{2b\epsilon}{3b} + \frac{\epsilon^2}{9b^2} \\ &\leq b^2 + \frac{2\epsilon}{3} + \frac{2\epsilon}{9} && \text{(since } \epsilon \leq 2\text{)} \\ &< b^2 + \epsilon \\ &= 2. \end{aligned}$$

This means that $x \in S$, which is a contradiction since $x > b$. This proves the claim.

We see from the claim that S is bounded above, for example by 2. By the completeness of $\mathbb{R}$, we may set $a := \sup S$.

Since a is an upper bound for S, we have $a^2 \geq 2$. If $a^2 \neq 2$ then $a^2 > 2$, and we may set $\epsilon := a^2 - 2 > 0$ and $b := a - \frac{\epsilon}{2a}$. Then

$$b^2 = a^2 - 2a\frac{\epsilon}{2a} + \frac{\epsilon^2}{4a^2} = 2 + \frac{\epsilon^2}{4a^2} \geq 2.$$

Thus, by the claim, b is an upper bound for S, but since $b < a$, this contradicts that a is the *least* upper bound. □

The following may seem obvious, but it is an important consequence of completeness.

Theorem 1.3.13 (The Archimedean Property)**.** *The set $\mathbb{N}_{\geq 1}$ is not bounded above.*

Proof. Suppose for a contradiction that $\mathbb{N}_{\geq 1}$ were bounded above. Then by completeness, it would have a least upper bound, $a := \sup \mathbb{N}_{\geq 1}$. Since a is the least upper bound, $a - 1$ is not an upper bound, so that there exists some $m \in \mathbb{N}_{\geq 1}$ such that

$$m > a - 1.$$

But then $m + 1 \in \mathbb{N}_{\geq 1}$ and $m + 1 > a$, contradicting that a is an upper bound. □

Exercise 1.3.1. Let $T := \{1/n : n \in \mathbb{N}_{\geq 1}\}$.

(a) Show that the set T doesn't contain any element which is a lower bound for T. *In other words, we are not allowed to write* $\min T$.

(b) Determine $\inf T$ (and prove your answer is correct).

Exercise 1.3.2. Show that every finite set is bounded, and contains its bounds.

Exercise 1.3.3. Here we show that suprema are unique (when they exist). Let F be an ordered field, let S be a subset of F and suppose that a and a' are both least upper bounds for S. Prove that $a = a'$.

Exercise 1.3.4. Prove Proposition 1.3.11.

Exercise 1.3.5. Find the $\inf, \sup, \min$, and max of the following sets, or show that they don't exist.

(a) $\mathbb{Z}$;

(b) $\mathbb{N}_{\geq 1}$;

Exercise 1.3.6. (a) Define $S := \{x \in \mathbb{Q} : x^2 \leq 2\}$. Prove carefully (along the lines of the proof of Proposition 1.3.12); that $a := \sup S$ satisfies $a^2 = 2$. (b) Prove (without using the fact mentioned in Interesting Fact 1.3.10) that $\mathbb{Q}$ is not complete.

Exercise 1.3.7. Define $p(x) := x^2 - x$ and $S := \{x > 0 : p(x) \leq 1\}$. Prove, along the lines of the proof of Proposition 1.3.12 (and definitely without using quadratic formula) that $a := \sup S$ exists and satisfies $p(a) = 1$.

Exercise 1.3.8. Let $S \subseteq \mathbb{Z}$ be a nonempty set which is bounded above, and set $p := \sup S$. Prove that $p \in \mathbb{Z}$ (i.e., that $p = \max S$).

1.4 The absolute value and distances between numbers

The *absolute value* of a real number $a \in \mathbb{R}$ is defined by

$$|a| := \begin{cases} a, & \text{if } a \geq 0; \\ -a, & \text{if } a < 0. \end{cases}$$

In other words, $|a| = \max\{a, -a\}$.

Here are the basic properties of $|\cdot| : \mathbb{R} \to [0, \infty)$.

Proposition 1.4.1. *Let $x, y \in \mathbb{R}$. Then*

(i) $|-x| = |x|$,

(ii) $-|x| \leq x \leq |x|$,

(iii) $|xy| = |x| \cdot |y|$,

(iv) $|x + y| \leq |x| + |y|$ *(*triangle inequality*), and*

(v) $\big||x| - |y|\big| \leq |x - y|$.

Proof. Exercise 1.4.1. □

Definition 1.4.2. *Let $x, y \in \mathbb{R}$. The* distance *between x and y is*

$$d(x, y) := |x - y|.$$

Here are the basic properties of the distance $d : \mathbb{R} \times \mathbb{R} \to [0, \infty)$.

Proposition 1.4.3. *Let $x, y, z \in \mathbb{R}$. Then*

(i) $d(x, y) = d(y, x)$, *(d is symmetric)*

(ii) $d(x, y) = 0$ *iff* $x = y$, *and* *(distinct points are separated)*

(iii) $d(x, z) \leq d(x, y) + d(y, z)$. *(triangle inequality)*

Proof. Exercise 1.4.2. □

Exercise 1.4.1. Prove Proposition 1.4.1.

Exercise 1.4.2. Prove Proposition 1.4.3.

Exercise 1.4.3. [TBB08, Exercise 1.10.3] Let $x, L \in \mathbb{R}$ and let $\epsilon > 0$. Show that the following are equivalent:

(i) $|x - L| < \epsilon$,

(ii) $L - \epsilon < x < L + \epsilon$.

These conditions will appear prominently in the definitions of limits.

Exercise 1.4.4. [Sav17, Exercise 1.6.7] Let $S \subseteq \mathbb{R}$ be a bounded nonempty set. Prove that

$$\sup\{|a - b| : a, b \in S\} = \sup S - \inf S.$$

Chapter 2

Sequences

2.1 Sequences and boundedness

Definition 2.1.1. *A* sequence *of real numbers is a function* $f : \mathbb{N}_{\geq 1} \to \mathbb{R}$.

The above is the formal definition of a sequence. However, in practice, we think of a sequence $f : \mathbb{N}_{\geq 1} \to \mathbb{R}$ as an infinite list of numbers

$$f(1), f(2), f(3), \dots.$$

As a convention, we rarely write a sequence as a function, and instead we use the following two ways of writing sequences:

$$(a_n)_{n=1}^{\infty} \quad \text{or} \quad (a_1, a_2, a_3, \dots).$$

In both cases, the sequence corresponds to a function $f : \mathbb{N}_{\geq 1} \to \mathbb{R}$ by $a_n = f(n)$.

Definition 2.1.2. *A sequence* $(a_n)_{n=1}^{\infty}$ *is* bounded *if the set* $\{a_n : n \in \mathbb{N}_{\geq 1}\} \subseteq \mathbb{R}$ *is bounded. Likewise, we may define* $(a_n)_{n=1}^{\infty}$ *to be* bounded above *or* bounded below*, by the corresponding property for* $\{a_n : n \in \mathbb{N}_{\geq 1}\}$.

Note that in many textbooks (including the ones recommended in the syllabus), sequences are denoted by $\{a_n\}_{n=1}^{\infty}$ or even $\{a_n\}$. We emphasize that ***sequences are not sets***, for two reasons:

(i) The order in a sequence matters. Sets, by contrast, are not ordered: $\{1, 2\} = \{2, 1\}$.

(ii) Repeats in a sequence matter. Sets, by contrast, do not see repeats: $\{1, 1\} = \{1\}$.

Exercise 2.1.1. Determine which of the following sequences are bounded.

(a) $(1, 2, 3, 4, \dots)$

(b) $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$

(c) $(a_n)_{n=1}^{\infty}$ defined by

$$a_n := \begin{cases} k, & \text{if } n = 2^k \text{for some } k \in \mathbb{N}_{\geq 1}; \\ 0, & \text{otherwise.} \end{cases}$$

(d) $(a_n)_{n=1}^{\infty}$ defined by

$$\begin{aligned} a_1 &:= 1, \\ a_2 &:= 2, \\ a_n &:= \frac{a_{n-1} + n a_{n-2}}{n+1}, \quad \text{for } n \geq 3. \end{aligned}$$

2.2 Convergence

One of the primary things that we are interested in with sequences is their "long-term behaviour": that is, what a_n looks like when n is large. To this end, the first thing we might ask is: does a_n get close to some particular number, as n gets large? This is what convergence of sequences means. Here is the formal definition:

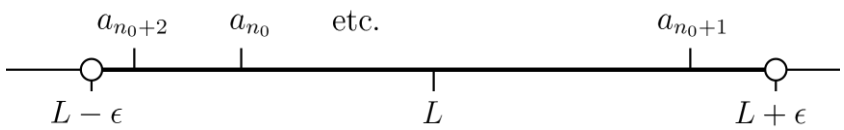
Definition 2.2.1. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers and let $L \in \mathbb{R}$. We say that $(a_n)_{n=1}^{\infty}$* converges *to L if for every $\epsilon > 0$ there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that for all $n \geq n_0$,*

$$|a_n - L| < \epsilon.$$

In this case, we may write

$$L = \lim_{n\to\infty} a_n \quad \text{or} \quad a_n \to L \text{ as } n \to \infty.$$

If a sequence does not converge to any real number, then we say it diverges.



Example 2.2.2. The sequence $(a_n)_{n=1}^{\infty} = \left(\frac{1}{n}\right)_{n=1}^{\infty}$ converges to 0. To prove this, we work from the definition.

Let $\epsilon > 0$ be given. We must prove that there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that for all $n \geq n_0$,

$$|a_n - 0| < \epsilon.$$

By the Archimedean Property (Theorem 1.3.13) there exists n_0 such that $n_0 > \frac{1}{\epsilon}$. Then if $n \geq n_0$,

$$\frac{1}{n} \leq \frac{1}{n_0} < \epsilon.$$

Since $\frac{1}{n} > 0$, we have $|a_n - 0| = |\frac{1}{n}| < \epsilon$.

Example 2.2.3. Let us prove that

$$\lim_{n\to\infty} \frac{2n^2+1}{n^2+n} = 2.$$

To prove this, we first rearrange

$$\begin{aligned}
\left|\frac{2n^2+1}{n^2+n} - 2\right| &= \left|\frac{2n^2+1-2(n^2+n)}{n^2+n}\right| \\
&= \left|\frac{1-2n}{n^2+n}\right| \\
&\leq \frac{1}{n^2+n} + \frac{2n}{n^2+n} \\
&= \frac{1}{n^2+n} + \frac{2}{n+1} \\
&\leq \frac{1}{n} + \frac{2}{n}.
\end{aligned}$$

Now, given $\epsilon > 0$, pick n_0 such that $n_0 > \frac{3}{\epsilon}$. Then if $n \geq n_0$,

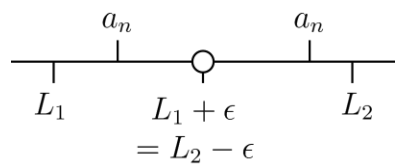
$$\left|\frac{2n^2+1}{n^2+n} - 2\right| \leq \frac{3}{n} \leq \frac{3}{n_0} < \epsilon.$$

Proposition 2.2.4 (Uniqueness of limits)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence and let $L_1, L_2 \in \mathbb{R}$. If*

$$\lim_{n\to\infty} a_n = L_1 \quad \textit{and} \quad \lim_{n\to\infty} a_n = L_2$$

then $L_1 = L_2$.

Proof. Suppose for a contradiction that $L_1 \neq L_2$, so without loss of generality, $L_1 < L_2$. Then define $\epsilon := \frac{L_2 - L_1}{2}$.



Since $\lim_{n\to\infty} a_n = L_1$, there exists n_0 such that for all $n \geq n_0$,

$$L_1 - \epsilon < a_n < L_1 + \epsilon.^{1}$$

Using the second inequality and the definition of ϵ, we have for $n \geq n_0$, that

$$a_n < L_1 + \epsilon = \frac{L_1 + L_2}{2}.$$

Likewise, since $\lim_{n\to\infty} a_n = L_2$, there exists m_0 such that for all $n \geq m_0$,

$$L_2 - \epsilon < a_n < L_2 + \epsilon,$$

and from the first inequality we get for $n \geq m_0$,

$$a_n > L_2 - \epsilon = \frac{L_1 + L_2}{2}.$$

Take $n := \max\{n_0, m_0\}$, so that $n \geq n_0$ and $n \geq m_0$. Then we have

$$a_n < \frac{L_1 + L_2}{2} < a_n,$$

a contradiction. □

Complementing the definition of limits is the following notion of divergence to $\pm\infty$.

Definition 2.2.5. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. We say that $(a_n)_{n=1}^{\infty}$* diverges to ∞ *if for every $R > 0$ there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that for all $n \geq n_0$,*

$$a_n > R.$$

Likewise, we say that $(a_n)_{n=1}^{\infty}$ diverges to $-\infty$ *if for every $R > 0$ there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that for all $n \geq n_0$,*

$$a_n < -R.$$

If $(a_n)_{n=1}^{\infty}$ diverges to ∞, we may write

$$\infty = \lim_{n\to\infty} a_n \quad \textit{or} \quad a_n \to \infty \textit{ as } n \to \infty.$$

If it diverges to $-\infty$, we may write

$$-\infty = \lim_{n\to\infty} a_n \quad \textit{or} \quad a_n \to -\infty \textit{ as } n \to \infty.$$

[1] We are using Exercise 1.4.3 here.

Again, we caution that ∞ is not a real number, so we must be careful not to treat it as one when we write something like $\lim_{n\to\infty} a_n = \infty$.

Example 2.2.6. $\lim_{n\to\infty} n = \infty$. To prove this, given R, we just choose any $n_0 > R$.

Example 2.2.7. The sequence $((-1)^n)_{n=1}^{\infty}$ diverges, but it does not diverge to ∞ or $-\infty$ (Exercise 2.2.1). Note that this sequence is in fact bounded.

Proposition 2.2.8. *Let $(a_n)_{n=1}^{\infty}$ be a sequence which converges to some number $L \in \mathbb{R}$. Then $(a_n)_{n=1}^{\infty}$ is bounded.*

Proof. Since $\lim_{n\to\infty} a_n = L$, using $\epsilon := 1$, there exist n_0 such that for all $n \geq n_0$,

$$|a_n - L| < 1.$$

Take

$$M := \max\{a_1, a_2, \dots, a_{n_0-1}, L+1\}.$$

(This maximum exists since the set is finite.)
Claim. M is an upper bound for the set $\{a_n : n \in \mathbb{N}_{\geq 1}\}$.

To see this, let $n \in \mathbb{N}_{\geq 1}$, and we must show $a_n \leq M$. If $n < n_0$ then a_n is among the list $a_1, \dots, a_{n_0-1}$, so that $a_n \leq M$ by the definition of M. On the other hand, if $n \geq n_0$ then by the choice of n_0, we have $a_n < L + 1 \leq M$. This proves the claim.

Setting

$$M' := \min\{a_1, \dots, a_{n_0-1}, L-1\},$$

then the same reasoning shows that M' is a lower bound for the set $\{a_n : n \in \mathbb{N}_{\geq 1}\}$. Thus the sequence is bounded both above and below, as required. □

A couple notes.

- The previous proposition can be reformulated as: if a sequence is not bounded then it diverges. This can be used to show that a sequence which diverges to $\pm\infty$ does not converge (a fact which is suggested by the terminology "*diverges* to $\pm\infty$").
- The converse of the previous proposition does not hold: a sequence which is bounded might not converge (Example 2.2.7).

Exercise 2.2.1. Prove that the sequence $((-1)^n)_{n=1}^{\infty}$ from Example 2.2.7 does not converge.
[*Begin by supposing it does converge to some $L \in \mathbb{R}$, and reach a contradiction.*]

Exercise 2.2.2. Prove that $\left(\frac{2}{\sqrt{n}}\right)_{n=1}^{\infty}$ converges to 0. Prove this directly from the definition, rather than using any theorems or rules (for example from previous calculus courses).

Exercise 2.2.3. Suppose that $(a_n)_{n=1}^{\infty}$ is a sequence that converges to some $L \in \mathbb{R}$, and define a sequence $(b_n)_{n=1}^{\infty}$ by

$$b_n := a_{2n}, \quad n \in \mathbb{N}_{\geq 1}.$$

Prove that $(b_n)_{n=1}^{\infty}$ also converges to L.

Exercise 2.2.4. Suppose that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences that both converge to some $L \in \mathbb{R}$, and define a sequence

$$(c_n)_{n=1}^{\infty} = (a_1, b_1, a_2, b_2, a_3, b_3, \dots).$$

Prove that $(c_n)_{n=1}^{\infty}$ converges to L.

Exercise 2.2.5. [TBB08, Exercise 2.4.15 and 2.4.16] If $(a_n)_{n=1}^{\infty}$ is a sequence of nonnegative real numbers converging to $L \geq 0$, prove that the sequence $(\sqrt{a_n})_{n=1}^{\infty}$ converges to $\sqrt{L}$. [*Hint. You might find it easiest to treat the cases $L = 0$ and $L > 0$ separately.*]

Exercise 2.2.6. Let $(a_n)_{n=1}^{\infty}$ be a sequence of nonzero numbers. Prove that if $\lim_{n\to\infty} a_n = \infty$ then $\lim_{n\to\infty} \frac{1}{a_n} = 0$. Is the converse true?

Exercise 2.2.7. Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence and let $(b_n)_{n=1}^{\infty}$ be a sequence that converges to 0. Prove that $(a_n b_n)_{n=1}^{\infty}$ converges to 0.

2.3 Properties of limits

In this section, we will establish many of the tools which one likes to apply in everyday limit computations.

Proposition 2.3.1 (Algebra of limits)**.** *Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences converging to L_a and L_b respectively, and let $c \in \mathbb{R}$. Then*

(i) $(a_n + b_n)_{n=1}^{\infty}$ *converges to* $L_a + L_b$.

(ii) $(ca_n)_{n=1}^{\infty}$ *converges to* cL_a.

(iii) $(a_n b_n)_{n=1}^{\infty}$ *converges to* $L_a L_b$.

(iv) If $a_n \neq 0$ *for all* n *and* $L_a \neq 0$ *then* $\left(\frac{1}{a_n}\right)_{n=1}^{\infty}$ *converges to* $\frac{1}{L_a}$.

Remark 2.3.2. (i) and (ii) of the above proposition say that the set V of converging sequences forms a vector space, and that the map $(a_n)_{n=1}^{\infty} \mapsto \lim_{n\to\infty} a_n$ is a linear map from V to $\mathbb{R}$.

Proof. (i): Let $\epsilon > 0$ be given. Since $\lim_{n\to\infty} a_n = L_a$, there exists n_a such that for all $n \geq n_a$, $|a_n - L_a| < \frac{\epsilon}{2}$. Since $\lim_{n\to\infty} b_n = L_b$, there exists n_b such that for all $n \geq n_b$, $|b_n - L_b| < \frac{\epsilon}{2}$. Set $n_0 := \max\{n_a, n_b\}$. Then for $n \geq n_0$,

$$\begin{aligned} |(a_n + b_n) - (L_a + L_b)| &\leq |a_n - L_a| + |b_n - L_b| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

(since $n \geq n_a$ and $n \geq n_b$).

(ii)-(iv) are Exercise 2.3.5 □

Proposition 2.3.3. *Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be converging sequences. If*

$$a_n \leq b_n \quad \textit{for all } n$$

then

$$\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n.$$

Proof. This is very similar to the proof of Proposition 2.2.4. Suppose for a contradiction that $L_a > L_b$. Define $\epsilon := \frac{L_a - L_b}{2}$. Since $\lim_{n\to\infty} a_n = L_a$, there exists n_a such that for all $n \geq n_a$,

$$L_a + \epsilon > a_n > L_a - \epsilon = \frac{L_a + L_b}{2}.$$

Likewise, since $\lim_{n\to\infty} b_n = L_b$, there exists n_b such that for all $n \geq n_b$,

$$L_b - \epsilon < b_n < L_b + \epsilon = \frac{L_a + L_b}{2}.$$

Using $n := \max\{n_a, n_b\}$, we get $a_n > \frac{L_a+L_b}{2} > b_n$, a contradiction. □

Corollary 2.3.4. *Let $(a_n)_{n=1}^{\infty}$ be a converging sequence such that*

$$m \leq a_n \leq M \quad \textit{for all } n.$$

Then

$$m \leq \lim_{n\to\infty} a_n \leq M.$$

Proof. Since $\lim_{n\to\infty} M = M$, we apply the previous proposition with $(b_n)_{n=1}^{\infty} := (M, M, M, \dots)$ to get the second inequality. Similarly for the first. □

Theorem 2.3.5 (Squeeze Theorem)**.** *Let $(a_n)_{n=1}^{\infty}, (b_n)_{n=1}^{\infty}, (c_n)_{n=1}^{\infty}$ be sequences such that:*

(i) $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ converge to the same number L, and

(ii) $a_n \leq b_n \leq c_n$ for all n.

Then $(b_n)_{n=1}^{\infty}$ also converges to L.

Proof. Let $\epsilon > 0$ be given. Using the definition of convergence for both $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ and taking the maximum n_0 from both, there exists n_0 such that for all $n \geq n_0$,

$$L - \epsilon < a_n < L + \epsilon \quad \text{and} \quad L - \epsilon < c_n < L + \epsilon.$$

Thus,

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon$$

for all $n \geq n_0$, as required. □

Example 2.3.6. Let us show that $\lim_{n\to\infty} \frac{\sin(n)}{n} = 0$. We use $a_n := -\frac{1}{n}$, $b_n := \frac{\sin(n)}{n}$, and $c_n := \frac{1}{n}$. Since $-1 \leq \sin(x) \leq 1$ for all x, it follows that

$$a_n \leq b_n \leq c_n \quad \text{for all } n.$$

Moreover, by Example 2.2.2, $c_n \to 0$, and then using Algebra of Limits, $a_n \to -0 = 0$. Hence the Squeeze Theorem applied and shows that $b_n \to 0$.

Here is a variant on the Squeeze Theorem for divergence to $\pm\infty$.

Proposition 2.3.7. *Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences such that*

$$a_n \leq b_n \quad \textit{for all } n.$$

Then

(i) If $\lim_{n\to\infty} a_n = \infty$ *then* $\lim_{n\to\infty} b_n = \infty$.

(ii) If $\lim_{n\to\infty} b_n = -\infty$ *then* $\lim_{n\to\infty} a_n = -\infty$.

Proof. Exercise 2.3.1 □

Example 2.3.8. Fix $r \in \mathbb{R}$. The sequence $(r^n)_{n=1}^{\infty}$ is an important sequence called the *geometric sequence*. Let us analyze its convergence behaviour.

The cases $r = 0$ and $r = 1$ are clear, as in these cases, the sequence is constant and so the limit exists (and equals r).

If $r > 1$, then set $x := r - 1 > 0$. By the Binomial Theorem, we have

$$r^n = (1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots + x^n \geq nx.$$

Since $nx \to \infty$, it follows by Proposition 2.3.7 that $r^n \to \infty$.

If $r \in (0,1)$, then set $s := r^{-1}$, so that $r^n = \frac{1}{s^n}$. Since $s^n \to \infty$, it follows by Exercise 2.2.6 that $r^n \to 0$.

If $r \in (-1,0)$ then set $s := |r| \in (0,1)$. We have

$$-s^n \leq r^n \leq s^n$$

for all n, and by the previous case, $s^n \to 0$. Hence $-s^n \to 0$ also and then by the Squeeze Theorem, $r^n \to 0$.

If $r = -1$ then we already saw in Example 2.2.7 that the sequence $(r^n)_{n=1}^{\infty}$ diverges.

Finally, if $r < -1$ then since $|r|^n \to \infty$, we see that the sequence $(r^n)_{n=1}^{\infty}$ is bounded neither above (due to the even terms) nor below (due to the odd terms). It diverges, but doesn't diverge to $\pm\infty$.

Here is an important fact.

Proposition 2.3.9. *For any real number* $\alpha \in \mathbb{R}$*, there exists a sequence* $(a_n)_{n=1}^{\infty}$ *of rational numbers (i.e.,* $a_n \in \mathbb{Q}$ *for all* n*) such that*

$$\lim_{n\to\infty} a_n = \alpha.$$

Proof. We will construct a_n such that

$$\alpha - \frac{1}{n} \leq a_n \leq \alpha. \tag{2.1}$$

It will follow from the Squeeze Theorem (Theorem 2.3.5) that

$$\lim_{n\to\infty} a_n = \alpha.$$

We may find an integer $p \in \mathbb{Z}$ such that

$$n\alpha - 1 \leq p \leq n\alpha.$$

(For example, let p be the supremum of $\{k \in \mathbb{Z} : k \leq n\alpha\}$, a set which is bounded above by $n\alpha$, and is nonempty by the Archimedean Property. By Exercise 1.3.8, $p \in \mathbb{Z}$. Also, we have $p \leq n\alpha + 1$, or else $p - 1$ would be a smaller upper bound for the set, and on the other hand, $p \geq n\alpha - 1$, or else $p + 1$ would be in the set, contradicting that p is an upper bound.)

Setting $a_n := \frac{p}{n}$, the inequalities (2.1) clearly hold. □

Exercise 2.3.1. Prove Proposition 2.3.7.

Exercise 2.3.2. Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two convergent sequences, such that $a_n < b_n$ for all n. Must it hold that

$$\lim_{n\to\infty} a_n < \lim_{n\to\infty} b_n?$$

Give a proof or counterexample.

Exercise 2.3.3. [TBB08, Exercise 2.7.5] Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences. Which of the following is true?

(a) If $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are both divergent, then so is $(a_n + b_n)_{n=1}^{\infty}$.

(b) If $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are both divergent, then so is $(a_n b_n)_{n=1}^{\infty}$.

(c) If $(a_n)_{n=1}^{\infty}$ and $(a_n + b_n)_{n=1}^{\infty}$ are both convergent, then so is $(b_n)_{n=1}^{\infty}$.

(d) If $(a_n)_{n=1}^{\infty}$ and $(a_n b_n)_{n=1}^{\infty}$ are both convergent, then so is $(b_n)_{n=1}^{\infty}$.

(e) If $(a_n)_{n=1}^{\infty}$ is convergent, then so is $\left(\frac{1}{a_n}\right)_{n=1}^{\infty}$.

(f) If $(a_n)_{n=1}^{\infty}$ is convergent, then so is $((a_n)^2)_{n=1}^{\infty}$.

(g) If $((a_n)^2)_{n=1}^{\infty}$ is convergent, then so is $(a_n)_{n=1}^{\infty}$.

(h) If $((a_n)^2)_{n=1}^{\infty}$ is convergent and $a_n \geq 0$ for all n, then so is $(a_n)_{n=1}^{\infty}$.

Exercise 2.3.4. [TBB08, Exercise 2.8.9] Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive numbers. Prove that if $\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$ exists and is strictly less than 1 ($\lim_{n\to\infty} \frac{a_{n+1}}{a_n} < 1$) then

$$\lim_{n\to\infty} a_n = 0.$$

Exercise 2.3.5. Prove parts (ii)-(iv) of Proposition 2.3.1

Exercise 2.3.6. (Big-O notation as $n \to \infty$) For sequences $(a_n)_{n=1}^{\infty}$, and $(b_n)_{n=1}^{\infty}$ (with $b_n \geq 0$), we write

$$a_n = O(b_n) \quad (\text{as } n \to \infty)$$

to mean that there exists N and a constant $K > 0$ such that $|a_n| \leq K c_n$ for all n. (This notation is very popular, e.g., in complexity theory in computer science.)

Check that the following hold with this definition.

(i) If $a_n = O(b_n)$ and $c_n = O(b_n)$ then $a_n + c_n = O(b_n)$.

(ii) If $a_n = O(b_n)$ and $c_n = O(d_n)$ then $a_n c_n = O(b_n d_n)$.

(iii) If $a_n = O(b_n)$ and $b_n = O(c_n)$ then $a_n = O(c_n)$.

(iv) If $a_n = O(b_n)$ and $b_n \to 0$ then $a_n \to 0$.

(v) If p is any polynomial and $a > 1$ then $p(n) = O(a^n)$ and $\frac{1}{p(n)} = O(a^{-n})$.

2.4 Monotone convergence criterion

Here we study sequences which are *monotone*, meaning that they always tend in the same direction (increasing or decreasing).

Definition 2.4.1. *Let $(a_n)_{n=1}^{\infty}$ be a sequence.*

(i) We say $(a_n)_{n=1}^{\infty}$ is (weakly) increasing *if*

$$a_1 \leq a_2 \leq a_3 \leq \cdots .$$

(ii) We say $(a_n)_{n=1}^{\infty}$ is strictly increasing *if*

$$a_1 < a_2 < a_3 < \cdots .$$

(iii) We say $(a_n)_{n=1}^{\infty}$ is (weakly) decreasing *if*

$$a_1 \geq a_2 \geq a_3 \geq \cdots .$$

(iv) We say $(a_n)_{n=1}^{\infty}$ is strictly decreasing *if*

$$a_1 > a_2 > a_3 > \cdots .$$

(v) Finally, we say $(a_n)_{n=1}^{\infty}$ is monotone *if it is either increasing or decreasing.*

Note that in some textbooks (including [TBB08]), "increasing" is used where we use "strictly increasing", and "nondecreasing" is used where we use "increasing".[2] In our context, the concept of (weakly) increasing and decreasing sequences is most important, and strictness is secondary.

The following is a powerful result about monotone sequences: the converge exactly when they are bounded.

Theorem 2.4.2 (Monotone Convergence Criterion)**.** *Let $(a_n)_{n=1}^{\infty}$ be a monotone sequence. Then it converges if and only if it is bounded.*

Specifically, if $(a_n)_{n=1}^{\infty}$ is bounded and increasing, then it converges to $\sup\{a_n : n \in \mathbb{N}_{\geq 1}\}$, whereas if it is bounded and decreasing, then it diverges to $\inf\{a_n : n \in \mathbb{N}_{\geq 1}\}$.

Proof. If the sequence converges then it is bounded by Proposition 2.2.8. For the converse, we will do the case that $(a_n)_{n=1}^{\infty}$ is bounded and increasing. The decreasing case is exactly the same (or alternatively, it can be recovered from the increasing case by considering the sequence $(-a_n)_{n=1}^{\infty}$).

Set $L := \sup\{a_n : n \in \mathbb{N}_{\geq 1}\}$. To show convergence, let $\epsilon > 0$ be given. Since L is the least upper bound for $\{a_n : n \in \mathbb{N}_{\geq 1}\}$, $L - \epsilon$ is not an upper bound, so there exists n_0 such that

$$L - \epsilon < a_{n_0}.$$

This is our n_0; now suppose $n \geq n_0$. Then since the sequence is increasing,

$$a_n \geq a_{n_0} > L - \epsilon.$$

On the other hand, since L is an upper bound, $a_n \leq L$. Altogether,

$$L - \epsilon < a_n \leq L < L + \epsilon,$$

as required. □

[2] This use of "nondecreasing" is confusing. It is natural to think that "nondecreasing" means "not decreasing", but there are sequences which are neither increasing nor decreasing, such as $((-1)^n)_{n=1}^{\infty}$.

Example 2.4.3. Consider the sequence $\left(\frac{1}{\sqrt{n}}\right)_{n=1}^{\infty}$. This sequence is decreasing, since for all n, we have

$$n \leq n+1 \Rightarrow \sqrt{n} \leq \sqrt{n+1} \Rightarrow \frac{1}{\sqrt{n}} \geq \frac{1}{\sqrt{n+1}}.$$

It is also bounded below by 0. By the Monotone Convergence Criterion, it follows that this sequence converges to its infimum, which is 0.

Exercise 2.4.1. [Sav17, Exercise 2.3.1] Suppose that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences such that $(a_n)_{n=1}^{\infty}$ is increasing, $a_n \leq b_n$ for all n, and $(b_n)_{n=1}^{\infty}$ converges. Prove that $(a_n)_{n=1}^{\infty}$ converges.

Exercise 2.4.2. [TBB08, Exercise 2.9.2] Define a sequence $(a_n)_{n=1}^{\infty}$ by

$$\begin{aligned} a_1 &:= 1, \\ a_{n+1} &:= \sqrt{1+a_n}, \quad n \geq 1. \end{aligned}$$

Prove that this sequence is increasing, and determine whether it converges.
[*Hint. Define* $\gamma := \frac{1+\sqrt{5}}{2}$*, so that* $1+\gamma = \gamma^2$*. Use induction to prove that* $a_n \leq \gamma$ *for all* n*, and then use this to prove the sequence is increasing.*]

Exercise 2.4.3. Let $(a_n)_{n=1}^{\infty}$, $(b_n)_{n=1}^{\infty}$, and $(c_n)_{n=1}^{\infty}$ be sequences such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n.$$

(a) If $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are both increasing, must $(b_n)_{n=1}^{\infty}$ be?

(b) If $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are both bounded, must $(b_n)_{n=1}^{\infty}$ be?

(c) If $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are both convergenc, must $(b_n)_{n=1}^{\infty}$ be?

2.5 Subsequences

Given a sequence

$$(a_1, a_2, a_3, \dots)$$

we may want to forget parts of the sequence and only take certain terms. What we get when we do this is a *subsequence*. Here is the formal definition.

Definition 2.5.1. *Let* $(a_n)_{n=1}^{\infty}$ *be a sequence. A* subsequence *is a sequence of the form* $(a_{n_k})_{k=1}^{\infty}$*, where* $(n_k)_{k=1}^{\infty}$ *is a strictly increasing sequence with* $n_k \in \mathbb{N}_{\geq 1}$ *for all* k*.*

One needs to be careful not to get confused with the notation of subsequences. Note that we typically index a subsequence with a different variable (k instead of n in the above definition). Since the subsequence arises *from* the sequence, we typically don't write something like $(b_n)_{n=1}^{\infty}$ for a subsequence.

If we think of a sequence as a list, a subsequence arises by erasing some of the entries of the list (specifically, erasing all of the entries that aren't in $\{n_k : k \in \mathbb{N}_{\geq 1}\}$).

Since we require the indices of the subsequence, $(n_k)_{k=1}^{\infty}$, to be strictly increasing, the following are ***not*** examples of subsequences:

$$(a_1, a_1, a_1, \dots),$$
$$(a_2, a_1, a_4, a_3, a_6, a_5, \dots).$$

Example 2.5.2. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the sequence of *even terms* $(a_{2n})_{n=1}^{\infty}$ is a subsequence. (We could write this sequence as $(a_{n_k})_{k=1}^{\infty}$ by setting $n_k := 2k$.)

Example 2.5.3. Define $(a_n)_{n=1}^{\infty}$ by

$$a_n := n + \frac{1}{n}.$$

Here are a couple subsequences of this sequence:

(i) Taking $n_k := 2^k$, we have

$$(a_{n_k})_{k=1}^{\infty} = (1+1, 2+2^{-1}, 2^2+2^{-2}, 2^3+2^{-3}, \dots).$$

(ii) Taking $n_k := k+1$, we have

$$(a_{n_k})_{k=1}^{\infty} = \left(k+1+\frac{1}{k+1}\right)_{k=1}^{\infty}.$$

Proposition 2.5.4. *If $(a_n)_{n=1}^{\infty}$ converges to L, then any subsequence also converges to L.*

Proof. Exercise 2.5.1. □

Proposition 2.5.5. *Every sequence contains a monotone subsequence.*

Proof. Let $(a_n)_{n=1}^{\infty}$ be a sequence. We begin by trying to construct an decreasing sequence in a naïve way: we look for points m of the sequence where a_m is bigger than all later terms (i.e., $a_m \geq a_n$ for all $n \geq m$). We'll call such indices m "turn-back points". The idea is that, if there are infinitely many of these, then we will get an decreasing sequence by taking them all; if there are only finitely many, then there must be a increasing subsequence. Let us explain: we have two cases.
Case 1. There are infinitely many turn-back points. In this case, we may take a sequence $n_1 < n_2 < \cdots$ such that each n_k is a turn-back point. In particular, we have $a_{n_1} \geq a_{n_2} \geq \cdots$, so the subsequence $(a_{n_k})_{k=1}^{\infty}$ is decreasing.
Case 2. There are only finitely many turn-back points. In this case, let M be the last turn-back point, so that every $n > M$ is not a turn-back point. Take some $n_1 > M$; since it is not a turn-back point, there is some $n_2 > n_1$ such that $a_{n_1} < a_{n_2}$. Likewise, n_2 is not a turn-back point, so there is some $n_3 > n_2$ such that $a_{n_2} < a_{n_3}$. Continuing in this way, we obtain a subsequence $(a_{n_k})_{k=1}^{\infty}$ that is strictly increasing. □

Here is an important corollary of the previous proposition.

Corollary 2.5.6 (Bolzano–Weierstrass Theorem)**.** *Every bounded sequence has a convergent subsequence.*

Proof. Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence. By the previous proposition, it has a subsequence $(a_{n_k})_{k=1}^{\infty}$ which is monotone. This subsequence is, of course, also bounded. Hence by the Monotone Convergence Criterion, $(a_{n_k})_{k=1}^{\infty}$ converges. □

Exercise 2.5.1. Prove Proposition 2.5.4.

Exercise 2.5.2. [TBB08, Exercise 2.11.3] Suppose that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences and $(a_{n_k})_{k=1}^{\infty}$, $(b_{m_k})_{k=1}^{\infty}$ (respectively) are subsequences. Must $(a_{n_k}+b_{m_k})_{k=1}^{\infty}$ be a subsequence of $(a_n+b_n)_{n=1}^{\infty}$?

Exercise 2.5.3. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Prove that $(a_n)_{n=1}^{\infty}$ is not bounded above if and only if it has a subsequence $(a_{n_k})_{k=1}^{\infty}$ which diverges to ∞.

Exercise 2.5.4. [TBB08, Exercise 2.11.11] Give an example of a sequence $(a_n)_{n=1}^{\infty}$ such that:

(i) For every natural number $k \in \mathbb{N}_{\geq 1}$, there is a subsequence of $(a_n)_{n=1}^{\infty}$ converging to k, and

(ii) For every $\alpha \in \mathbb{R}$, if α is a limit of a subsequence of $(a_n)_{n=1}^{\infty}$, then $\alpha \in \mathbb{N}_{\geq 1}$.

Exercise 2.5.5. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Suppose that for every positive integer M, there exists a subsequence $(a_{n_k})_{k=1}^{\infty}$ such that $\lim_{k\to\infty} a_{n_k} = \frac{1}{M}$. Prove that there is a subsequence $(a_{m_k})_{k=1}^{\infty}$ such that $\lim_{k\to\infty} a_{m_k} = 0$. Is it possible to even arrange that $(a_{m_k})_{k=1}^{\infty}$ is monotone?

2.6 Cauchy sequences

It is useful (especially in developing further theory) to characterize when a sequence converges, in a way that doesn't require knowing what the sequence converges to. This is what we will do in this section. Note that the Monotone Convergence Theorem does this, in the special case of monotone sequences: it says that a monotone sequence converges if and only if it is bounded; the condition "it is bounded" makes no reference to any (potential) value for the limit. But that theorem is limited to monotone sequences, and we already know that the condition "it is bounded" will not characterize convergence for non-monotone sequences.

Definition 2.6.1. *A sequence $(a_n)_{n=1}^{\infty}$ is* Cauchy *if for all $\epsilon > 0$ there exists n_0 such that for all $m, n \geq n_0$,*

$$|a_m - a_n| < \epsilon.$$

Theorem 2.6.2 (Cauchy Convergence Criterion)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Then it converges if and only if it is Cauchy.*

Proof. $\Rightarrow$: Assume that $(a_n)_{n=1}^{\infty}$ converges to some value, $L \in \mathbb{R}$. We will show that the definition of a Cauchy sequence holds; so let $\epsilon > 0$ be given. By the definition of convergence with $\epsilon/2$ in place of ϵ, there exists n_0 such that for all $n \geq n_0$,

$$|a_n - L| < \frac{\epsilon}{2}.$$

Now, if $m, n \geq n_0$ then

$$\begin{aligned} |a_m - a_n| &\leq |a_m - L| + |L - a_n| && (\Delta\text{-inequality}) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

as required.

$\Leftarrow$: Assume that $(a_n)_{n=1}^{\infty}$ is Cauchy. First, we must establish that $(a_n)_{n=1}^{\infty}$ is bounded; this is Exercise 2.6.1. Now, we may apply the Bolzano–Weierstrass Theorem (Corollary 2.5.6) to get a subsequence $(a_{n_k})_{k=1}^{\infty}$ which converges to some value, $L \in \mathbb{R}$. Let us show that $(a_n)_{n=1}^{\infty}$ converges to L.

Given $\epsilon > 0$, by the definition of a Cauchy sequence, pick n_0 such that for all $m, n \geq n_0$,

$$|a_m - a_n| < \frac{\epsilon}{2}.$$

Next, since $\lim_{k\to\infty} a_{n_k} = L$, we may also find k_0 such that for all $k \geq k_0$,

$$|a_{n_k} - L| < \frac{\epsilon}{2}.$$

Let us now fix $k_1 \geq k_0$ such that $n_k \geq n_0$; this is possible since the sequence $(n_k)_{k=1}^{\infty}$ is strictly increasing.

For $n \geq n_0$, we now have

$$\begin{aligned} |a_n - L| &\leq |a_n - a_{n_k}| + |a_{n_k} - L| && (\Delta\text{-inequality}) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

as required. □

Exercise 2.6.1. Finish the proof of Theorem 2.6.2, by proving (*without using that theorem*) that every Cauchy sequence is bounded.

Exercise 2.6.2. [TBB08, Exercise 2.12.1] Show directly that the sequence $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is Cauchy.

Exercise 2.6.3. (cf. [TBB08, Example 2.42]) Let $(\gamma_n)_{n=1}^{\infty}$ be a sequence of numbers in $(0, 1)$. Define a sequence $(a_n)_{n=1}^{\infty}$ by $a_1 := 1$, $a_2 := 2$, and

$$a_n := \gamma_n a_{n-2} + (1 - \gamma_n) a_{n-1}, \quad \text{for } n \geq 3.$$

(a) Prove that

$$|a_n - a_{n-1}| = \gamma_n \gamma_{n-1} \cdots \gamma_3$$

for all $n \geq 3$.

(b) Prove that for all $m \geq n$,

$$|a_m - a_n| \leq |a_{n+1} - a_n|.$$

(c) Give an example of a sequence $(\gamma_n)_{n=1}^{\infty}$ for which the associated sequence $(a_n)_{n=1}^{\infty}$ does not converge.

(d) Prove that if

$$\liminf_{n\to\infty} \gamma_n < 1$$

then $(a_n)_{n=1}^{\infty}$ does converge.

Exercise 2.6.4. [TBB08, Exercise 2.12.5] Give an example of a sequence $(a_n)_{n=1}^{\infty}$ which is not Cauchy, yet such that for every $\epsilon > 0$, there exists n_0 such that for all $n \geq n_0$,

$$|a_{n+1} - a_n| < \epsilon.$$

[*Hint. Suppose that* $(b_n)_{n=1}^{\infty}$ *is another sequence and* a_n *is defined by* $a_n := b_1 + \cdots + b_n$*. What does this condition tell us about the* b_n*?*]

2.7 lim sup and lim inf

Definition 2.7.1. *Let* $(a_n)_{n=1}^{\infty}$ *be a sequence of real numbers. The* limit superior *of* $(a_n)_{n=1}^{\infty}$ *is*

$$\limsup_{n\to\infty} a_n := \inf\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\}.$$

The limit inferior *of* $(a_n)_{n=1}^{\infty}$ *is*

$$\liminf_{n\to\infty} a_n := \sup\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \geq \beta \ \forall n \geq n_0\}.$$

In other words, the limit superior is the infimum of the set of *eventual* upper bounds for $(a_n)_{n=1}^{\infty}$, and similarly for the limit inferior. Using the notational conventions we introduced around inf and sup, we write the following:

- If $(a_n)_{n=1}^{\infty}$ is not bounded above then (since we have no eventual upper bounds), $\limsup_{n\to\infty} a_n = +\infty$.
- If $(a_n)_{n=1}^{\infty}$ is not bounded below then (since we have no eventual lower bounds), $\liminf_{n\to\infty} a_n = -\infty$.
- If $(a_n)_{n=1}^{\infty}$ diverges to $-\infty$ then (since every number, no matter how negative, will be an eventual upper bound), $\limsup_{n\to\infty} a_n = -\infty$.
- If $(a_n)_{n=1}^{\infty}$ diverges to ∞ then (since every number, no matter how large, will be an eventual lower bound), $\liminf_{n\to\infty} a_n = +\infty$.

Example 2.7.2. Define $a_n := \frac{1}{n} + (-1)^n$. Let's first compute $\limsup_{n\to\infty} a_n$.

First, we see that any number $\beta > 1$ will be an eventual upper bound: as long as $n \geq n_0 := \frac{1}{\beta-1}$, we have that $a_n \leq \beta$. On the other hand, 1 is not an eventual lower bound (there is no n_0 such that $a_n \leq 1$ for all $n \geq n_0$), since there are arbitrarily large n such that $a_n > 1$. Thus,

$$\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\} = (1, \infty)$$

and so

$$\limsup_{n\to\infty} a_n = \inf(1, \infty) = 1.$$

Now we compute $\liminf_{n\to\infty} a_n$. Here, -1 is clearly a lower bound, and so any number ≤ -1 is also a lower bound (whence an eventual lower bound). However, for any $\beta > -1$, β is not an eventual lower bound: for any n_0, we may find $n \geq n_0$ such that n is odd and $n \geq \frac{1}{\beta+1}$, and so $a_n < \beta$. Thus,

$$\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \geq \beta \ \forall n \geq n_0\} = (-\infty, -1]$$

and so

$$\liminf_{n\to\infty} a_n = \sup(-\infty, -1] = -1.$$

Proposition 2.7.3. *For any sequence* $(a_n)_{n=1}^{\infty}$,

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

Proof. If the sequence isn't bounded then either $\limsup_{n\to\infty} a_n = \infty$ or $\liminf_{n\to\infty} a_n = -\infty$, and in either case, the result is trivial. So let us assume that $(a_n)_{n=1}^{\infty}$ is bounded.

Consider the sets used to define lim sup and lim inf:

$$S := \{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\},$$
$$T := \{\alpha \in \mathbb{R} : \exists m_0 \text{ such that } a_n \geq \alpha \ \forall n \geq m_0\}.$$

If $\beta \in S$ and $\alpha \in T$ then there is some n_0 as in the definition of S and an m_0 as in the definition of T. Then with $n := \max\{m_0, n_0\}$, we get

$$\alpha \leq a_n \leq \beta.$$

Thus we have shown that every $\alpha \in T$ is below every $\beta \in S$ (abusing notation, we could write $T \leq S$).

Rewording this, we have that every $\alpha \in T$ is a lower bound for S, and thus for all $\alpha \in T$,

$$\alpha \leq \inf S = \limsup_{n\to\infty} a_n.$$

But then this shows that $\limsup_{n\to\infty} a_n$ is an upper bound for T, so that

$$\limsup_{n\to\infty} a_n \geq \sup T = \liminf_{n\to\infty} a_n,$$

as required. □

We now give another formula for lim sup and lim inf, which justifies the notation.

Proposition 2.7.4. *Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence of real numbers. Then*

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_n, a_{n+1}, a_{n+2}, \dots\} \qquad \textit{and} \tag{2.2}$$
$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf\{a_n, a_{n+1}, a_{n+2}, \dots\}.$$

Proof. We will prove the formula for lim sup; the other one can be proven by essentially the same argument.

For notational convenience, let us set

$$b_n := \sup\{a_n, a_{n+1}, a_{n+2}, \dots\}.$$

Note that since $\{a_n, a_{n+1}, a_{n+2}, \dots\} \supseteq \{a_{n+1}, a_{n+2}, \dots\}$, it follows that

$$b_n \geq b_{n+1} \quad \text{for each } n.$$

Thus the sequence $(b_n)_{n=1}^{\infty}$ is decreasing. It is also bounded, since $(a_n)_{n=1}^{\infty}$ is, so by the Monotone Convergence Criterion, it converges.

Set $L := \lim_{n\to\infty} b_n$. We will prove $\limsup_{n\to\infty} a_n = L$ by two-way inequality.

$\leq$: For each k, b_k is an eventual upper bound: setting $n_0 := k$ we have that $b_k \leq a_n$ for all $n \geq n_0$. Thus, by the definition of lim sup, we have $\limsup_{n\to\infty} a_n \leq b_k$. Consequently by Corollary 2.3.4 (with $m := \limsup_{n\to\infty} a_n$),

$$\limsup_{n\to\infty} a_n \leq \lim_{k\to\infty} b_k = L.$$

$\geq$: Let β be an eventual upper bound for $(a_n)_{n\to\infty}$, so that there exists n_0 such that $a_n \leq \beta$ for all $n \geq n_0$. Then for all $n \geq n_0$, β is an upper bound for $\{a_n, a_{n+1}, a_{n+2}, \dots\}$. Hence,

$$\beta \geq \sup\{a_n, a_{n+1}, \dots\} = b_n.$$

Since this holds for all $n \geq n_0$, it follows from Corollary 2.3.4 (with $M = \beta$) that β is greater than or equal to the limit of $(b_{n_0}, b_{n_0+1}, \dots)$, which is L (by Proposition 2.5.4):

$$\beta \geq L.$$

Thus, L is a lower bound for the set $\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\}$ used in the definition of lim sup, so

$$L \leq \inf\{\beta \in \mathbb{R} : \exists n_0 \text{ such that } a_n \leq \beta \ \forall n \geq n_0\} = \limsup_{n\to\infty} a_n.$$

□

Remark 2.7.5. In the above proof, if we assume only that $(a_n)_{n=1}^{\infty}$ is bounded above, then the b_n are still real numbers, and the same arguments still work (although $\lim_{n\to\infty} b_n = -\infty = \limsup_{n\to\infty} a_n$ is possible), and so (2.2) still holds. In the case that $(a_n)_{n=1}^{\infty}$ is not bounded above, this means in the notation of the above proof, $b_n = \infty$ for all n. However, if we agree that $\lim_{n\to\infty} \infty = \infty$, then under this convention, again holds.

In conclusion, adopting the right conventions, the hypothesis that $(a_n)_{n=1}^{\infty}$ is bounded is *not needed* in the above proposition.

Theorem 2.7.6. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Then $(a_n)_{n=1}^{\infty}$ converges if and only if $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$ and this value is finite. In this case,*

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

Proof. First, suppose that the sequence converges, and set $L := \lim_{n\to\infty} a_n$. Given $\epsilon > 0$ there exists n_0 such that for all $n \geq n_0$,

$$L - \epsilon < a_n < L + \epsilon.$$

Hence,

$$L - \epsilon \leq \inf\{a_{n_0}, a_{n_0+1}, \dots\}$$

and so from the formula in Proposition 2.7.4,

$$L - \epsilon \leq \liminf_{n\to\infty} a_n.$$

Since ϵ is arbitrary, we find $L \leq \liminf_{n\to\infty} a_n$. Likewise we may infer that $L \geq \limsup_{n\to\infty} a_n$. Hence,

$$\begin{aligned} L &\leq \liminf_{n\to\infty} a_n \\ &\leq \limsup_{n\to\infty} a_n && \text{(Proposition 2.7.3)} \\ &\leq L. \end{aligned}$$

We conclude that

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

This shows one direction of the "if and only if", as well as the final statement of the theorem.

Conversely, suppose that

$$\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n,$$

and that this value is finite. Set $L := \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$.

Let $\epsilon > 0$ be given. We may find n_0 such that

$$\sup\{a_{n_0}, a_{n_0+1}, \dots\} \in (L - \epsilon, L + \epsilon)$$

and

$$\inf\{a_{n_0}, a_{n_0+1}, \dots\} \in (L - \epsilon, L + \epsilon).$$

It follows from the first of these that for all $n \geq n_0$,

$$L + \epsilon > \sup\{a_{n_0}, a_{n_0+1}, \dots\} \geq a_n,$$

and likewise, $L - \epsilon < a_n$. Thus, for $n \geq n_0$, we have $|a_n - L| < \epsilon$, as required. $\square$

Exercise 2.7.1. [TBB08, Exercise 2.13.2] Compute lim sup and lim inf for each of the following sequences.

(a) $((-1)^n n)_{n=1}^{\infty}$,

(b) $(\sin(n\pi/8))_{n=1}^{\infty}$,

(c) $(n\sin(n\pi/8))_{n=1}^{\infty}$,

(d) $\left(\frac{(n+1)\sin(n\pi/8)}{n}\right)_{n=1}^{\infty}$,

(e) $(1+(-1)^n)_{n=1}^{\infty}$,

(f) $(r_n)_{n=1}^{\infty}$, consisting of all the rational numbers in the interval $(0,1)$, arranged in some order. *(Does the order matter?)*

Exercise 2.7.2. [TBB08, Exercise 2.13.3] Find a sequence $(a_n)_{n=1}^{\infty}$ of *rational* numbers such that:

(a) $\limsup\limits_{n\to\infty} a_n = \sqrt{2}$ and $\liminf\limits_{n\to\infty} a_n = -\sqrt{2}$.

(b) $\limsup\limits_{n\to\infty} a_n = \infty$ and $\liminf\limits_{n\to\infty} a_n = -\sqrt{2}$.

(c) $\limsup\limits_{n\to\infty} a_n = \pi$ and $\liminf\limits_{n\to\infty} a_n = e$.

Exercise 2.7.3. [TBB08, Exercise 2.13.7] Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence, set $L := \limsup\limits_{n\to\infty} a_n$, and let $\epsilon > 0$. Prove the following.

(a) $a_n > L + \epsilon$ for only finitely many $n \in \mathbb{N}_{\geq 1}$, and

(b) $a_n > L - \epsilon$ for infinitely many $n \in \mathbb{N}_{\geq 1}$.

Exercise 2.7.4. [TBB08, Exercise 2.13.9] Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two bounded sequences.

(a) Prove that $\limsup\limits_{n\to\infty}(a_n + b_n) \leq \limsup\limits_{n\to\infty} a_n + \limsup\limits_{n\to\infty} b_n$.

(b) Give an example where

$$\limsup_{n\to\infty}(a_n + b_n) \neq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Exercise 2.7.5. Suppose that $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence. Prove directly (without using the Cauchy Convergence Criterion, though you are allowed to use that $(a_n)_{n=1}^{\infty}$ is bounded) that $\limsup\limits_{n\to\infty} a_n = \liminf\limits_{n\to\infty} a_n$. *If we then use Theorem 2.7.6, we thus get an alternative proof of the Cauchy Convergence Criterion (Theorem 2.6.2).*

Exercise 2.7.6. Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence. Prove that there is a subsequence $(a_{n_k})_{k=1}^{\infty}$ which converges to $\limsup\limits_{n\to\infty} a_n$.

Chapter 3

Series

This chapter is about infinite sums, also called *series*. Finite sums make sense in any field; however, to sum infinitely many elements requires taking a limit, so depends on the theory of sequences developed in the last chapter.

3.1 Definition and basic properties

A *series* (of real numbers) is an infinite sum

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

where $(a_n)_{n=1}^{\infty}$ is a sequence of real numbers.

Since addition in $\mathbb{R}$ is (a priori) only defined for finitely many numbers at a time[1] we need to define what an infinite sum means for us.

Definition 3.1.1. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. For $N \in \mathbb{N}_{\geq 1}$, define*

$$s_N := \sum_{n=1}^{N} a_n = a_1 + \cdots + a_N,$$

called the Nth partial sum *of the series $\sum_{n=1}^{\infty} a_n$. We say that the series $\sum_{n=1}^{\infty} a_n$* converges *(to L) if the sequence $(s_N)_{N=1}^{\infty}$ converges (to L), and in this case, we may also write*

$$\sum_{n=1}^{\infty} a_n = L.$$

If a series does not converge, we say it diverges. *If $(s_N)_{n=1}^{\infty}$ diverges to $\pm\infty$, we write*

$$\sum_{n=1}^{\infty} a_n = \pm\infty.$$

[1] In fact, recall that we only defined addition for two numbers at a time. But by iterating we can of course do finitely many numbers: $\sum_{n=1}^{N} a_n = a_1 + a_2 + \cdots + a_n$ formally means $(\cdots((a_1 + a_2) + a_3) + \cdots + a_{n-1}) + a_n$.

In other words,

$$\sum_{n=1}^{\infty} a_n := \lim_{N\to\infty} \sum_{n=1}^{N} a_n,$$

provided that the limit on the right exists.

We use theory from the previous topic (sequences) to establish results about series.

Proposition 3.1.2. *Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences and let $c \in \mathbb{R}$. Suppose that the series*

$$\sum_{n=1}^{\infty} a_n \quad \textit{and} \quad \sum_{n=1}^{\infty} b_n$$

both converge.

(i) $\sum_{n=1}^{\infty}(a_n + b_n)$ *converges, and*

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

(ii) $\sum_{n=1}^{\infty} ca_n$ *converges, and*

$$\sum_{n=1}^{\infty} ca_n = c\sum_{n=1}^{\infty} a_n.$$

Proof. Exercise 3.1.1. □

The above says that the set

$$V := \{(a_n)_{n=1}^{\infty} : \sum_{n=1}^{\infty} a_n \text{ converges}\}$$

is a vector space and the function $V \to \mathbb{R}$ defined by $(a_n)_{n=1}^{\infty} \mapsto \sum_{n=1}^{\infty} a_n$ is a linear map.

Proposition 3.1.3. *Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences such that $a_n \leq b_n$ for all n. If the series*

$$\sum_{n=1}^{\infty} a_n \quad \textit{and} \quad \sum_{n=1}^{\infty} b_n$$

both converge, then

$$\sum_{n=1}^{\infty} a_n \leq \sum_{n=1}^{\infty} b_n.$$

Proof. Exercise 3.1.2. □

Often, as in the following proposition, we start an infinite sum at a different index than 1. This is no problem: we can rewrite such a series as one starting at 1, for example as

$$\sum_{n=m}^{\infty} a_n = \sum_{n=1}^{\infty} b_n$$

where $b_n = a_{m+n-1}$. (This is needed to make sense of such an infinite sum from a theoretical perspective, but in practice, when it is clear what is going on, we don't need to do this.)

Proposition 3.1.4. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers and let $m \in \mathbb{N}_{\geq 1}$. Then*

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if

$$\sum_{n=m}^{\infty} a_n$$

converges.

Proof. Exercise 3.1.3. □

Example 3.1.5. Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Note that

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Therefore, we may simplify the partial sum

$$\begin{aligned} s_N &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+1}\right) \\ &= 1 - \frac{1}{N+1}, \end{aligned}$$

by cancelling terms in the middle. Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{N\to\infty} \left(1 - \frac{1}{N+1}\right) = 1.$$

A series in which the partial sums simplify in this way is called a *telescoping series*.

Typically, series are not telescoping, so we want more systematic techniques for determining whether a series converges.

Example 3.1.6. The *harmonic series* is

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

Consider the partial sum

$$\begin{aligned} s_{2^k} &= 1 + \frac{1}{2} + \cdots + \frac{1}{2^k} \\ &= 1 + \frac{1}{2} + \overbrace{\frac{1}{3} + \frac{1}{4}}^{\text{each term } \geq 1/4} + \overbrace{\frac{1}{5} + \cdots + \frac{1}{8}}^{\text{each term } \geq 1/8} + \cdots + \overbrace{\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}}^{\text{each term } \geq 1/2^k} \end{aligned}$$

We have 2 terms which are $\geq \frac{1}{4}$, so their sum is at least $\frac{1}{2}$. Following this, we have 4 terms which are $\geq \frac{1}{8}$, so their sum is at least $\frac{1}{2}$. Continuing, we can see that each grouping sums to at least $\frac{1}{2}$. There are $k-1$ such groupings, so we have

$$s_{2^k} \geq 1 + \frac{1}{2} + (k-1)\frac{1}{2} = 1 + \frac{k}{2}.$$

It follows that $(s_N)_{N=1}^{\infty}$ is not bounded above, so it diverges. In fact, the sequence $(s_N)_{N=1}^{\infty}$ is increasing, so we see that $s_N \to \infty$ and thus

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

The previous example is very important, since $\frac{1}{n} \to 0$. Intuitively, one might have expected that a series $\sum\limits_{n=1}^{\infty} a_n$ converges if and only if its terms converge to 0; while the harmonic series shows that one direction of this is not true, the intuition is correct in the other direction, as the next result shows.

Proposition 3.1.7. *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. If the series*

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Proof. Set $L := \sum\limits_{n=1}^{\infty} a_n$. Consider the partial sum

$$s_N = \sum_{n=1}^{N} a_n,$$

so that $\lim\limits_{N\to\infty} s_N = L$. But then we also have $\lim\limits_{N\to\infty} s_{N-1} = L$, so

$$0 = L - L = \lim_{N\to\infty} (s_N - s_{N-1}) = \lim_{N\to\infty} a_N.$$

□

It was mentioned before the above proposition, but bears repeating: the converse is not true. *If a sequence $(a_n)_{n=1}^{\infty}$ converges to 0, it does not follow that the series $\sum\limits_{n=1}^{\infty} a_n$ converges.*

By taking the contrapositive of Proposition 3.1.7, we have a useful test for divergence of a series: *The Divergence Test.* If a sequence $(a_n)_{n=1}^{\infty}$ does not converge to 0, then the series

$$\sum_{n=1}^{\infty} a_n$$

does not converge.

Example 3.1.8. A *geometric series* is one of the form

$$1 + r + r^2 + r^3 + \cdots$$

where $r \in \mathbb{R}$. When $|r| \geq 1$, the sequence $(r^n)_{n=0}^{\infty}$ does not converge to 0, so by the Divergence Test, it follows that the corresponding geometric series diverges.

On the other hand, for $|r| < 1$, the sequence $(r^n)_{n=0}^{\infty}$ does converge to 0, so the corresponding geometric series at least has a chance to converge. We rewrite the partial sum

$$s_N = 1 + r + r^2 + \cdots + r^N = \frac{1 - r^{N+1}}{1 - r}.$$

From this we see that the series does converge,

$$\sum_{n=0}^{\infty} r^n = \lim_{N\to\infty} \frac{1 - r^{N+1}}{1 - r} = \frac{1}{1 - r}.$$

Exercise 3.1.1. Prove Proposition 3.1.2.

Exercise 3.1.2. Prove Proposition 3.1.3.

Exercise 3.1.3. Prove Proposition 3.1.4. When they converge, what is the relationship between $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=m}^{\infty} a_n$?

Exercise 3.1.4. [TBB08, Exercise 3.4.3] If $\sum\limits_{n=1}^{\infty}(a_n + b_n)$ converges, what can you say about the series

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n?$$

Exercise 3.1.5. [TBB08, Exercise 3.4.4] If $\sum\limits_{n=1}^{\infty}(a_n + b_n)$ diverges, what can you say about the series

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n?$$

Exercise 3.1.6. [Leb16, Exercise 2.5.4] Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

(a) Prove that if $\sum\limits_{n=1}^{\infty} a_n$ converges, then so does $\sum\limits_{n=1}^{\infty}(a_{2n-1} + a_{2n})$.

(b) Give an example where the converse of (a) doesn't hold, that is, such that $\sum\limits_{n=1}^{\infty}(a_{2n-1} + a_{2n})$ converges but $\sum\limits_{n=1}^{\infty} a_n$ does not.

This exercise shows that we must be cautious when we write a series $\sum\limits_{n=1}^{\infty} a_n$ *as*

$$a_1 + a_2 + \cdots .$$

3.2 Convergence tests

Here we will establish a number of results which are useful for proving convergence of series in various different settings.

Proposition 3.2.1 (Boundedness Test)**.** *Let* $(a_n)_{n=1}^{\infty}$ *be a sequence of real numbers. Suppose that:*

(i) $a_n \geq 0$ *for all* n*, and*

(ii) There is a bound $M \in \mathbb{R}$ *on the partial sums, so that*

$$\sum_{n=1}^{N} a_n \leq M$$

for all $N \in \mathbb{N}_{\geq 1}$*.*

Then $\sum\limits_{n=1}^{\infty} a_n$ *converges, and in fact* $\sum\limits_{n=1}^{\infty} a_n \leq M$*.*

Proof. Since $a_n \geq 0$, the partial sums $(s_N)_{N=1}^{\infty}$ satisfy

$$s_N \leq s_{N+1} \quad \text{for all } N.$$

In other words, $(s_N)_{N=1}^{\infty}$ is an increasing sequence. The second condition ensures that this sequence is bounded above, Therefore, by the Monotone Convergence Criterion (Theorem 2.4.2), it converges. Hence, $\sum\limits_{n=1}^{\infty} a_n$ converges. □

Proposition 3.2.2 (Comparison Test)**.** *Let* $(a_n)_{n=1}^{\infty}$ *and* $(b_n)_{n=1}^{\infty}$ *be sequences such that:*

$$0 \leq a_n \leq b_n \quad \textit{for all } n.$$

Then:

(i) if $\sum\limits_{n=1}^{\infty} b_n$ *converges, then so does* $\sum\limits_{n=1}^{\infty} a_n$*.*

(ii) if $\sum\limits_{n=1}^{\infty} a_n$ *diverges, then so does* $\sum\limits_{n=1}^{\infty} b_n$*.*

Warning: it is easy to get confused about the hypothesis of the Comparison Test. If $0 \leq a_n \leq b_n$, and $\sum\limits_{n=1}^{\infty} a_n$ converges, then we cannot conclude anything about $\sum\limits_{n=1}^{\infty} b_n$. Likewise, if $\sum\limits_{n=1}^{\infty} b_n$ diverges, we cannot conclude anything about $\sum\limits_{n=1}^{\infty} a_n$.

Proof. (ii) is the contrapositive of (i). We will prove (i) now, using the Boundedness Test. Hypothesis (i) of the Boundedness Test is true since it is a hypothesis here.

Set $M := \sum\limits_{n=1}^{\infty} b_n$. Since the sequence

$$\left(\sum_{n=1}^{N} b_n\right)_{n=1}^{\infty}$$

is increasing and converges to M, we have that M is the supremum of this sequence, and in particular,

$$\sum_{n=1}^{N} b_n \leq M$$

for all N. Therefore,

$$\sum_{n=1}^{N} a_n \leq \sum_{n=1}^{N} b_n \leq M.$$

This verifies hypothesis (ii) of the Boundedness Test, so by that test we conclude that $\sum_{n=1}^{\infty} a_n$ converges. $\square$

Proposition 3.2.3 (Absolute Convergence Test)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. If the series*

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then so does the series

$$\sum_{n=1}^{\infty} a_n.$$

Proof. Write

$$(a_n)_+ := \max\{a_n, 0\} \quad \text{and}$$
$$(a_n)_- := \max\{-a_n, 0\}.$$

Then $0 \leq (a_n)_+ \leq |a_n|$, so by the Comparison Test (i) (with $(a_n)_+$ in place of a_n and $|a_n|$ in place of b_n), $\sum_{n=1}^{\infty}(a_n)_+$ converges. By the same argument, the series $\sum_{n=1}^{\infty}(a_n)_-$ converges. Finally, we observe that $a_n = (a_n)_+ - (a_n)_-$, so by linearity,

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}(a_n)_+ - \sum_{n=1}^{\infty}(a_n)_-$$

converges. $\square$

Interesting fact 3.2.4. One calls a series $\sum_{n=1}^{\infty} a_n$ *absolutely convergent* when the series $\sum_{n=1}^{\infty} |a_n|$ converges. It is nontrivial result (by Dirichlet and Riemann) that a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent if and only if any rearrangement of it converges to the same value, i.e.,

$$\sum_{n=1}^{\infty} a_{s(n)} = \sum_{n=1}^{\infty} a_n$$

for any bijection $s : \mathbb{N}_{\geq 1} \to \mathbb{N}_{\geq 1}$.

As an illustration of this, consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \cdots .$$

This series converges to some value $L \geq \frac{1}{2}$ (by the Alternating Series Test, Proposition 3.2.9 below; the estimate $\frac{1}{2}$ comes by looking at the second partial sum). *If rearrangements were allowed*, then

we could do the following

$$\begin{aligned}
L &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots \\
&= 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{7} + \cdots + \frac{1}{2k+1} - \frac{1}{4k+2} - \frac{1}{4k+3} + \cdots \\
&= \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \cdots \\
&= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots \\
&= \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \cdots\right) \\
&= \frac{L}{2}.
\end{aligned}$$

Since $L \neq 0$, this is a contradiction. This shows that rearranging the alternating harmonic series *can* change its value, confirming a special case of the Dirichlet–Riemann result, since we already know that the alternating harmonic series does not converge absolutely, by Example 3.1.6.

Proposition 3.2.5 (Ratio Test)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of nonzero real numbers.*

(i) If

$$\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$$

then $\sum_{n=1}^{\infty} a_n$ converges (absolutely).

(ii) If

$$\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$$

then $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. (i): Suppose

$$q := \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1.$$

We wish to show that $\sum_{n=1}^{\infty} |a_n|$ converges, so we will replace a_n with $|a_n|$ and show that $\sum_{n=1}^{\infty} a_n$ converges. This allows us to assume that $a_n \geq 0$ for all n.

Pick $r \in (q, 1)$. Then by the definition of lim sup, r is an eventual upper bound for $\frac{a_{n+1}}{a_n}$. That is, there exists n_0 such that

$$\frac{a_{n+1}}{a_n} \leq r$$

for all $n \geq n_0$, which we rearrange as

$$a_{n+1} \leq a_n r.$$

Thus we have,

$$\begin{aligned}
a_{n_0+1} &\leq a_{n_0} r, \\
a_{n_0+2} &\leq a_{n_0+1} r \leq a_{n_0} r^2,
\end{aligned}$$

and continuing in this way, $a_{n_0+k} \leq a_{n_0} r^k$ for all $k \in \mathbb{N}_{\geq 0}$. Since $r \in (0,1)$, the geometric series $\sum_{k=0}^{\infty} r^k$ converges (Example 3.1.8). Hence by the comparison test, $\sum_{k=0}^{\infty} a_{n_0+k}$ converges, which is the same as saying

$$\sum_{n=n_0}^{\infty} a_n \text{ converges.}$$

By Proposition 3.1.4, it follows that $\sum_{n=1}^{\infty} a_n$ converges.

(ii): Suppose

$$q := \liminf_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| > 1.$$

Then by the definition of lim inf, since $1 < q$, 1 is an eventual lower bound for $\frac{|a_{n+1}|}{|a_n|}$, so there exists n_0 such that

$$\frac{|a_{n+1}|}{|a_n|} \geq 1$$

for all $n \geq n_0$, which we rearrange as

$$|a_{n+1}| \geq |a_n|.$$

Similarly to what we did in part (i), from this we get $|a_{n+k}| \geq |a_{n_0}|$ for all $k \in \mathbb{N}_{\geq 0}$. We conclude that $(a_n)_{n=1}^{\infty}$ doesn't converge to 0, so by the Divergence Test, $\sum_{n=1}^{\infty} a_n$ diverges. □

Remark 3.2.6. In many examples of series you will see, the limit $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ will exist (so will be equal to the lim inf and the lim sup). However, it often happens that

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = 1,$$

and in this case, we cannot conclude anything from the Ratio Test. To see why, recall the two series from Examples 3.1.5 and 3.1.6:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1,$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

However, in both cases, the ratio $\frac{a_{n+1}}{a_n}$ converges to 1.

For an example where $\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} > 1 > \liminf_{n\to\infty} \frac{a_{n+1}}{a_n}$, define

$$a_n := \begin{cases} 1, & n \text{ even;} \\ 2, & n \text{ odd.} \end{cases}$$

Then

$$\frac{a_{n+1}}{a_n} = \begin{cases} 2, & n \text{ even;} \\ \frac{1}{2}, & n \text{ odd.} \end{cases}$$

Of course in this case, $\sum_{n=1}^{\infty} a_n$ diverges (by the Divergence Test). See also Exercise 3.2.8.

Proposition 3.2.7 (Root Test)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.*

(i) If

$$\limsup_{n\to\infty} \sqrt[n]{|a_n|} < 1$$

then $\sum_{n=1}^{\infty} a_n$ converges (absolutely).

(ii) If

$$\limsup_{n\to\infty} \sqrt[n]{|a_n|} > 1$$

then $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. This proof is similar to the proof of the Ratio Test.

(i): Suppose

$$q := \limsup_{n\to\infty} \sqrt[n]{|a_n|} < 1.$$

We wish to show that $\sum_{n=1}^{\infty} |a_n|$ converges, so we replace a_n with $|a_n|$ and we'll show that $\sum_{n=1}^{\infty} a_n$ converges.

Pick $r \in (q, 1)$, and using the definition of lim sup, there exists n_0 such that

$$\sqrt[n]{a_n} \leq r \quad \text{for all } n \geq n_0.$$

In other words,

$$0 \leq a_n \leq r^n$$

for all $n \geq n_0$. Since the geometric series $\sum_{n=n_0}^{\infty} r^n$ converges, we conclude by the Comparison Test that so does $\sum_{n=n_0}^{\infty} a_n$. Thus by Proposition 3.1.4, $\sum_{n=1}^{\infty} a_n$ converges.

(ii) Suppose

$$q := \limsup_{n\to\infty} \sqrt[n]{|a_n|} > 1.$$

Using the definition of lim sup, there are infinitely many n such that

$$\sqrt[n]{|a_n|} \geq 1.$$

For such n, we have

$$|a_n| \geq 1.$$

We conclude that $(a_n)_{n=1}^{\infty}$ doesn't converge to 0, so by the Divergence Test, $\sum_{n=1}^{\infty} a_n$ diverges. □

Remark 3.2.8. Although $\sqrt[n]{|a_n|}$ will often converge in examples you will see, it might converge to 1, in which case the Root Test tells us nothing. Again this can be see through the examples

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1,$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

However, in both cases, the $\sqrt[n]{a_n}$ converges to 1.

Our next test concerns *alternating series*, which is a series in which the signs of the terms alternate between positive and negative. We set up an alternating series by taking a sequence $(a_n)_{n=1}^{\infty}$ of positive numbers and forming $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$.

Proposition 3.2.9 (Alternating Series Test)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose that:*

(i) $(a_n)_{n=1}^{\infty}$ *is an decreasing sequence, and*

(ii) $\lim_{n\to\infty} a_n = 0$.

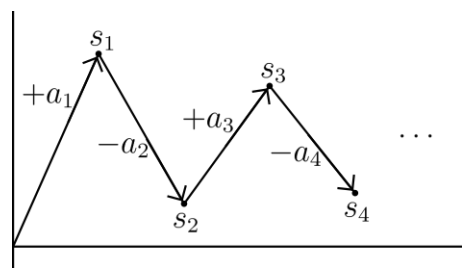
Then

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n$$

converges. Moreover, for any N,

$$\sum_{n=1}^{2N}(-1)^{n+1}a_n \le \sum_{n=1}^{\infty}(-1)^{n+1}a_n \le \sum_{n=1}^{2N-1}(-1)^{n+1}a_n.$$

Proof. Note that $a_n \ge 0$ for all n, because the sequence is decreasing and converges to 0.



Set

$$s_N := \sum_{n=1}^{N}(-1)^{n+1}a_n.$$

Since $(a_n)_{n=1}^{\infty}$ is a decreasing sequence, we have

$$\begin{aligned} s_{N+2} &= s_N + a_{N+1} - a_{N+2} \ge s_N \quad \text{if } N \text{ is even.} \\ s_{N+2} &= s_N - a_{N+1} + a_{N+2} \le s_N \quad \text{if } N \text{ is odd.} \end{aligned}$$

Thus, the subsequence of even terms, $(s_{2N})_{N=1}^{\infty}$ is an increasing sequence, and the subsequence of odd terms, $(s_{2N-1})_{N=1}^{\infty}$ is decreasing. To show that the sequence of even terms $(s_{2N})_{N=1}^{\infty}$ is bounded, observe that for each N, since $a_{2N+1} \ge 0$,

$$s_{2N} \le s_{2N} + a_{2N+1} = s_{2N+1} \le s_1.$$

So, s_1 is an upper bound for $(s_{2N})_{N=1}^{\infty}$. By the Monotone Convergence Criterion (Theorem 2.4.2), $(s_{2N})_{N=1}^{\infty}$ converges to $L := \sup\{s_{2N} : N \in \mathbb{N}_{\ge 1}\}$.

Next, for the subsequence of odd terms, since $s_{2N} = s_{2N-1} - a_{2N}$, we get

$$\lim_{N\to\infty} s_{2N-1} = \lim_{N\to\infty} s_{2N} + a_{2N} = \lim_{N\to\infty} s_{2N} + \lim_{N\to\infty} a_{2N} = L + 0.$$

By Exercise 2.2.4, since both the even and odd subsequences converge to the same value, we conclude that

$$\lim_{N\to\infty} s_N = L,$$

i.e., $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges and equals L.

Finally, since L is the supremum of $\{s_{2N} : N \in \mathbb{N}_{\geq 1}\}$, we have

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n = L \geq s_{2N} = \sum_{n=1}^{2N}(-1)^{n+1}a_n.$$

By a similar argument as above, we can show that $L = \inf\{s_{2N-1} : N \in \mathbb{N}_{\geq 1}\}$, and from this,

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n \leq \sum_{n=1}^{2N-1}(-1)^{n+1}a_n.$$

□

In the Alternating Series Test, it is crucial that the sequence $(a_n)_{n=1}^{\infty}$ is decreasing – see Exercise 3.2.3.

The last test that we will state uses the integral. Since we haven't formally defined the integral, we will not be able to prove this result yet; we will prove it later (Exercise 7.4.4 when we study integration. However, it is a very powerful test, particularly as one can often use the Fundamental Theorem of Calculus to easily analyse convergence of an integral.

Proposition 3.2.10 (Integral Test)**.** *Let $f : [1, \infty) \to \mathbb{R}$ be a function. Suppose that:*

(i) $f(x) \geq 0$ for all $x \in [1, \infty)$, and

(ii) f is decreasing: $f(x) \geq f(y)$ whenever $x \leq y$.

Then the series

$$\sum_{n=1}^{\infty} f(n)$$

converges if and only if the improper integral

$$\int_1^{\infty} f(x)\,dx$$

converges.

Exercise 3.2.1. For each of the following, determine whether the series converges.

(a) $\sum_{n=1}^{\infty} \frac{n}{2^n}$.

(b) $\sum_{n=1}^{\infty} \frac{n^2-3}{(n+2)(n+5)}$.

(c) $\sum_{n=1}^{\infty} \frac{(-1)^n n^2-3}{(n+2)(n+5)}$.

(d) $\sum_{n=1}^{\infty} \frac{(n^2+3)^n}{(2n^2-1)^n}$.

(e) $\sum_{n=1}^{\infty} \frac{\sin(n)}{n^2+n}$.

(f) $\sum_{n=1}^{\infty} \frac{n^2+2}{n^4+4}$.

(g) $\sum_{n=1}^{\infty} \frac{n^3+3}{n^6+6}$.

(h) $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

Exercise 3.2.2. [Leb16, Exercise 2.5.7] *(Difficult, and this exercise probably belongs better in Section 3.1)* Suppose that $(a_n)_{n=1}^{\infty}$ is a decreasing sequence. If the series $\sum_{n=1}^{\infty} a_n$ converges, prove that $\lim_{n\to\infty} na_n = 0$.

Exercise 3.2.3. Give an example of a sequence $(a_n)_{n=1}^{\infty}$ such that $a_n \geq 0$ for all n and $\lim_{n\to\infty} a_n = 0$, such that the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

does *not* converge.

Exercise 3.2.4. [TBB08, Exercise 3.5.1] Suppose that $(a_n)_{n=1}^{\infty}$ is a sequence of real numbers such that $a_n \geq 0$ for all n. If we know that $\sum_{n=1}^{\infty} a_n$ converges, does it follow that $\sum_{n=1}^{\infty} a_n^2$ converges?

Exercise 3.2.5. [TBB08, Exercise 3.4.7] Suppose that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences of real numbers such that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge. Does it follow that $\sum_{n=1}^{\infty} a_n b_n$ converges? What about if we assume in addition that $a_n \geq 0$ for all n?

Exercise 3.2.6. Suppose that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences of real numbers such that:

(i) $b_n > 0$ for all n,

(ii) $\sum_{n=1}^{\infty} b_n$ converges, and

(iii) The sequence $\left(\frac{a_n}{b_n}\right)_{n=1}^{\infty}$ converges.

Prove that $\sum_{n=1}^{\infty} a_n$ converges.

Exercise 3.2.7. [TBB08, Exercise 3.5.11] Show that a series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent if and only if every "subseries" $\sum_{k=1}^{\infty} a_{n_k}$ converges.

Exercise 3.2.8. Produce an example of a sequence $(a_n)_{n=1}^{\infty}$ of numbers in $(0, \infty)$ such that

$$\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} > 1, \quad \liminf_{n\to\infty} \frac{a_{n+1}}{a_n} < 1,$$

and the series $\sum_{n=1}^{\infty} a_n$ converges. Can you even arrange that $\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} = \infty$?

3.3 Cauchy convergence criterion for series

The following is an important theoretical device, although for examples of sequences you may encounter, it is rarely the right tool to prove convergence.

Proposition 3.3.1 (Cauchy Convergence Criterion)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Then $\sum_{n=1}^{\infty}$ converges if and only if, for every $\epsilon > 0$ there exists N_0 such that, for all $N \geq M \geq N_0$,*

$$\left| \sum_{n=M}^{N} a_n \right| < \epsilon.$$

Proof. Again we use the partial sums

$$s_N := \sum_{n=1}^{N} a_n.$$

By the Cauchy Convergence Criterion for sequences (Theorem 2.6.2), $(s_N)_{N=1}^{\infty}$ converges (i.e., $\sum_{n=1}^{\infty} a_n$ converges) if and only if $(s_N)_{N=1}^{\infty}$ is Cauchy. Since

$$s_N - s_{M-1} = \sum_{n=M}^{N} a_n,$$

it is easy to see that $(s_N)_{N=1}^{\infty}$ is Cauchy if and only if, for every $\epsilon > 0$ there exists N_0 such that for all $N \geq M \geq N_0$,

$$\left| \sum_{n=M}^{N} a_n \right| < \epsilon.$$

□

Exercise 3.3.1. Let $(a_n)_{n=1}^{\infty}$, $(b_n)_{n=1}^{\infty}$, and $(c_n)_{n=1}^{\infty}$ are sequences such that for all n,

$$a_n \leq b_n \leq c_n.$$

Prove that if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ converge, then so does $\sum_{n=1}^{\infty} b_n$. Can this be proven using the Squeeze Theorem (Theorem 2.3.5)?

Chapter 4

Topological concepts

In this brief chapter we develop notions of open and closed subsets of the reals, as well as limit points.

4.1 Open and closed sets

Definition 4.1.1. *Let $A \subseteq \mathbb{R}$ be a set.*

(i) A is open *if for every $x \in A$, there exists $\epsilon > 0$ such that the open interval $(x - \epsilon, x + \epsilon)$ is contained in A.*

(ii) A is closed *if its complement,*

$$\mathbb{R} \setminus A = \{x \in \mathbb{R} : x \notin A\}$$

is open.

It is *not* the case that a set is either open or closed. For example, the set $(0, 1]$ is neither open (since $1 \in (0, 1]$ but $(0 - \epsilon, 0 + \epsilon)$ is not a subset of $(0, 1]$ for any $\epsilon > 0$) nor closed (similar reasoning with $0 \in \mathbb{R} \setminus (0, 1]$).

Example 4.1.2. Any open interval is an open set.

First consider a bounded open interval (a, b). Given $x \in (a, b)$ we have $x > a$ and $x < b$, so we may set $\epsilon := \min\{x - a, b - x\}$. Then we see that

$$a \leq x - \epsilon, \quad x + \epsilon \leq b$$

and so $(x - \epsilon, x + \epsilon) \subseteq (a, b)$.

For an unbounded open interval (a, ∞) or $(-\infty, b)$, similar reasoning applies.

Likewise, unbounded closed interval $[a, \infty)$ or $(-\infty, b]$ is closed, since each is the complement of an unbounded open interval ($(-\infty, a)$ or (b, ∞) respectively).

Proposition 4.1.3 (Permanence properties of open sets)**.** *(i) The sets $\emptyset$ and $\mathbb{R}$ are open.*

(ii) For any finite collection of open sets, $U_1, \ldots, U_m \subseteq \mathbb{R}$, their intersection,

$$U_1 \cap \cdots \cap U_m,$$

is open.

(iii) For any arbitrary collection of open sets $\{U_\alpha : \alpha \in I\}$, their union,

$$\bigcup_{\alpha \in I} U_\alpha,$$

is open.

Proof. Parts (i) and (ii) are Exercise 4.1.1.

(iii): Set

$$U := \bigcup_{\alpha \in I} U_\alpha,$$

and let $x \in U$. By definition of union, this means that $x \in U_\alpha$ for some $\alpha \in I$. Since U_α is open, there is some $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq U_\alpha$. Since $U_\alpha \subseteq U$, it follows that

$$(x - \epsilon, x + \epsilon) \subseteq U,$$

as required. □

Example 4.1.4. Although we have that finite intersections of open sets are open, it is not true that infinite intersections of open sets are open. For example,

$$\{0\} = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right),$$

but $\{0\}$ isn't an open set because it contains no open interval centred at 0.

Example 4.1.5. For any $a, b \in \mathbb{R}$, the bounded closed interval $[a, b]$ is closed, since

$$\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty),$$

which is open by the previous example and since unions of open sets are open.

Proposition 4.1.6 (Sequential characterization of closedness)**.** *Let $F \subseteq \mathbb{R}$. Then F is closed if and only if, for every sequence $(a_n)_{n=1}^{\infty}$ in F (i.e., $a_n \in F$ for all n), if $(a_n)_{n=1}^{\infty}$ converges then*

$$\lim_{n \to \infty} a_n \in F.$$

Proof. $\Rightarrow$: We assume that F is closed, i.e., $\mathbb{R} \setminus F$ is open. Let $(a_n)_{n=1}^{\infty}$ be a sequence in F which converges, and suppose for a contradiction that $L := \lim_{n \to \infty} a_n$ is *not* in F. That means $L \in \mathbb{R} \setminus F$. By openness, there exists $\epsilon > 0$ such that $(L - \epsilon, L + \epsilon) \subseteq \mathbb{R} \setminus F$. Using the definition of convergence, we may find n such that $|a_n - L| < \epsilon$. But this means that

$$a_n \in (L - \epsilon, L + \epsilon) \subseteq \mathbb{R} \setminus F,$$

so that $a_n \notin F$, which is a contradiction.

$\Leftarrow$: We assume that for every sequence in F that converges, the limit is in F. We need to show that $\mathbb{R} \setminus F$ is open.

Take $x \in \mathbb{R} \setminus F$, and suppose for a contradiction that there is no $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subseteq \mathbb{R} \setminus F$. Then for each $n \in \mathbb{N}_{\geq 1}$, since $\left(x - \frac{1}{n}, x + \frac{1}{n}\right)$ is not contained in $\mathbb{R} \setminus F$, this means that there must be some number in this interval and in F. Let a_n be such a point, so $a_n \in F$ and $|a_n - x| < \frac{1}{n}$.

Choosing such a_n for each n, we arrive at a sequence $(a_n)_{n=1}^{\infty}$ in F. Since $0 \leq |a_n - x| < \frac{1}{n}$, we get $\lim_{n \to \infty} |a_n - x| = 0$ by the Squeeze Theorem. In other words,

$$\lim_{n \to \infty} a_n = x.$$

By hypothesis, it follows that $x \in F$, which is a contradiction. □

Definition 4.1.7. *Let $A \subseteq \mathbb{R}$ be a set and let $a \in \mathbb{R}$.*

(i) *a is an* interior point *of A if there exists $\epsilon > 0$ such that $(a-\epsilon, a+\epsilon) \subseteq A$. The* interior *of A is*

$$A^\circ := \{x \in \mathbb{R} : x \text{ is an interior point}\}.$$

(ii) *a is an* accumulation point *of A if there is a sequence $(a_n)_{n=1}^\infty$ in A such that $a = \lim_{n\to\infty} a_n$. The* closure *of A is*

$$\overline{A} := \{x \in \mathbb{R} : x \text{ is an accumulation point}\}.$$

(iii) *a is a* boundary point *of A if it is an accumulation point of A and it is not an interior point. The* boundary *of A is*

$$\partial A := \{x \in \mathbb{R} : x \text{ is a boundary point}\} = \overline{A} \setminus A^\circ.$$

(iv) *a is an* isolated point *of A if there exists $\epsilon > 0$ such that $(a-\epsilon, a+\epsilon) \cap A = \{a\}$.*

(v) *a is a* limit point *of A if it is an accumulation point and it is not an isolated point of A.*

Example 4.1.8. Let $A := (0,1] \cup \{2\}$.

- The interior points of A are all points in $(0,1)$:

$$A^\circ = (0,1).$$

- The accumulation points of A are all points in $[0,1] \cup \{2\}$:

$$\overline{A} = [0,1] \cup \{2\}.$$

- The boundary points of A are 0, 1, and 2:

$$\partial A = \{0,1,2\}.$$

- A has only one isolated point, namely 2.
- The limit points of A are all points in $[0,1]$.

Example 4.1.9. Let $A := \mathbb{Q}$.

- $A^\circ = \emptyset$.
- $\overline{A} = \partial A = \mathbb{R}$.
- A has no isolated points.
- Every point of $\mathbb{R}$ is a limit point of A.

Example 4.1.10. Let $A := \mathbb{Z}$.

- $A^\circ = \emptyset$.
- $\overline{A} = \partial A = \mathbb{Z}$.

- Every point of A is an isolated point.
- A has no limit points.

Exercise 4.1.1. Prove parts (i)-(ii) of Proposition 4.1.3.

Exercise 4.1.2. Prove the following facts about closed sets.

(a) $\emptyset, \mathbb{R}$ are closed sets.

(b) For any finite collection of closed sets, $F_1, \ldots, F_m \subseteq \mathbb{R}$, their union,

$$F_1 \cup \cdots \cup F_m,$$

is closed.

(c) For any arbitrary collection of closed sets $\{F_\alpha : \alpha \in I\}$, their intersection,

$$\bigcap_{\alpha \in I} F_\alpha,$$

is closed.

Exercise 4.1.3. [Sav17, Exercise 4.3.2 (b)] Prove that any finite set is closed.

Exercise 4.1.4. Let $(a_n)_{n=1}^\infty$ be a sequence in $\mathbb{R}$ which converges to $L \in \mathbb{R}$. Prove that the set

$$\{a_n : n \in \mathbb{N}_{\geq 1}\} \cup \{L\}$$

is closed.

Exercise 4.1.5. Let $A \subseteq \mathbb{R}$. Suppose that $a \in A$, $b \notin A$, and $a < b$.

(a) Define

$$T := \{t \in [a, b] : t \in A\}.$$

Show that $t_0 := \sup T$ exists.

(b) Show that t_0 from part (a) is in the boundary of A.

(c) Show that the only sets in $\mathbb{R}$ whose boundary is $\emptyset$ are $\emptyset$ and $\mathbb{R}$.

Exercise 4.1.6. Let $A \subseteq \mathbb{R}$.

(a) Prove that the closure, $\overline{A}$, is closed.

(b) Prove that the interior, A°, is open.

(c) Prove that the boundary, ∂A, is closed.

(d) Prove that $A^\circ = \mathbb{R} \setminus \overline{(\mathbb{R} \setminus A)}$.

Exercise 4.1.7. Let $(a_n)_{n=1}^\infty$ be a sequence in $\mathbb{R}$, and define $A := \{a_n : n \in \mathbb{N}_{\geq 1}\}$. Show that if $x \in \mathbb{R}$ is a limit point of A, then there is a subsequence $(a_{n_k})_{k=1}^\infty$ which converges to x.

Exercise 4.1.8. Let $A \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$. Prove that the following are equivalent.

(i) a is a limit point of A.

(ii) For every $\epsilon > 0$,

$$(a - \epsilon, a + \epsilon) \cap A \setminus \{a\} \neq \emptyset.$$

(iii) There exists a sequence $(a_n)_{n=1}^\infty$ in $A \setminus \{a\}$ which converges to a.

Exercise 4.1.9. Construct a set $A \subseteq \mathbb{R}$ which has some limit points, but such that none of its limit points are in A itself.

4.2 Compactness

Definition 4.2.1. *Let $K \subseteq \mathbb{R}$ be a set. We say that K is* (sequentially) compact *if every sequence $(a_n)_{n=1}^{\infty}$ in K has a subsequence that converges to a point $a \in K$.*

In this course, we will frequently abbreviate "sequential compactness" to just "compactness". However, there is a different concept which is called "compactness", which is equivalent to sequential compactness for subsets of $\mathbb{R}$ (and more generally for "metric spaces" – see MAT3120), though the two aren't equivalent in some more general settings. This is discussed further in Interesting Fact 4.2.7.

Example 4.2.2. Let F be a finite set,

$$F = \{x_1, \dots, x_k\}.$$

Then F is compact. To see this, let $(a_n)_{n=1}^{\infty}$ be a sequence in K. Then there must be some i such that $a_n = x_i$ for infinitely many n. Consequently, we can realize the constant sequence

$$(x_i, x_i, \dots)$$

as a subsequence $(a_{n_k})_{k=1}^{\infty}$.

Example 4.2.3. The set $\mathbb{R}$ is not compact. To see this, take the sequence $(1, 2, 3, 4, \dots)$. This sequence diverges to ∞, and it is not hard to see that any subsequence of it also diverges to ∞. Hence it has no subsequence which converges.

Example 4.2.4. The set $A := (0, 1]$ is not compact. To see this, take the sequence $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ in A. This sequence converges to 0, and hence so does every subsequence of it (Proposition 2.5.4). Hence, it has no subsequence converging to a point *in* A.

Theorem 4.2.5 (Heine–Borel Theorem)**.** *Let K be a subset of $\mathbb{R}$. Then K is compact if and only if K is closed and bounded.*

Proof. $\Rightarrow$: Suppose that K is compact. To see that K is closed, suppose for a contradiction that it is not closed. By Proposition 4.1.6, it follows that there is some sequence $(a_n)_{n=1}^{\infty}$ in K which converges, yet such that

$$\lim_{n\to\infty} a_n \notin K.$$

Then by Proposition 2.5.4, every subsequence of $(a_n)_{n=1}^{\infty}$ converges to the same point, and hence no subsequence converges to a point in K. This contradicts that K is compact.

Next, to see that K is bounded, again suppose for a contradiction that it is not. Then for each $n \in \mathbb{N}_{\geq 1}$, we may find some $a_n \in K$ such that $|a_n| \geq n$. This gives us a sequence $(a_n)_{n=1}^{\infty}$. Any subsequence $(a_{n_k})_{k=1}^{\infty}$ will be unbounded, and therefore won't converge. Again, this contradicts that K is compact.

$\Leftarrow$: We assume that K is closed and bounded, and we let $(a_n)_{n=1}^{\infty}$ be a sequence from K. Since K is bounded, so is the sequence $(a_n)_{n=1}^{\infty}$. Therefore by the Bolzano–Weierstrass Theorem (Corollary 2.5.6), there is a subsequence $(a_{n_k})_{k=1}^{\infty}$ which converges to some point $L \in \mathbb{R}$. Since K is closed and the sequence $(a_{n_k})_{k=1}^{\infty}$ is in K, it follows from Proposition 4.1.6 that $L \in K$. Thus $(a_n)_{n=1}^{\infty}$ has a subsequence converging to a point in K, as required. $\square$

Proposition 4.2.6 (Permanence properties of compact sets)**.** *(i) For any finite collection of compact sets, $K_1, \dots, K_m \subseteq \mathbb{R}$, their union,*

$$K_1 \cup \dots \cup K_m,$$

is compact.

(ii) For any arbitrary collection of compact sets $\{K_\alpha : \alpha \in I\}$, their intersection,

$$\bigcap_{\alpha \in I} K_\alpha,$$

is compact.

Proof. Exercise 4.2.1 □

Interesting fact 4.2.7. The *correct* definition of compactness is as follows: $K \subseteq \mathbb{R}$ is compact if for any family $\{U_i\}_{i \in I}$ of open sets in $\mathbb{R}$ (indexed by any set I) such that

$$K \subseteq \bigcup_{i \in I} U_i$$

($\{U_i\}_{i \in I}$ is an "open cover" of K), there are $i_1, \ldots, i_m \in I$ such that

$$K \subseteq U_{i_1} \cup \cdots \cup U_{i_m}$$

($\{U_{i_1}, \ldots, U_{i_m}\}$ is a "finite subcover"). This definition makes sense in a much more general setting of *topological spaces*. *Metric spaces* are certain special topological spaces, which include all subsets of $\mathbb{R}$ (and much more), and for these, sequential compactness is equivalent to compactness, a result which you will see in MAT 3120. The Heine–Borel Theorem, however, does not generalize to this setting. In the more general setting of topological spaces, compactness and sequential compactness are different notions, neither implying the other.

Exercise 4.2.1. Prove Proposition 4.2.6

Exercise 4.2.2. Let $K \subseteq \mathbb{R}$ be a compact set. Let $f : K \to \mathbb{R}$ be a function. Suppose that for every $x \in K$ there exists an open set U containing x and some $M \in \mathbb{R}$ such that for all $y \in U$,

$$f(y) \leq M.$$

Prove that $f(K) = \{f(x) : x \in K\}$ is bounded above.
[*Hint. Proof by contradiction. Suppose it is not bounded above, use this to construct a sequence* $(a_n)_{n=1}^{\infty}$ *in* K *with certain properties, and then use a convergent subsequence.*]

Exercise 4.2.3. Let $K \subseteq \mathbb{R}$. Prove that the following are equivalent.

(i) K is compact.

(ii) Every infinite set $E \subseteq K$ has a limit point z which is in K.

Exercise 4.2.4. Let $K \subseteq \mathbb{R}$ be a compact set. Prove that $\min(K)$ and $\max(K)$ exist (recall that this means that K contains a lower and upper bound for itself).
[*Hint. Use the Heine–Borel Theorem!*]

Chapter 5

Continuous functions

In this chapter, we develop another notion of a limit: here it is for a function defined on a subset of the real numbers, and the concept concerns the behaviour of the function as the variable tends to a particular point. This concept enables us to talk about continuous functions; it will also (in the next chapter) play a key role in establishing differentiation.

5.1 The limit of a function

Definition 5.1.1. *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point of X. Let $f : X \to \mathbb{R}$ and let $L \in \mathbb{R}$. We write*

$$\lim_{x \to a} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that, if $x \in X \setminus \{a\}$ and $|x - a| < \delta$ then $|f(x) - L| < \epsilon$.

The idea in the above definition is that when x is close (but not equal) to a, then $f(x)$ should be close (and possibly equal) to L. It makes sense that we restrict to a being a limit point, because that is exactly the condition that tells us that there are points (arbitrarily) close to a inside X (Exercise 4.1.8). Recall that a limit point a may or may not be inside X itself; however, if $a \in X$, then the value $f(a)$ does not matter at all to the limit $\lim_{x \to a} f(x)$ (or whether this limit exists).

Example 5.1.2. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := 2x + 5$. Then for any $a \in \mathbb{R}$, we have $\lim_{x \to a} f(x) = 2a + 5$.

To see this, set $L := 2a + 5$, and let $\epsilon > 0$ be given. Set $\delta := \frac{\epsilon}{2}$. Then if $x \in \mathbb{R} \setminus \{a\}$ and $|x - a| < \delta = \frac{\epsilon}{2}$, then

$$|f(x) - L| = |2x + 5 - (2a + 5)| = 2|x - a| < 2\frac{\epsilon}{2} = \epsilon.$$

Example 5.1.3. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) := x^2$. Then for any $a \in \mathbb{R}$, we have $\lim_{x \to a} g(x) = a^2$.

To see this, set $L := a^2$ and let $\epsilon > 0$ be given. Now, for any $x \in \mathbb{R}$, we have

$$|g(x) - L| = |x^2 - a^2| = |x - a| \cdot |x + a|.$$

We want to make this $< \epsilon$. We can make the term $|x - a|$ arbitrarily small, but for the $|x + a|$ term, it might not be so small but we can at least try to bound it by something fixed. Indeed, if $|x - a| < 1$ then

$$|x + a| = |x - a + 2a| \leq |x - a| + 2|a| < 1 + 2|a|.$$

So, we now see that we want $|x - a| < \frac{\epsilon}{1+2|a|}$.

Now we set $\delta := \min\left\{1, \frac{\epsilon}{1+2|a|}\right\}$. If $|x-a| < \delta$ then

$$\begin{aligned}|g(x) - L| &= |x-a| \cdot |x+1| \\ &\leq |x-a|(1+2|a|) \\ &< \frac{\epsilon}{1+2|a|}(1+2|a|) \\ &= \epsilon,\end{aligned}$$

as required.

We next establish uniqueness of limits of a function, in much the same way as we did for sequences.

Proposition 5.1.4. *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point. Let $f : X \to \mathbb{R}$ and let $L, L' \in \mathbb{R}$. If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} f(x) = L'$ then $L = L'$.*

Proof. Suppose for a contradiction that $L \neq L'$. Set $\epsilon := \frac{|L-L'|}{2}$. Using the definition of limit twice simultaneously, we may find $\delta > 0$ such that for all $x \in X \setminus \{a\}$ with $|x-a| < \delta$,

$$|f(x) - L| < \epsilon \quad \text{and} |f(x) - L'| < \epsilon.$$

Since a is a limit point of X, we may find $x \in X \cap (x-\delta, x+\delta) \setminus \{a\}$ (Exercise 4.1.8); that is, $x \in X$, $x \neq a$, and $|x-a| < \delta$.

Consequently, $|f(x) - L| < \epsilon$ and $|f(x) - L'| < \epsilon$. But then by using the Triangle Inequality,

$$|L - L'| \leq |L - f(x)| + |f(x) - L'| < \epsilon + \epsilon = |L - L'|,$$

a contradiction. □

Next, we shall connect limits of a function with limits of a sequence. This will be instrumental towards quickly proving basic properties (such as algebra of limits).

Proposition 5.1.5 (Sequential Characterization of Limits)**.** *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point. Let $f : X \to \mathbb{R}$ and let $L \in \mathbb{R}$. Then $\lim_{x\to a} f(x) = L$ if and only if for every sequence $(a_n)_{n=1}^{\infty}$ in $X \setminus \{a\}$ which converges to a, we have*

$$\lim_{n\to\infty} f(a_n) = L.$$

Proof. $\Rightarrow$: We assume that $\lim_{x\to a} f(x) = L$. To show that $\lim_{n\to\infty} f(a_n) = L$, let $\epsilon > 0$ be given. Choose $\delta > 0$ such that if $x \in X \setminus \{a\}$ and $|x-a| < \delta$ then

$$|f(x) - L| < \epsilon.$$

Next, since $\lim_{n\to\infty} a_n = a$, there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that if $n \geq n_0$ then $|a_n - a| < \delta$.

Then for $n \geq n_0$, we have $a_n \in X \setminus \{a\}$ and $|a_n - a| < \delta$, and thus,

$$|f(a_n) - L| < \epsilon,$$

as required.

$\Leftarrow$: We now assume that for every sequence $(a_n)_{n=1}^{\infty}$ in $X \setminus \{a\}$ which converges to a, we have $\lim_{n\to\infty} f(a_n) = L$.

Suppose for a contradiction that it is *not* true that $\lim_{x\to a} f(x) = L$. This means that there is some $\epsilon > 0$ such that for every $\delta > 0$, there exists $x \in X \setminus \{a\}$ such that $|x - a| < \delta$ and $|f(x) - L| \geq \epsilon$. We will now construct a sequence using this.

Namely, for each $n \in \mathbb{N}_{\geq 1}$, using $\delta := \frac{1}{n}$, we may choose some $a_n \in X \setminus \{a\}$ such that

$$|a_n - a| < \frac{1}{n} \quad \text{and} \quad |f(a_n) - L| \geq \epsilon.$$

This gives us a sequence $(a_n)_{n=1}^{\infty}$ in $X \setminus \{a\}$. Since $0 \leq |a_n - a| < \frac{1}{n}$ for all n, the Squeeze Theorem ensures that $\lim_{n\to\infty} |a_n - a| = 0$, i.e., $\lim_{n\to\infty} a_n = a$.

By hypothesis, it follows that $\lim_{n\to\infty} f(a_n) = L$. But this is a contradiction since we have $|f(a_n) - L| \geq \epsilon$ for all n. □

Proposition 5.1.6 (Algebra of Limits)**.** *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point. Let $f, g : X \to \mathbb{R}$ be functions which all have limits at a. Let $c \in \mathbb{R}$. Then:*

(i) $\lim_{x\to a}(f(x) + g(x)) = \left(\lim_{x\to a} f(x)\right) + \left(\lim_{x\to a} g(x)\right)$.

(ii) $\lim_{x\to a}(cf(x)) = c \lim_{x\to a} f(x)$.

(iii) $\lim_{x\to a}(f(x)g(x)) = \left(\lim_{x\to a} f(x)\right)\left(\lim_{x\to a} g(x)\right)$

(iv) If $f(x) \neq 0$ for all $x \in X$ and $\lim_{x\to a} f(x) \neq 0$ then $\lim_{x\to a} \frac{1}{f(x)} = \frac{1}{\lim_{x\to a} f(x)}$.

Proof. Exercise 5.1.1. (Use the Sequential Characterization of Limits and Proposition 2.3.1.) □

Theorem 5.1.7 (Squeeze Theorem)**.** *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point. Let $f, g, h : X \to \mathbb{R}$ be functions which satisfy*

$$f(x) \leq g(x) \leq h(x) \quad \textit{for all } x \in X.$$

If

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$

as well.

Proof. Exercise 5.1.2. □

The order structure in $\mathbb{R}$ allows us to talk of one-sided limits when we have a function of one variable.

Definition 5.1.8. *Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$ be a function and let $L \in \mathbb{R}$. If $a \in \mathbb{R}$ is a limit point of $X \cap (a, \infty)$, then we write*

$$\lim_{x\to a^+} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that, if $x \in X$, $x > a$, and

$$|x - a| < \delta$$

then

$$|f(x) - L| < \epsilon.$$

Likewise, if $a \in \mathbb{R}$ is a limit point of $X \cap (-\infty, a)$, then we write

$$\lim_{x \to a^-} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that, if $x \in X$, $x < a$, and

$$|x - a| < \delta$$

then

$$|f(x) - L| < \epsilon.$$

Example 5.1.9. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} -1, & x < 0; \\ 0, & x = 0; \\ 1, & x > 0. \end{cases}$$

Then one can see that

$$\lim_{x \to 0^+} f(x) = 1$$

and

$$\lim_{x \to 0^-} f(x) = -1.$$

Exercise 5.1.1. Prove Proposition 5.1.6.

Exercise 5.1.2. Prove Theorem 5.1.7.

Exercise 5.1.3. Define $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ by

$$f(x) := \frac{x+1}{x+3}.$$

Prove directly using the definition that

$$\lim_{x \to 0} f(x) = \frac{1}{3}.$$

Exercise 5.1.4. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function, $L \in \mathbb{R}$, and

$$\lim_{n \to \infty} f\left(\frac{1}{n}\right) = L.$$

Does it follow that $\lim_{x \to 0^+} f(x) = L$? (Give a proof or counterexample.)

Exercise 5.1.5. Let $(a_n)_{n=1}^{\infty}$ be a sequence in $\mathbb{R}$ and let $L \in \mathbb{R}$. Define a function

$$f : \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\} \to \mathbb{R}$$

by

$$f\left(\frac{1}{n}\right) = a_n.$$

Prove that $\lim_{n \to \infty} a_n = L$ if and only if $\lim_{x \to 0} f(x) = L$.

Exercise 5.1.6. Prove that the following limits do not exist.

(i) $\lim_{x\to 0} \frac{1}{x}$.

(ii) $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$.

Exercise 5.1.7. Let $X \subseteq \mathbb{R}$, let $a \in \mathbb{R}$ be a limit point of $X \cap (a, \infty)$. Let $f : X \to \mathbb{R}$ and let $L \in \mathbb{R}$. Define $g := f|_{X\cap(a,\infty)} : X \cap (a, \infty) \to \mathbb{R}$. Prove that

$$\lim_{x\to a^+} f(x) = L \quad \text{iff} \quad \lim_{x\to a} g(x) = L.$$

Exercise 5.1.8. (cf. [Sav17, Exercise 5.1.8]) Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point of both $X \cap (a, \infty)$ and $X \cap (-\infty, a)$. Let $f : X \to \mathbb{R}$ be a function and let $L \in \mathbb{R}$. Prove that

$$\lim_{x\to a} f(x) = L$$

if and only if both

$$\lim_{x\to a^+} f(x) = L \quad \text{and} \quad \lim_{x\to a^-} f(x) = L.$$

5.2 Continuity

We now formulate the notion of a continuous function by making use of the concept of a limit.

Definition 5.2.1. *Let $X \subseteq \mathbb{R}$ and let $a \in X$ be a point which is not isolated. Let $f : X \to \mathbb{R}$. We say f is* continuous *at a if*

$$\lim_{x\to a} f(x) = f(a).$$

If $a \in X$ is an isolated point, then we say that any function $f : X \to \mathbb{R}$ is continuous at a.

Making use of the definition of limit, we find that $f : X \to \mathbb{R}$ is continuous at $a \in X$ if for any $\epsilon > 0$ there exists $\delta > 0$ such that, if $x \in X$ and $|a-x| < \delta$ then $|f(a)-f(x)| < \epsilon$. This reformulation says that f is continuous if nearby points get sent to nearby images; it works whether or not a is an isolated point.

Example 5.2.2. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := 2x + 5$. Then using Example 5.1.2, we have for any $a \in \mathbb{R}$,

$$\lim_{x\to a} f(x) = f(a),$$

and thus f is continuous at every $a \in \mathbb{R}$.

Example 5.2.3. Define $f : [0, \infty) \to \mathbb{R}$ by

$$f(x) := \begin{cases} \sin\left(\frac{1}{x}\right), & \text{if } x > 0; \\ 0, & \text{if } x = 0. \end{cases}$$

Then f is not continuous at 0. To see this, we need to show that $\lim_{x\to 0} f(x) = 0$ does not hold. Consider the sequence

$$(a_n)_{n=1}^{\infty} = \left(\frac{1}{\pi(2n + 0.5)}\right)_{n=1}^{\infty}$$

which converges to 0. Note that the corresponding sequence $(f(a_n))_{n=1}^{\infty}$ is the constant sequence

$$(1, 1, \dots),$$

which does not converge to 0.

Example 5.2.4. Define $g : [0, \infty) \to \mathbb{R}$ by

$$g(x) := \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x > 0; \\ 0, & \text{if } x = 0. \end{cases}$$

This function oscillates similarly to the previous example, but the magnitude of the oscillation is tempered by the factor of x. We shall prove that this function *is* continuous at 0.

For this, consider functions $f, h : [0, \infty) \to \mathbb{R}$ given by $f(x) := x, h(x) := -x$. Then we have $f(x) \leq g(x) \leq h(x)$ for all x, and

$$\lim_{x \to \infty} f(x) = 0 = \lim_{x \to \infty} h(x).$$

Therefore, by the Squeeze Theorem, $\lim_{x \to \infty} g(x) = 0 = g(0)$, as required.

Proposition 5.2.5. *Let $X, Y \subseteq \mathbb{R}$, let $f : X \to Y \subseteq \mathbb{R}$ and $g : Y \to \mathbb{R}$ be functions, and let $a \in X$. Suppose that f is continuous at a and g is continuous at $f(a)$. Then $g \circ f : X \to \mathbb{R}$ is continuous at a.*

Proof. Let $\epsilon > 0$ be given. Since g is continuous at $f(a)$, there exists $\eta > 0$ (we're using η instead of δ) such that for all $y \in Y$, if $|y - f(a)| < \eta$ then $|g(y) - g(f(a))| < \epsilon$. Next, since f is continuous at a, using η in place of ϵ in that definition, we obtain $\delta > 0$ such that for all $x \in X$, if $|x - a| < \delta$ then $|f(x) - f(a)| < \eta$.

Now, putting these together, if $x \in X$ and $|x - a| < \delta$ then $|f(x) - f(a)| < \eta$ and so (viewing $f(x)$ as y),

$$|g(f(x)) - g(f(a))| < \epsilon,$$

as required. □

Proposition 5.2.6. *Let $X \subseteq \mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point. Let $f, g : X \to \mathbb{R}$ be functions which are all continuous at a. Let $c \in \mathbb{R}$. Then:*

(i) $f + g$ is continuous at a.

(ii) $c \cdot f$ is continuous at a.

(iii) $f \cdot g$ is continuous at a.

(iv) If $f(x) \neq 0$ for all $x \in X$ then $\frac{1}{f}$ is continuous at a.

Proof. These follow from Proposition 5.1.6. □

Exercise 5.2.1. [Leb16, Exercise 3.2.3] Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q}; \\ x^2, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that f is continuous at 1 and discontinuous at 2.

Exercise 5.2.2. Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $a \in X$. Let $r > 0$ and define

$$g := f|_{X \cap (a-r, a+r)} : X \cap (a - r, a + r) \to \mathbb{R}.$$

Prove that f is continuous at a if and only if g is continuous at a.

Exercise 5.2.3. [Sav17, Exercise 5.2.3] Let $f : \mathbb{R} \to \mathbb{R}$. Recall that for $B \subseteq \mathbb{R}$ the *preimage* of B under f is

$$f^{-1}(B) := \{x \in \mathbb{R} : f(x) \in B\}.$$

Prove that $f(x)$ is continuous for all $x \in \mathbb{R}$ if and only if for every open set U of $\mathbb{R}$, the preimage $f^{-1}(U)$ is open. Prove also that f is continuous if and only if for every closed set F of $\mathbb{R}$, the preimage $f^{-1}(F)$ is closed.

Exercise 5.2.4. Define the *Dirichlet function* $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} 0, & \text{if } x \notin \mathbb{Q}; \\ \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ in lowest terms (with } p \in \mathbb{Z} \text{ and } q \in \mathbb{N}_{\geq 1}). \end{cases}$$

Prove that for any $a \in \mathbb{R}$, $\lim_{x \to a} f(x) = 0$. From this, determine the points at which f is continuous.

5.3 Properties of continuous functions

In the previous section, we defined what it means for a function to be continuous at a single point. Here we consider functions that are "globally" continuous, or continuous at every point of their domain. Such functions have some strong and useful properties, as we shall see.

Definition 5.3.1. *Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function. We say that f is* continuous (on X) *if f is continuous at a for every $a \in X$.*

Theorem 5.3.2. *Let $K \subseteq \mathbb{R}$ be compact and let $f : K \to \mathbb{R}$ be a continuous function. Then its image, $f(K)$, is also compact.*

Proof. Let $(y_n)_{n=1}^{\infty}$ be a sequence in $f(K)$; we need to find a subsequence that converges to a point in $f(K)$. By definition of $f(K)$, we can find $x_n \in K$ for each n such that $y_n = f(x_n)$. Since K is compact, there is a subsequence $(x_{n_k})_{k=1}^{\infty}$ (of the sequence $(x_n)_{n=1}^{\infty}$) which converges to a point $a \in K$. Since f is continuous at a, we have

$$\lim_{x \to a} f(x) = f(a),$$

and therefore by the Sequential Characterization of Limits (Proposition 5.1.5), $\lim_{k \to \infty} f(x_{n_k}) = f(a)$. This says that the subsequence $(y_{n_k})_{k=1}^{\infty}$ converges to the point $f(a) \in f(K)$, as required. □

This has an important consequence for the problem of optimizing functions.

Corollary 5.3.3 (Extreme Value Theorem)**.** *Let $K \subset \mathbb{R}$ be compact and nonempty, and let $f : K \to \mathbb{R}$ be a continuous function. Then there exists $x_{\min}, x_{\max} \in K$ such that for all $x \in K$,*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

In other words, the image of f is bounded above and below, and it attains its bounds.

Proof. By the previous theorem, $f(K)$ is compact, By Exercise 4.2.4, $\min(f(K))$ and $\max(f(K))$ exist: that is, there are $y_{\min}, y_{\max} \in f(K)$ such that

$$y_{\min} \leq z \leq y_{\max}$$

for all $z \in f(K)$. We may take $x_{\min}, x_{\max} \in K$ such that

$$f(x_{\min}) = y_{\min} \quad \text{and} \quad f(x_{\max}) = y_{\max}.$$

Then it follows that for all $x \in K$,

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

□

Remark 5.3.4. If $X \subseteq \mathbb{R}$ is not compact, then it is possible that $f(X)$ is not bounded, or that it is bounded but it does not attain its bounds. Here are some examples.

Define $f : (0, 1] \to \mathbb{R}$ by $f(x) := \frac{1}{x}$. Then $f((0, 1]) = [1, \infty)$ is not bounded above.

Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) := \frac{x^2}{x^2+1}$. Then $g(\mathbb{R}) = [0, 1)$, which is bounded, but has no maximum.

Theorem 5.3.5 (Intermediate Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Let $y \in \mathbb{R}$ be any value between $f(a)$ and $f(b)$. Then there exists $z \in [a, b]$ such that $f(z) = y$.*

Proof. We may assume without loss of generality that $f(a) \leq f(b)$, since in the other case, we may replace f with $-f$ (and y with $-y$). Define

$$S := \{x \in [a, b] : f(x) \leq y\}.$$

This set is nonempty (it contains a) and bounded above (by b). Therefore, $z := \sup(S)$ exists.

We need only prove that $f(z) = y$.

To show $f(z) \geq y$, assume for a contradiction that $f(z) < y$. Then using $\epsilon := y - f(z)$ we can find δ such that for all $x \in [a, b]$ with $|x - z| < \delta$, we have $|f(x) - f(z)| < \epsilon$. We may find $x \in [a, b]$ with $|x - z| < \delta$ and $x > z$ (since $z \neq b$), and for such x,

$$f(x) < f(z) + \epsilon = f(z) + y - f(z) = y.$$

Hence, $x \in S$, contradicting that z is an upper bound.

Next, to show $f(z) \leq y$, assume for a contradiction that $f(z) > y$. Then using $\epsilon := f(z) - y$ we can find δ such that for all $x \in [a, b]$ with $|x - z| < \delta$, we have $|f(x) - f(z)| < \epsilon$. Then for all $x \in (z - \delta, z]$, we have

$$f(x) > f(z) - \epsilon = f(z) - (f(z) - y) = y,$$

so that $x \notin S$. Since z is an upper bound for S, this shows that $z - \delta$ is also an upper bound. But that contradicts that z is the *least* upper bound. □

The Intermediate Value Theorem is extremely powerful, since it gives us criteria (often easy to check) for an equation to have a solution. For example, for any $p \in \mathbb{N}_{\geq 0}$ and $\alpha > 0$, the function $f : [0, \infty) \to \mathbb{R}$ defined by

$$f(x) := x^p - \alpha$$

is continuous. We have $f(0) = -\alpha < 0$ and (since $n^p \to \infty$ as $n \to \infty$) we can find $b \in [0, \infty)$ such that $f(b) > 0$. Then the Intermediate Value Theorem immediately tells us that there exists $z \in [0, \infty)$ such that $f(z) = 0$, i.e., $z^p = \alpha$. In this way, we are immediately able to define p^{th} roots.

Proposition 5.3.6. *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then $f([a, b]) = [c, d]$ for some $c, d \in \mathbb{R}$.*

Proof. By the Extreme Value Theorem, there exist $x_{\min}, x_{\max} \in [a, b]$ such that

$$f([a, b]) \in [f(x_{\min}), f(x_{\max})].$$

Set $c := f(x_{\min})$ and $d := f(x_{\max})$.

We thus have one containment, $f([a, b]) \subseteq [c, d]$, and we only need to prove the other containment. Let $y \in [c, d]$. Then by the Intermediate Value Theorem, there exists z between $x_{\min}$ and $x_{\max}$ such that $f(z) = y$. Thus, $y \in f([a, b])$, as required. □

Exercise 5.3.1. [TBB08, Exercise 5.7.1] Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ such that for all $a \in \mathbb{R}$, f is not continuous at a, yet such that the conclusion of the Extreme Value Theorem holds, i.e., there exists $x_{\min}, x_{\max} \in \mathbb{R}$ such that for all $x \in \mathbb{R}$,

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

Exercise 5.3.2. Let $f : \mathbb{R} \to \mathbb{R}$ be a *periodic function*, i.e., such that there exists $P > 0$ such that $f(P + x) = f(x)$ for all $x \in \mathbb{R}$. Suppose that f is continuous. Prove that the conclusion of the Extreme Value Theorem holds, i.e., there exists $x_{\min}, x_{\max} \in \mathbb{R}$ such that for all $x \in \mathbb{R}$,

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}).$$

Exercise 5.3.3. Let $f : [a, b] \to [a, b]$ be a continuous function. Prove that there exists $y \in [a, b]$ such that $f(y) = y$ (such y is called a *fixed point*).

Exercise 5.3.4. Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a continuous function. Prove that $f(I)$ is an interval.

Exercise 5.3.5. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Suppose there exists $x_+, x_- \in \mathbb{R}$ such that

$$f(x_-) < 0 < f(x_+).$$

Prove that there exists $x_0 \in \mathbb{R}$ such that $f(x_0) = 0$. For a bigger challenge, prove that there are infinitely many such points.

Exercise 5.3.6. [Leb16, Exercise 3.3.3] Let $f : (0, 1) \to \mathbb{R}$ be a continuous function such that $\lim_{x\to 0} f(x) = \lim_{x\to 1} f(x)$. Prove that f attains a minimum or f attains a maximum (but possibly not both): that is, either there exists $x_{\min} \in (0, 1)$ such that $f(x_{\min}) \leq f(x)$ for all $x \in (0, 1)$, or there exists $x_{\max} \in (0, 1)$ such that $f(x) \leq f(x_{\max})$ for all $x \in (0, 1)$.

Exercise 5.3.7. (Big-O notation as $x \to 0$) Suppose that $A \subseteq \mathbb{R}$ is a set such that $0 \in \overline{A}$, and let $f : A \to \mathbb{R}$ and $g : A \to [0, \infty)$. We write

$$f(x) = O(g(x)) \quad (\text{as } x \to 0)$$

to mean that there exists $\delta > 0$ and a constant $K > 0$ such that $|f(x)| \leq Kg(x)$ for all $x \in A \cap (-\delta, \delta) \setminus \{0\}$.

Check that the following hold with this definition.

(i) If $f(x) = O(g(x))$ and $h(x) = O(g(x))$ then $f(x) + h(x) = O(g(x))$.

(ii) If $f(x) = O(g(x))$ and $h(x) = O(k(x))$ then $f(x)h(x) = O(g(x)k(x))$.

(iii) If $f(x) = O(g(x))$ and $g(x) = O(h(x))$ then $a_n = O(h(x))$.

(iv) If $f(x) = O(g(x))$ and $\lim_{x\to 0} g(x) = 0$ then $\lim_{x\to 0} f(x) = 0$.

5.4 Inverses of continuous functions

Recall that a function $f : A \to B$ is *injective* or *one-to-one* if for all $x, y \in A$,

$$f(x) = f(y) \Rightarrow x = y.$$

A function $f : A \to B$ is *surjective* or *onto* if

$$f(A) = B.$$

When f is both injective and surjective, it is called *bijective*, and it follows that it has an *inverse*: a function $f^{-1} : B \to A$ such that

$$f^{-1} \circ f = \mathrm{id}_A, \quad f \circ f^{-1} = \mathrm{id}_B.$$

Namely, for $y \in B$, one defines

$$f^{-1}(y) := x$$

where $x \in A$ is the unique element satisfying $f(x) = y$.

Often we are given an injective function $f : A \to B$, and by replacing the codomain B with $f(B)$, we get a function $f : A \to f(A)$ which is surjective for free, and thus bijective.

When we start with a continuous injective function $f : A \to \mathbb{R}$, it would be nice if its inverse f^{-1} were also continuous; unfortunately, this is not true in this level of generality, and we need to put restrictions on the domain A.

Example 5.4.1. Define $f : [0, 1) \cup [2, 3] \to \mathbb{R}$ by

$$f(t) := \begin{cases} t, & t \in [0, 1); \\ t - 1, & t \in [2, 3]. \end{cases}$$

This function is injective, so it has an inverse $f^{-1} : [0, 2] \to \mathbb{R}$ which is given by

$$f^{-1}(t) = \begin{cases} t, & t \in [0, 1); \\ t + 1, & t \in [1, 2]. \end{cases}$$

The function f is continuous but f^{-1} is not.

For positive results, we will focus on maps from an interval into $\mathbb{R}$; however see also Exercise 5.4.2 for a different general result.

Definition 5.4.2. *Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function.*

(i) We say f is (weakly) increasing *if for $x, y \in X$,*

$$x \leq y \implies f(x) \leq f(y).$$

(ii) We say f is strictly increasing *if for $x, y \in X$,*

$$x < y \implies f(x) < f(y).$$

(iii) We say f is (weakly) decreasing *if for $x, y \in X$,*

$$x \leq y \implies f(x) \geq f(y).$$

(iv) We say f is strictly decreasing *if for $x, y \in X$,*

$$x < y \implies f(x) > f(y).$$

(Note that when $X = \mathbb{N}_{\geq 1}$ then the above concepts for a function $f : \mathbb{N}_{\geq 1} \to \mathbb{R}$ agree with those already defined in Definition 2.4.1.)

Lemma 5.4.3. *Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a continuous function. The following are equivalent.*

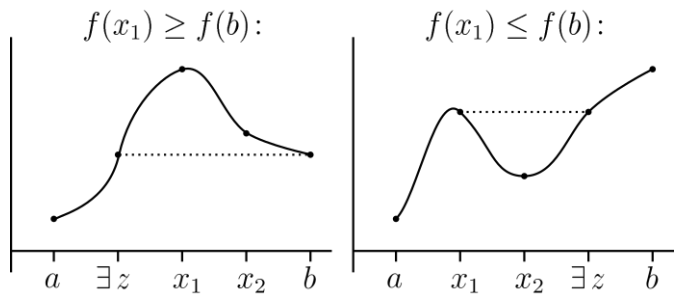
(i) f is either strictly increasing or strictly decreasing.

(ii) f is injective.

Proof. (i) $\Rightarrow$ (ii): This is immediate from the definitions.

(ii) $\Rightarrow$ (i): Assume without loss of generality that $f(a) < f(b)$, and let's prove that f is strictly increasing. If it is not then there exists $x_1, x_2 \in [a, b]$ such that $x_1 < x_2$ and $f(x_1) \geq f(x_2)$.

Consider two cases:



If $f(x_1) \geq f(b)$ then we have

$$f(a) \leq f(b) \leq f(x_1),$$

so by applying the Intermediate Value Theorem to $f|_{[a,x_1]}$ (with $y := f(b)$), there exists $z \in [a, x_1]$ such that $f(z) = f(b)$. Since f is injective, this would mean that $z = b$, but $z \leq x_1 < x_2 \leq b$, so this is a contradiction.

If $f(x_1) \leq f(b)$, then we have

$$f(x_2) \leq f(x_1) \leq f(b),$$

so by applying the Intermediate Value Theorem to $f|_{[x_2,b]}$ (with $y := f(x_1)$), there exists $z \in [x_2, b]$ such that $f(z) = f(x_1)$. Again, since f is injective, this would mean that $z = x_1$, but $z \geq x_2 > x_1$, so this is a contradiction. $\square$

Theorem 5.4.4. *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be an injective continuous function. Then $f^{-1} : f(I) \to \mathbb{R}$ is continuous.*

Proof. On every bounded closed subinterval $[a, b]$ of I, f is either strictly increasing or strictly decreasing. From this it is clear that f is either strictly increasing or strictly decreasing on all of I. Assume, without loss of generality, that f is strictly increasing.

Let $b \in f(I)$; to show that f^{-1} is continuous at b, we must show $\lim_{y \to b} f^{-1}(y) = f^{-1}(b)$. So, let $\epsilon > 0$ be given.

Set $a := f^{-1}(b)$ (so that $f(a) = b$). Let us assume that $a \in I^\circ$ – in the other case, a is an endpoint of the interval I and then we will need to adjust the following argument slightly. Since there is no harm in decreasing ϵ, we may further assume that $a - \epsilon, a + \epsilon \in I$. Since f is strictly increasing, we have

$$f(a - \epsilon) < f(a) < f(a + \epsilon).$$

We then set

$$\delta := \min\{f(a) - f(a - \epsilon), f(a + \epsilon) - f(a)\} > 0.$$

We are done once we prove the following claim.

Claim. If $y \in f(I)$ and $|y - b| < \delta$, then $|f^{-1}(y) - f^{-1}(b)| < \epsilon$.

To prove this claim, suppose it is false, so either

$$f^{-1}(y) \leq f^{-1}(b) - \epsilon \quad \text{or} \quad f^{-1}(y) \geq f^{-1}(b) + \epsilon.$$

Set $x := f^{-1}(y)$ (so that $f(x) = y$), so the above reads

$$x \leq a - \epsilon \quad \text{or} \quad x \geq a + \epsilon.$$

In the first case, since f is strictly increasing, we get

$$y = f(x) \leq f(a - \epsilon) \leq f(a) - \delta = b - \delta.$$

In the second case, we similarly obtain

$$y = f(x) \geq f(a + \epsilon) \geq f(a) + \delta = b + \delta.$$

In either case, we get a contradiction to the assumption that $|y - b| < \delta$.

This concludes the proof of the claim. □

Exercise 5.4.1. Let $p \in \mathbb{N}_{\geq 2}$. Prove that the function $f : [0, \infty) \to \mathbb{R}$ given by $f(x) := \sqrt[p]{x}$ is continuous.

Exercise 5.4.2. Let $K \subseteq \mathbb{R}$ be a compact set and let $f : K \to \mathbb{R}$ be a continuous injective function. Prove that $f^{-1} : K \to \mathbb{R}$ is continuous.
[*Hint. Generalize Exercise 5.2.3 to say that if $X \subseteq \mathbb{R}$ is closed then a function $f : X \to \mathbb{R}$ is continuous if and only if for every closed set F of $\mathbb{R}$, the preimage $f^{-1}(F)$ is closed.*]

5.5 Uniform continuity

Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function. If we unwind the definition of "f is a continuous function", it says:

For any $x \in X$ and $\epsilon > 0$ there exists $\delta > 0$ such that if $y \in X$ and $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.

In this definition, the δ is allowed to depend on ϵ and on x. It is crucial that δ is allowed to depend on ϵ, to capture the idea that "close points get sent to close images" (see Exercise 5.5.3). However, it is less clear, at first, what role the dependence on x plays.

Here we explore this subtlety, by first introducing a variant on continuity where δ is not allowed to depend on x.

Definition 5.5.1. *Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function. We say that f is* uniformly continuous *(on X) if for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x, y \in X$, if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$.*

Example 5.5.2. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := |x|$. This is uniformly continuous.

To see this, given $\epsilon > 0$, we set $\delta := \epsilon$. Then if $x, y \in \mathbb{R}$ and $|x - y| < \delta$, then

$$\big||x| - |y|\big| \leq |x - y| < \delta = \epsilon.$$

Example 5.5.3. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := x^2$ Then f is continuous (by Proposition 5.2.6 (iii)) but not uniformly continuous.

To see that it is not uniformly continuous, suppose for a contradiction that it is. Then using $\epsilon := 1$ there would be some $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|x - y| < \delta$ then $|f(x) - f(y)| < 1$.

However, we may take

$$x := \frac{1}{\delta}, \quad y := x + \frac{\delta}{2},$$

so that $|x - y| = \frac{\delta}{2} < \delta$. However,

$$\begin{aligned} |f(x) - f(y)| &= y^2 - x^2 \\ &= (y - x)(y + x) \\ &\geq \frac{\delta}{2}(x + x) \\ &= \frac{\delta}{2} \cdot \frac{2}{\delta} = 1, \end{aligned}$$

a contradiction.

The obstruction to uniform continuity in the previous example came from points that are large. In fact, in the next example, we should that if we eliminate large points from the domain then the function becomes uniformly continuous.

Example 5.5.4. Consider the restriction of the previous example to $[a, b]$ where $a, b \in \mathbb{R}$ with $a < b$. That is, $f : [a, b] \to \mathbb{R}$ given by $f(x) := x^2$. This function is uniformly continuous.

To see this, let $\epsilon > 0$ be given. Set $M := \max\{|a|, |b|\}$, so that M is an upper bound for $\{|x| : x \in [a, b]\}$. Take $\delta := \frac{\epsilon}{2M}$. Then for $x, y \in [a, b]$ with $|x - y| < \delta$, we have

$$\begin{aligned} |f(x) - f(y)| &= |x^2 - y^2| \\ &= |x - y||x + y| \\ &\leq |x - y|(|x| + |y|) \\ &< \delta(M + M) \\ &= \epsilon, \end{aligned}$$

as required.

In fact, the previous example is a special case of a general phenomenon: continuous functions on compact sets are *automatically* uniformly continuous.

Theorem 5.5.5. *Let $K \subseteq \mathbb{R}$ be compact and let $f : K \to \mathbb{R}$ be continuous. Then f is uniformly continuous.*

Proof. We prove this by contradiction: suppose that f is not uniformly continuous, so there exists $\epsilon > 0$ such that for every $\delta > 0$ there exist $x, y \in K$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon$.

In particular, for each n, we may choose some $x_n, y_n \in X$ such that $|x_n - y_n| < \frac{1}{n}$ yet $|f(x_n) - f(y_n)| \geq \epsilon$. This gives us two sequences, $(x_n)_{n=1}^{\infty}$ and $(y_n)_{n=1}^{\infty}$, in K.

Since K is compact, there is a subsequence $(x_{n_k})_{k=1}^{\infty}$ which converges to a point $a \in K$. The subsequence $(y_{n_k})_{k=1}^{\infty}$ must also converge to a, since

$$|y_{n_k} - a| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - a| \to 0,$$

and so by the Squeeze Theorem, $|y_{n_k} - a| \to 0$.

Since f is continuous and by using the Sequential Characterization of Limits (Proposition 5.1.5), it follows that

$$\lim_{k\to\infty} f(x_{n_k}) = f(a) = \lim_{k\to\infty} f(y_{n_k}),$$

and in particular,

$$\lim_{k\to\infty} |f(x_{n_k}) - f(y_{n_k})| = 0.$$

But this is a contradiction since $|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon$ for all k. □

Exercise 5.5.1. Determine which of the following functions are uniformly continuous.

(a) $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) := \frac{1}{x}$.

(b) $f : [1, \infty) \to \mathbb{R}$ defined by $f(x) := \frac{1}{x}$.

(c) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := \frac{1}{1+x^2}$

(d) $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) := \sin\left(\frac{1}{x}\right)$.

(e) $f : [1, \infty) \to \mathbb{R}$ defined by $f(x) := \ln(x)$.

(f) $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) := \sin(x^2)$.

Exercise 5.5.2. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Prove that for any bounded set $X \subset \mathbb{R}$, the restricted function $f|_X$ is uniformly continuous.

Exercise 5.5.3. Consider modifying the definition of continuity so that δ may not depend on ϵ (although it can depend on x): for any x there exists $\delta > 0$ such that for any $\epsilon > 0$, if $y \in X$ and $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$. Show that if $X = \mathbb{R}$ and $f : X \to \mathbb{R}$ satisfies this definition then f must be a constant function.

Exercise 5.5.4. Let $X = X_1 \cup X_2 \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function. Suppose that $f|_{X_1}$ and $f|_{X_2}$ are both uniformly continuous. Must f be uniformly continuous? [*Hint. Consider the case* $X_1 = (-\infty, 0)$ *and* $X_2 = (0, \infty)$.]

Exercise 5.5.5. Let $X \subseteq \mathbb{R}$ be bounded and let $f : X \to \mathbb{R}$ be uniformly continuous. Prove that $f(X)$ is bounded.

Exercise 5.5.6. Let $f : \mathbb{R} \to \mathbb{R}$ be a function and let $L \in \mathbb{R}$. Let us write $\lim_{|x|\to\infty} f(x) = L$ if for every $\epsilon > 0$ there exists M such that if $x \in \mathbb{R}$ and $|x| \geq M$ then $|f(x) - L| < \epsilon$.

Prove that if f is continuous and $\lim_{|x|\to\infty} f(x) = L$ then f is uniformly continuous.

Exercise 5.5.7. Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be a function. f is *Lipschitz* if there is some $M > 0$ such that for all $x, y \in X$,

$$|f(x) - f(y)| \leq M|x - y|.$$

(i) Prove that if f is Lipschitz then it is uniformly continuous.

(ii) Prove that $f : [0, 1] \to \mathbb{R}$ defined by $f(t) := \sqrt{t}$ is uniformly continuous but *not* Lipschitz.

(iii) Prove that $f : [1, \infty) \to \mathbb{R}$ defined by $f(t) := \sqrt{t}$ is Lipschitz.

Exercise 5.5.8. (Challenging!) Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$ be uniformly continuous. Prove that there is a continuous function $g : \overline{X} \to \mathbb{R}$ such that $g|_X = f$.
[*For a point* $a \in \overline{X} \setminus X$, *take a sequence* $(x_n)_{n=1}^{\infty}$ *in* X *that converges to* a. *Prove that* $(f(x_n))_{n=1}^{\infty}$ *is Cauchy, so define* $g(a) := \lim_{n\to\infty} f(x_n)$. *Prove that this makes* g *a well-defined function and that it is continuous.*]

5.6 Infinite limits and limits at infinity

It is often useful to have the concept of ∞ at our disposal when speaking of limits. We have a few definitions related to this concept.

Definition 5.6.1. *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$.*

- *If $m = 1$ and $a \in \mathbb{R}$ is a limit point of A, then we write $\lim_{x \to a} f(x) = \infty$ if for every $R > 0$ there exists $\delta > 0$ such that if $x \in A \setminus \{a\}$ and $|x - a| < \delta$ then*
$$f(x) > R.$$
- *Similarly, if $m = 1$ and $a \in \mathbb{R}$ is a limit point of A, then we write $\lim_{x \to a} f(x) = -\infty$ if for every $R > 0$ there exists $\delta > 0$ such that if $x \in A \setminus \{a\}$ and $|x - a| < \delta$ then*
$$f(x) < -R.$$
- *If $d = 1$, A is not bounded above, and $L \in \mathbb{R}$, we write $\lim_{x \to \infty} f(x) = L$ if for every $\epsilon > 0$ there exists $R > 0$ such that if $x \in A$ and $x > R$ then*
$$|f(x) - L| < \epsilon.$$
- *Similarly, if $d = 1$, A is not bounded below, and $L \in \mathbb{R}$, we write $\lim_{x \to -\infty} f(x) = L$ if for every $\epsilon > 0$ there exists $R > 0$ such that if $x \in A$ and $x < -R$ then*
$$|f(x) - L| < \epsilon.$$
- *If A is not bounded and $L \in \mathbb{R}$, we write $\lim_{|x| \to \infty} f(x) = L$ if for every $\epsilon > 0$ there exists $R > 0$ such that if $x \in A$ and $|x| > R$ then*
$$|f(x) - L| < \epsilon.$$

One can also define things such as $\lim_{|x| \to \infty} f(x) = \infty$, by splicing these definitions appropriately (Exercise 5.6.1).

Let $f : \mathbb{N}_{\geq 1} \to \mathbb{R}$, and form the associated sequence $(a_n)_{n=1}^{\infty}$ (that is, $a_n = f(n)$). Observe that for $L \in \mathbb{R}$,
$$\lim_{x \to \infty} f(x) = L \quad \text{iff} \quad \lim_{n \to \infty} a_n = L.$$

Exercise 5.6.1. Splice the definitions in order to define $\lim_{x \to \infty} f(x) = \infty$, $\lim_{x \to -\infty} f(x) = \infty$, $\lim_{x \to \infty} f(x) = -\infty$, $\lim_{x \to -\infty} f(x) = -\infty$, $\lim_{|x| \to \infty} f(x) = -\infty$, and $\lim_{|x| \to -\infty} f(x) = -\infty$.

Exercise 5.6.2. Let $A \subseteq \mathbb{R}$, $B \subseteq \mathbb{R}$, and let $f : A \to \mathbb{R}$, $g : B \to \mathbb{R}$, such that $f(A) \subseteq B$. Let $a \in \mathbb{R}$ be a limit point of A. Suppose that $\lim_{x \to a} |f(x)| = \infty$ (this makes sense since $x \mapsto |f(x)|$ is a real-valued function) and $\lim_{|y| \to \infty} g(y) = L \in \mathbb{R}$. Prove that
$$\lim_{x \to a} g(f(x)) = L.$$

Exercise 5.6.3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Suppose that $\lim_{|x| \to \infty} f(x)$ exists. Prove that f is bounded.
[*Use the Extreme Value Theorem.*]

Chapter 6

Differentiation

We will now use the concept of a limit in order to define differentiability and the derivative of a real-variable function. Differentiation is a fundamental operation in calculus; the machinery built in the previous chapter will allow us to rigorously establish the key properties of the derivative.

6.1 The derivative

Definition 6.1.1. *Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$ be a function, let $a \in X$ be a non-isolated point. We write*

$$f'(a) := \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

(either as a real number, as $\pm\infty$, or as "not existing"). When this limit exists and is finite, we say f is differentiable *at a. If all points of X are non-isolated and f is differentiable at every point of X, then we say f is* differentiable *on X. In this case, we get a function $f' : X \to \mathbb{R}$, also written as $\frac{df}{dx}$.*

As you have seen in Calculus, $f'(a)$ is the slope of the tangent line to the graph of $y = f(x)$ at the point $(a, f(a))$.

Example 6.1.2. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2$. Then for $a \in \mathbb{R}$,

$$\begin{aligned} f'(a) &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \to a} \frac{x^2 - a^2}{x - a} \\ &= \lim_{x \to a} \frac{(x + a)(x - a)}{x - a} \\ &= \lim_{x \to a} x + a = 2a. \end{aligned}$$

Thus f is differentiable on $\mathbb{R}$.

Example 6.1.3. Define $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ by

$$f(x) := \frac{1}{x}.$$

Then for $a \in \mathbb{R} \setminus \{0\}$,

$$\begin{aligned} f'(a) &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a} \\ &= \lim_{x \to a} \frac{a - x}{xa} \cdot \frac{1}{x - a} \\ &= \lim_{x \to a} \frac{-1}{xa} \\ &= \frac{-1}{a^2}. \end{aligned}$$

Example 6.1.4. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := |x|$. If $a > 0$, note that for x close to a we have $f(x) = x$. Thus,

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{x \to a} \frac{x - a}{x - a} = 1.$$

If $a < 0$, for x close to a we have $f(x) = -x$, and thus,

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{x \to a} \frac{-x - (-a)}{x - a} = -1.$$

However for $a = 0$, we have

$$\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{x - 0}{x - 0} = 1,$$

whereas

$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{-x - 0}{x - 0} = -1.$$

Since these one-sided limits are different, $f'(0)$ does not exist (Exercise 5.1.8).

Proposition 6.1.5. *Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$ be a function, let $a \in X$ be a non-isolated point. If f is differentiable at a then f is continuous at a.*

Proof. We need to show that $\lim_{x \to a} f(x) = f(a)$. We have

$$\begin{aligned} \lim_{x \to a} f(x) - f(a) &= \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \cdot (x - a) \right) \\ &= \left(\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \right) \left(\lim_{x \to a} x - a \right) && \text{(Algebra of Limits)} \\ &= f'(a) \cdot 0 = 0. \end{aligned}$$

In the above, we were able to use Algebra of Limits (Proposition 5.1.6) because the limits exist.

We therefore have $\lim_{x \to a} f(x) = f(a)$, as required. □

The following example shows that, even if f is differentiable everywhere, the function f' might not be continuous.

Example 6.1.6. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0; \\ 0, & x = 0. \end{cases}$$

This function is differentiable everywhere; for $a \neq 0$, this will follow from the rules of differentiation (Chain Rule, Product Rule, Quotient Rule), and we have

$$\begin{aligned} f'(a) &= 2a \sin\left(\frac{1}{a}\right) + a^2 \cos\left(\frac{1}{a}\right) \cdot \frac{-1}{a^2} \\ &= 2a \sin\left(\frac{1}{a}\right) - \cos\left(\frac{1}{a}\right). \end{aligned}$$

For $a = 0$, we have

$$\begin{aligned} f'(a) &= \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) \\ &= 0 \end{aligned}$$

by Example 5.2.4.

Similarly,

$$\begin{aligned} \lim_{x \to 0} f'(x) &= \lim_{x \to 0} 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) \\ &= 0 - \lim_{x \to 0} \cos\left(\frac{1}{x}\right), \end{aligned}$$

but this limit does not exist (just like Exercise 5.1.6 (ii)). Hence, f' is not continuous at 0.

If f is differentiable and f' is differentiable, we write the derivative of f' as f'' or $f^{(2)}$. Similarly we define $f''' = f^{(3)}$, and so on.

Exercise 6.1.1. Let $n \in \mathbb{N}_{\geq 1}$ and define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := x^n$. Directly compute $f'(a)$ for $a \in \mathbb{R}$.

Exercise 6.1.2. [TBB08, Exercise 7.2.1] Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$ be a function, let $a \in X$ be a non-isolated point. Show that

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Exercise 6.1.3. [TBB08, Exercise 7.2.3] Determine which of the following functions $f : \mathbb{R} \to \mathbb{R}$ are differentiable at 0.

(i) $f(x) := x|x|$.

(ii) $f(x) := \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0; \\ 0, & x = 0. \end{cases}$

(iii) $f(x) := \begin{cases} x^2, & x \in \mathbb{Q}; \\ 0, & x \notin \mathbb{Q}. \end{cases}$

Exercise 6.1.4. [Leb16, Exercise 4.1.9] Let $X \subseteq \mathbb{R}$ be a set in which all points are non-isolated. Suppose that $f : X \to \mathbb{R}$ is differentiable function which is Lipschitz (see the definition in Exercise 5.5.7). Prove that $f' : X \to \mathbb{R}$ is bounded.

Exercise 6.1.5. [TBB08, Exercise 7.2.13] Let $f : \mathbb{R} \to \mathbb{R}$ be strictly increasing and differentiable. Does this imply that $f'(x) \geq 0$ for all $x \in \mathbb{R}$? Does it imply that $f'(x) > 0$ for all $x \in \mathbb{R}$?

6.2 Computation rules for derivatives

In Calculus, you have seen a number of tools for computing derivatives. In this section, we shall prove these rules.

Proposition 6.2.1. *Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$ be functions, let $a \in X$ be a non-isolated point. Suppose that f and g are both differentiable at a, and let $c \in \mathbb{R}$. Then*

(i) $(cf)'(a) = c(f'(a))$ *and* $(f+g)'(a) = f'(a) + g'(a)$; (Linearity)

(ii) $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$. (Product Rule)

We don't do the Quotient Rule here; it can be proven using the Chain Rule and the Product Rule (Exercise 6.2.2).

Proof. Part (i) is Exercise 6.2.1

(iii): We compute

$$\begin{aligned}
\lim_{x\to a} \frac{(fg)(x) - (fg)(a)}{x-a} &= \lim_{x\to a} \frac{f(x)g(x) - f(a)g(a)}{x-a} \\
&= \lim_{x\to a} \frac{f(x)g(x) - f(a)g(x) + f(a)g(x) - f(a)g(a)}{x-a} \\
&= \left(\lim_{x\to a} \frac{f(x)-f(a)}{x-a}\right)\left(\lim_{x\to a} g(x)\right) + f(a)\left(\lim_{x\to a} \frac{g(x)-g(a)}{x-a}\right) \\
&= f'(a)g(a) + f(a)g'(a),
\end{aligned}$$

using Algebra of Limits in the third line, and that g is continuous at a (Proposition 6.1.5) for the last equality. □

Proposition 6.2.2 (Chain Rule)**.** *Let $X, Y \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$ and $g : Y \to \mathbb{R}$ be functions, let $a \in X$ be a non-isolated point. Suppose that $f(X) \subseteq Y$ and that $f(a)$ is a non-isolated point of Y. Suppose also that f is differentiable at a and g is differentiable at $f(a)$. Then $g \circ f$ is differentiable at a, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Proof. For $y \in Y$, define

$$h(y) := \begin{cases} \frac{g(y)-g(f(a))}{y-f(a)}, & y \neq f(a); \\ g'(f(a)), & y = f(a). \end{cases}$$

Since $\lim_{y\to f(a)} h(y) = g'(f(a)) = h(f(a))$, h is continuous at $f(a)$.

Claim. For all $x \in X \setminus \{a\}$,

$$\frac{g(f(x)) - g(f(a))}{x-a} = h(f(x))\frac{f(x)-f(a)}{x-a}.$$

To see this claim, we consider two cases. If $f(x) \neq f(a)$, then we have

$$\begin{aligned}
\frac{g(f(x)) - g(f(a))}{x-a} &= \frac{g(f(x)) - g(f(a))}{f(x)-f(a)} \cdot \frac{f(x)-f(a)}{x-a} \\
&= h(f(x))\frac{f(x)-f(a)}{x-a}.
\end{aligned}$$

On the other hand, if $f(x) = f(a)$, then both sides are 0, so the claim holds.

Now, using the claim, we compute

$$\begin{aligned}(g \circ f)'(a) &= \lim_{x \to a} \frac{g(f(x)) - g(f(a))}{x - a} \\ &= \lim_{x \to a} h(f(x)) \frac{f(x) - f(a)}{x - a} \\ &= h(f(a)) f'(a) \\ &= g'(f(a)) f'(a).\end{aligned}$$

In the second-last line we used the fact that $h \circ f$ is continuous at a, which is a consequence of Proposition 5.2.5. □

Proposition 6.2.3 (Inverse Rule)**.** *Let $X \subseteq \mathbb{R}$ be an interval, let $f : X \to \mathbb{R}$ be a continuous, injective function. Let $a \in X$. If f is differentiable at a and $f'(a) \neq 0$ then $f^{-1} : f(X) \to \mathbb{R}$ is differentiable at $f(a)$ and*

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}.$$

Proof. For convenience, set $b := f(a)$.

Similarly to the previous proof, we define $h : X \to \mathbb{R}$ by

$$h(x) := \begin{cases} \frac{f(x)-f(a)}{x-a}, & x \neq a; \\ f'(a), & x = a. \end{cases}$$

This function is continuous, and $h(a) \neq 0$. It follows that h is nonzero everywhere on some open interval containing a; so by shrinking X, we may assume that $h(x) \neq 0$ for all $x \in X$.

Now we observe that, for $y \in f(X) \setminus \{b\}$, we have

$$\begin{aligned}\frac{f^{-1}(y) - f^{-1}(b)}{y - b} &= \frac{f^{-1}(y) - a}{f(f^{-1}(y)) - f(a)} \\ &= \frac{1}{h(f^{-1}(y))}.\end{aligned}$$

Since h and f^{-1} are continuous (the latter by Theorem 5.4.4), so is their composition. Hence by the Algebra of Limits, we get

$$\lim_{y \to b} \frac{f^{-1}(y) - f^{-1}(b)}{y - b} = \frac{1}{\lim_{y \to b} h(f^{-1}(y))} = \frac{1}{h(f^{-1}(b))} = \frac{1}{h(a)} = \frac{1}{f'(a)}.$$

□

Exercise 6.2.1. Prove Proposition 6.2.1 (i).

Exercise 6.2.2. Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$ be a function, let $a \in X$ be a non-isolated point. Suppose that f and g are both differentiable at a, and that $g(x) \neq 0$ for all $x \in X$, and $g'(a) \neq 0$.

(a) Using the Chain Rule, prove that $\left(\frac{1}{g}\right)'(a) = \frac{-g'(a)}{g(a)^2}$.

(b) Using (a) and the Product Rule, prove the *Quotient Rule*: $\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}$.

Exercise 6.2.3. (cf. [Leb16, Exercise 4.1.11]) Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$ be a function, let $a \in X$ be a non-isolated point. Suppose that f is bounded and that $g(a) = g'(a) = 0$. Consider the function $h = fg$.

(a) Prove that $h'(a) = 0$.

(b) Is it possible to use the product rule to do part (a)?

Exercise 6.2.4. [TBB08, Exercise 7.3.17] Using the fact that the derivative of $\sin(x)$ is $\cos(x)$, find a formula for the derivative of $\sin^{-1}(x)$ (considered as a function $[-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

6.3 Optimizing differentiable functions

Derivatives are often used for optimizing functions: finding their maximum and/or minimum (extrema). The precise relationship between derivatives and extrema has to do with the concept of "local extrema": the minimum or maximum of a point in a (possibly) small neighbourhood.

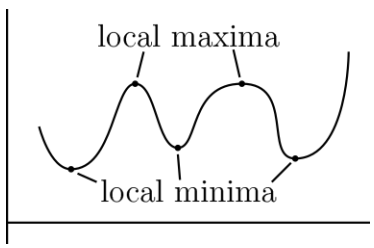
Definition 6.3.1. *Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $a \in X$ be an interior point.*

(i) a is a local minimum *of f if there exists $r > 0$ such that $(a - r, a + r) \subseteq X$ and*

$$f(a) \leq f(x) \quad \textit{for all } x \in (a - r, a + r).$$

(ii) a is a local maximum *of f if there exists $r > 0$ such that $(a - r, a + r) \subseteq X$ and*

$$f(a) \geq f(x) \quad \textit{for all } x \in (a - r, a + r).$$



Theorem 6.3.2. *Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $a \in X$ be an interior point. If f has a local maximum or local minimum at a and f is differentiable at a, then $f'(a) = 0$.*

Proof. Assume that f has a local minimum at a (the other case is almost identical). Let $r > 0$ be such that $(a - r, a + r) \subseteq X$ and

$$f(a) \leq f(x) \quad \text{for all } x \in (a - r, a + r).$$

Since f is differentiable, the limit

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists, and by the Sequential Characterization of Limits, for any sequence $(x_n)_{n=1}^{\infty}$ in X converging to a, we will have

$$f'(a) = \lim_{n \to \infty} \frac{f(x_n) - f(a)}{x_n - a}.$$

Now, let us first take some sequence $(x_n)_{n=1}^{\infty}$ in $(a, a+r)$ converging to a from the right: for example,

$$x_n := a + \frac{r}{n+1}.$$

Then we have $x_n - a > 0$ and $f(x_n) - f(a) \geq 0$. Thus,

$$\frac{f(x_n) - f(a)}{x_n - a} \geq 0,$$

which implies that

$$f'(a) = \lim_{n\to\infty} \frac{f(x_n) - f(a)}{x_n - a} \geq 0.$$

However, we may similarly take some sequence $(x_n)_{n=1}^{\infty}$ in $(a-r, a)$ converging to a from the left: for example,

$$x_n := a - \frac{r}{n+1}.$$

In this case we have $x_n - a < 0$ and $f(x_n) - f(a) \geq 0$, so that

$$\frac{f(x_n) - f(a)}{x_n - a} \leq 0,$$

and thus,

$$f'(a) = \lim_{n\to\infty} \frac{f(x_n) - f(a)}{x_n - a} \leq 0.$$

Putting these two inequalities together, we obtain $f'(a) = 0$ as required. □

Suppose $f : [a, b] \to \mathbb{R}$ is continuous. By the Extreme Value Theorem, it attains a maximum and a minimum. Let's say we want to find a point $x_{\max} \in [a, b]$ where it achieves its maximum; the point $x_{\max}$ will in particular be a local maximum, so if $x_{\max}$ is an interior point of $[a, b]$ and f is differentiable at $x_{\max}$, then by the above theorem, $f'(x_{\max}) = 0$. It is possible, however, that $x_{\max}$ doesn't satisfy the hypotheses of the above theorem (it might not be differentiable, or it might not be an interior point). Altogether, we get the following three possibilities:

(i) f is differentiable at $x_{\max}$ and $f'(x_{\max}) = 0$,

(ii) $x_{\max} = a$ or $x_{\max} = b$ (it is not an interior point), or

(iii) f is not differentiable at $x_{\max}$.

Exercise 6.3.1. Give examples of continuous functions $f : [a, b] \to \mathbb{R}$ which attain a maximum at a unique point $x_{\max}$ such that

(a) in the first example, f is not differentiable at $x_{\max}$,

(b) in the second example, $x_{\max}$ is a and f is differentiable at $x_{\max}$,

Exercise 6.3.2. [TBB08, Exercise 7.5.1] Give an example of a differentiable function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(0) = 0$ but 0 is not a local maximum nor minimum.

Exercise 6.3.3. [TBB08, Exercise 7.5.2] Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} x^4 \left(2 + \sin\left(\frac{1}{x}\right)\right), & x \neq 0; \\ 0, & x = 0. \end{cases}$$

(i) Prove that f is differentiable on $\mathbb{R}$.

(ii) Prove that f has a minimum at 0.

(iii) Prove that for every $r > 0$, $f'((-r, r))$ contains both (strictly) positive and negative numbers.

6.4 The Mean Value Theorem

The Mean Value Theorem tells us that the average change of a differentiable function is attained as the derivative of the function at some point. This will be derived from Rolle's Theorem, which is the special case in which the average change of a function is zero.

Theorem 6.4.1 (Rolle's Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b). If $f(a) = f(b)$ then there exists $x_0 \in (a, b)$ such that*

$$f'(x_0) = 0.$$

Proof. If f is constant, then $f'(x_0) = 0$ for all $x_0 \in (a, b)$. Otherwise, by the Extreme Value Theorem, let $x_{\min}, x_{\max}$ be the minimal and maximal points of f, so that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for all $x \in [a, b]$. Since x is not constant, we know that either

$$f(x_{\min}) < f(a) = f(b) \text{ or } f(x_{\max}) > f(a) = f(b),$$

(or both) and in particular, either $x_{\min} \in (a, b)$ or $x_{\max} \in (a, b)$. We set x_0 equal to one of these two points which is in (a, b).

Then f is differentiable at x_0, and since x_0 is a global extreme point for f, it follows from Theorem 6.3.2 that $f'(x_0) = 0$, as required. □

Next we do a generalization of the Mean Value Theorem. This generalization can be used to prove L'Hôpital's rule (Exercise 6.4.5).

Theorem 6.4.2 (Cauchy's Mean Value Theorem)**.** *Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous functions that are differentiable on (a, b). Then there exists $x_0 \in (a, b)$ such that*

$$(f(b) - f(a))g'(x_0) = (g(b) - g(a))f'(x_0).$$

Proof. Define $h : [a, b] \to \mathbb{R}$ by

$$h(x) := (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

Our strategy is to apply Rolle's Theorem to h.

The function h is continuous, and it is differentiable on (a, b) with

$$h'(x) = (f(b) - f(a))g'(x) - (g(b) - g(a))f'(x).$$

Also, we compute

$$\begin{aligned} h(a) &= (f(b) - f(a))g(a) - (g(b) - g(a))f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a). \end{aligned}$$

Likewise, $h(b) = f(b)g(a) - g(b)f(a)$.

Hence by Rolle's Theorem, there exists $x_0 \in (a, b)$ such that $h'(x_0) = 0$. Looking at the computation for $h'(x)$, it follows that

$$(f(b) - f(a))g'(x_0) = (g(b) - g(a))f'(x_0).$$

□

Corollary 6.4.3 (Mean Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be a continuous function that is differentiable on (a, b). Then there exists $x_0 \in (a, b)$ such that*

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define $g : [a, b] \to \mathbb{R}$ by $g(x) := x$, then apply Cauchy's Mean Value Theorem to f, g to get $x_0 \in (a, b)$ such that

$$(f(b) - f(a))g'(x_0) = (g(b) - g(a))f'(x_0).$$

Since $g'(x_0) = 1$ and $g(b) - g(a) = b - a$, this rearranges to become

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}.$$

□

Exercise 6.4.1. [TBB08, Exercise 7.6.4] Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function and there is some $c > 0$ such that $f'(x) \geq c$ for all $x \in \mathbb{R}$. Prove that $\lim_{x\to\infty} f(x) = \infty$.

Exercise 6.4.2. [TBB08, Exercise 7.6.5] Let $f : \mathbb{R} \to \mathbb{R}$ be a twice-differentiable function (i.e., $f'(x)$ and $f''(x)$ exist everywhere). Prove that if f has at least three zeroes (three distinct points x_1, x_2, x_3 such that $f(x_1) = f(x_2) = f(x_3) = 0$) then there exists $x_0 \in \mathbb{R}$ such that $f''(x_0) = 0$.

Exercise 6.4.3. Suppose that $f : [a, b] \to \mathbb{R}$ is a continuous function which is differentiable on (a, b), such that $f'(x) = 0$ for all $x \in (a, b)$. Prove that f is the constant function.

Exercise 6.4.4. Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a differentiable function. Recall from Exercise 6.1.4 that if f is Lipschitz (see Exercise 5.5.7 for the definition) then f' is bounded. Use the Mean Value Theorem to prove the converse: that if f' is not Lipschitz then f is not bounded.

Exercise 6.4.5. Let I be an interval containing a as an interior point, and let $f, g : I \to \mathbb{R}$ be continuous functions, such that $g(x) > 0 \forall x \in I \setminus \{a\}$, $g'(x) \neq 0 \forall x \in I \setminus \{a\}$, and $f(a) = g(a) = 0$.

(i) Let $\epsilon > 0$. Suppose that $b \in I$ satisfies

$$f(b) \geq g(b)(L + \epsilon).$$

Prove that there is some $x \in I$ such that

$$f'(x) \geq g'(x)(L + \epsilon).$$

[*Use Cauchy's Mean Value Theorem.*]

(ii) Prove that if

$$g(x)(L - \epsilon) < f(x) < g(x)(L + \epsilon) \quad \text{for all } x \in I.$$

then

$$g'(x)(L - \epsilon) < f'(x) < g'(x)(L + \epsilon) \quad \text{for all } x \in I.$$

(iii) Prove that if

$$L := \lim_{x \to a} \frac{f'(x)}{g'(x)} \text{ exists,}$$

then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

(This is one case of L'Hôpital's Rule.)

Exercise 6.4.6. [TBB08, Exercise 7.6.15] Let $f : [0, \infty) \to \mathbb{R}$ be a function which is differentiable, such that $f'(x)$ is decreasing and positive. By writing $f(N) - f(0)$ as $\sum_{n=1}^{N} f(n) - f(n-1)$, prove that the series

$$\sum_{n=1}^{\infty} f'(n)$$

converges if and only if f is bounded.
[*You may not use the Integral Test. In fact, this is essentially a restatement of the Integral Test through the Fundamental Theorem of Calculus, but the task here is to prove it by just using the Mean Value Theorem.*]

Chapter 7

Riemann integration

In this chapter we define integration and study its properties. Integrating a positive function can be thought of as measuring the area under its graph (and above the x-axis). We will prove some important properties of the integral, including the Fundamental Theorem of Calculus (relating integration to differentiation).

7.1 The Riemann integral

To formally define the integral, we proceed by estimating this area. Fix $a < b$.

A *partition* of an interval $[a, b]$ is a finite set $\{t_0, t_1, \ldots, t_n\}$ such that

$$a = t_0 < t_1 < \cdots < t_n = b.^1$$

A partition breaks up the interval $[a, b]$ into n subintervals

$$[t_0, t_1], [t_1, t_2], \ldots, [t_{n-1}, t_n].$$

Let $P = \{t_0, t_1, \ldots, t_n\}$ be a partition and let $f : [a, b] \to \mathbb{R}$ be a bounded function. For $i = 1, \ldots, n$, define

$$\begin{aligned} m_i(P, f) &:= \inf f([t_{i-1}, t_i]) = \inf\{f(t) : t \in [t_{i-1}, t_i]\}, \\ M_i(P, f) &:= \sup f([t_{i-1}, t_i]) = \sup\{f(t) : t \in [t_{i-1}, t_i]\}. \end{aligned}$$

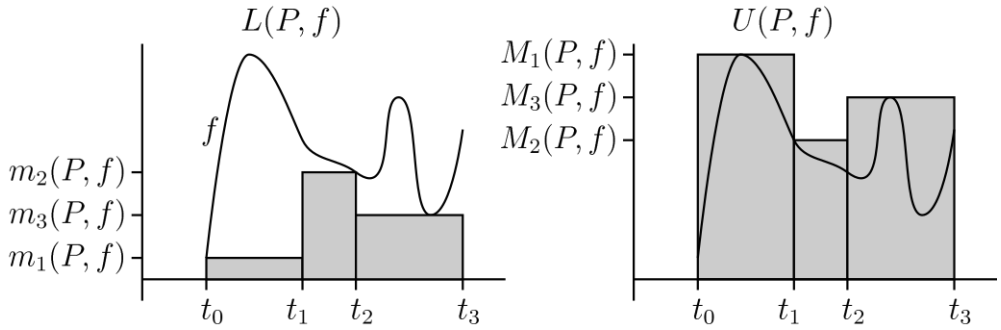
Definition 7.1.1. *Let $P = \{t_0, t_1, \ldots, t_n\}$ be a partition and let $f : [a, b] \to \mathbb{R}$ be a bounded function. The* lower Darboux sum *of f for P is*

$$L(P, f) := \sum_{i=1}^{n} m_i(P, f)(t_i - t_{i-1}).$$

The upper Darboux sum *of f for P is*

$$U(P, f) := \sum_{i=1}^{n} M_i(P, f)(t_i - t_{i-1}).$$

[1] Of course, sets are not ordered: $\{t_0, t_1, t_2, t_3\} = \{t_1, t_3, t_0, t_2\}$. However, when we write out a partition this way as $\{t_0, t_1, \ldots, t_n\}$, we will understand that the elements are in order (unless otherwise noted).



The idea is the following. Suppose that f is a positive function. We start with a partition P, which corresponds to a way of breaking up $[a,b]$ into intervals $[t_0,t_1],[t_1,t_2],\ldots,[t_{n-1},t_n]$. If we then ask for the smallest rectangles which have these intervals on one side, and which cover the graph of f, then the rectangles we will come up with are precisely

$$[t_0,t_1]\times[0,M_1(P,f)],\ldots,[t_{i-1},t_i]\times[0,M_i(P,f)],\ldots[t_{n-1},t_n]\times[0,M_n(P,f)].$$

The area of these rectangles is precisely the upper Darboux sum, and in this sense, the upper Darboux sum gives an estimation from above of the area under f. Likewise, the lower Darboux sum corresponds to the largest rectangles that can be fit under the graph of f, so the lower Darboux sum gives an estimation from below of the area under f.

Example 7.1.2. Let $f:[0,1]\to\mathbb{R}$ be given by $f(t):=t$, and let $P:=\left\{t_0:=0,t_1:=\frac{1}{3},t_2:=1\right\}$. Then

$$m_1(P,f)=0,\quad m_2(P,f)=\frac{1}{3},\quad M_1(P,f)=\frac{1}{3},\quad M_2(P,f)=1.$$

Hence,

$$L(P,f)=m_1(P,f)(t_1-t_0)+m_2(P,f)(t_2-t_1)=0\left(\frac{1}{3}-0\right)+\frac{1}{3}\left(1-\frac{1}{3}\right)=\frac{2}{9}$$

and

$$U(P,f)=M_1(P,f)(t_1-t_0)+M_2(P,f)(t_2-t_1)=\frac{1}{3}\left(\frac{1}{3}-0\right)+1\left(1-\frac{1}{3}\right)=\frac{7}{9}.$$

It is clear that $m_i(P,f)\le M_i(P,f)$ always holds, and therefore,

$$L(P,f)\le U(P,f)$$

for any partition P.

Definition 7.1.3. *Let P,P' be partitions. We say that P'* refines *P if $P\subseteq P'$ (as sets).*

Lemma 7.1.4. *Let $f:[a,b]\to\mathbb{R}$ be a bounded function and let P,P' be partitions of $[a,b]$ such that P' refines P. Then*

$$L(P,f)\le L(P',f)\quad\textit{and}\quad U(P',f)\le U(P,f).$$

Proof. Since P refines P', it is obtained from P by adding finitely many points. We will prove that if P' contains exactly one more point than P, then

$$L(P,f)\le L(P',f)\quad\text{and}\quad U(P',f)\le U(P,f).$$

The general case will then follow by induction, using this special case as the inductive step.

We will only prove $L(P, f) \leq L(P', f)$, as the case for upper sums is similar.
Let $P = \{t_0, t_1, \ldots, t_n\}$ and let $P' = \{t_0, \ldots, t_{k-1}, s, t_k, \ldots, t_n\}$, with

$$a = t_0 < t_1 < \cdots < t_{k-1} < s < t_k < \cdots < t_n = b.$$

Let us compare

$$L(P, f) = m_1(P, f)(t_1 - t_0) + \cdots + m_k(P, f)(t_k - t_{k-1}) + \cdots + m_n(P, f)(t_n - t_{n-1})$$

with

$$\begin{aligned} L(P', f) = m_1(P', f)(t_1 - t_0) + \cdots + m_k(P', f)(s - t_{k-1}) + m_{k+1}(P', f)(t_k - s) \\ + \cdots + m_{n+1}(P', f)(t_n - t_{n-1}). \end{aligned}$$

Note that for $i < k$, we have

$$m_i(P, f) = m_i(P', f),$$

and likewise, for $i > k$,

$$m_i(P, f) = m_{i+1}(P', f).$$

Hence,

$$L(P', f) - L(P, f) = m_k(P', f)(s - t_{k-1}) + m_{k+1}(P', f)(t_k - s) - m_k(P, f)(t_k - t_{k-1}).$$

Nett, observe that since $f([t_{k-1}, s]) \subseteq f([t_{k-1}, t_k]$, we have

$$m_k(P, f) = \inf f([t_{k-1}, t_k]) \leq \inf f([t_{k-1}, s]) = m_k(P', f),$$

and likewise,

$$m_k(P, f) \leq m_{k+1}(P', f).$$

Thus,

$$\begin{aligned} &L(P', f) - L(P, f) \\ &\quad = m_k(P', f)(s - t_{k-1}) + m_{k+1}(P', f)(t_k - s) - m_k(P, f)(t_k - t_{k-1}) \\ &\quad \geq m_k(P, f)(s - t_{k-1}) + m_k(P, f)(t_k - s) - m_k(P, f)(t_k - t_{k-1}) \\ &\quad = 0, \end{aligned}$$

as required. □

Corollary 7.1.5. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let P, P' be partitions of $[a, b]$. Then*

$$L(P, f) \leq U(P', f).$$

Proof. Let $P'' = P \cup P'$ (as sets), so that P'' is a partition of $[a, b]$ which refines both P and P'. Then

$$\begin{aligned} L(P, f) &\leq L(P'', f) && \text{(previous lemma)} \\ &\leq U(P'', f) \\ &\leq U(P', f) && \text{(previous lemma).} \end{aligned}$$

□

Definition 7.1.6. *A bounded function* $f : [a, b] \to \mathbb{R}$ *is* (Riemann) integrable *if*

$$\sup\{L(P, f) : P \text{ a partition of } [a, b]\} = \inf\{U(P, f) : P \text{ a partition of } [a, b]\}.$$

In this case, we set

$$\int_a^b f(t)\,dt \quad \text{or} \quad \int_a^b f$$

equal to this number, called the integral *of* f *(on* $[a, b]$*).*

In other words,

$$\int_a^b f(t)\,dt = I$$

if we can approximate I both from above and from below arbitrarily well by upper and lower Darboux sums (respectively).

If $[a, b] \subseteq A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$ is a function such that $f|_{[a,b]}$ is integrable, then we continue to write

$$\int_a^b f(t)\,dt$$

for the value of the integral of $f|_{[a,b]}$. We also write

$$\int_b^a f(t)\,dt = -\int_a^b f(t)\,dt.$$

To shorten notation, we will sometimes write

$$\sup_P L(P, f)$$

to mean

$$\sup\{L(P, f) : P \text{ a partition of } [a, b]\},$$

and similarly

$$\inf_P U(P, f)$$

for

$$\inf\{U(P, f) : P \text{ a partition of } [a, b]\}.$$

Note that by Corollary 7.1.5, it follows that

$$\boxed{\sup_P L(P, f) \leq \inf_P U(P, f).}$$

Example 7.1.7. Let $f : [0, 1] \to \mathbb{R}$ be defined by $f(t) := t$. This is integrable and $\int_0^1 f(t)\,dt = \frac{1}{2}$.

To see this, consider the partition $P = \{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n} = 1\}$. Then

$$m_i(f, P) = \inf f\left(\left[\frac{i-1}{n}, \frac{i}{n}\right]\right) = \inf\left[\frac{i-1}{n}, \frac{i}{n}\right] = \frac{i-1}{n},$$

and so

$$\begin{aligned} L(f,P) &= \sum_{i=1}^{n} m_i(f,P)\left(\frac{i}{n} - \frac{i-1}{n}\right) \\ &= \sum_{i=1}^{n} \frac{i-1}{n} \cdot \frac{1}{n} \\ &= \frac{1}{n^2}\sum_{i=1}^{n}(i-1) \\ &= \frac{n(n-1)}{2n^2} = \frac{n-1}{2n}. \end{aligned}$$

Similarly, $M_i(f,P) = \frac{i}{n}$ and so

$$\begin{aligned} U(f,P) &= \sum_{i=1}^{n} M_i(f,P)\left(\frac{i}{n} - \frac{i-1}{n}\right) \\ &= \sum_{i=1}^{n} \frac{i}{n} \cdot \frac{1}{n} \\ &= \frac{1}{n^2}\sum_{i=1}^{n} i \\ &= \frac{n(n+1)}{2n^2} = \frac{n+1}{2n}. \end{aligned}$$

It follows that

$$\sup_P L(P,f) \geq \sup\left\{\frac{n-1}{2n} : n \in \mathbb{N}_{\geq 1}\right\} = \frac{1}{2},$$

and on the other hand,

$$\inf_P U(P,f) \leq \inf\left\{\frac{n+1}{2n} : n \in \mathbb{N}_{\geq 1}\right\} = \frac{1}{2}.$$

Therefore, these must both equal $\frac{1}{2}$, which means that

$$\int_0^1 f(t)\,dt = \frac{1}{2}.$$

Example 7.1.8. Consider the function $f : [0,1] \to \mathbb{R}$ defined by

$$f(t) := \begin{cases} 0, & t \in \mathbb{Q}; \\ 1, & t \notin \mathbb{Q}. \end{cases}$$

For any partition P, we will show that

$$L(P,f) = 0 \quad \text{and} \quad U(P,f) = 1.$$

Consequently,

$$\sup_P L(P,f) = 0 \neq 1 = \inf_P U(P,f),$$

so that f is not integrable.

Let $P = \{t_0, \ldots, t_n\}$ be a partition. Then for each i, the set $[t_{i-1}, t_i]$ contains both rational and irrational numbers. Hence,

$$f([t_{i-1}, t_i]) = \{0, 1\},$$

so that $m_i(P, f) = 0$ and $M_i(P, f) = 1$. From this we get

$$\begin{aligned} U(P, f) &= \sum_{i=1}^{n} M_i(P, f)(t_i - t_{i-1}) \\ &= \sum_{i=1}^{n} (t_i - t_{i-1}) \\ &= t_n - t_0 && \text{(by telescoping)} \\ &= 1. \end{aligned}$$

On the other hand,

$$L(P, f) = \sum_{i=1}^{n} m_i(P, f)(t_i - t_{i-1}) = \sum_{i=1}^{n} 0(t_i - t_{i-1}) = 0.$$

Example 7.1.9. Consider the function $f : [0, 2] \to \mathbb{R}$ defined by

$$f(t) := \begin{cases} 0, & t = 1; \\ 1, & t \neq 1. \end{cases}$$

This is integrable and $\int_0^1 f(t)\, dx = 2$.

To see this, given $\epsilon > 0$, consider the partition

$$P := \{t_0 = 0, t_1 = 1 - \epsilon, t_2 = 1 + \epsilon, t_3 = 2\}.$$

We compute

$$m_1(P, f) = 1, m_2(P, f) = 0, m_3(P, f) = 1,$$

and therefore

$$\begin{aligned} L(f, P) &= \sum_{i=1}^{3} m_i(f, P)(t_i - t_{i-1}) \\ &= 1(1 - \epsilon - 0) + 0(1 + \epsilon - (1 - \epsilon)) + 1(2 - (1 + \epsilon)) \\ &= 1 - \epsilon + 1 - \epsilon = 2 - 2\epsilon. \end{aligned}$$

Therefore,

$$\sup_P L(f, P) \geq 2 - 2\epsilon,$$

and since $\epsilon > 0$ is arbitrary,

$$\sup_P L(f, P) \geq 2.$$

On the other hand, taking the "trivial" partition $P := \{0, 2\}$, we have $M_1(f, P) = 1$, and so

$$U(f, P) = M_1(f, P)(2 - 0) = 2,$$

and therefore

$$\inf_P U(f,P) \le 2.$$

Hence, $\sup_P L(f,P) = \inf_P U(f,P) = 2$, so f is integrable and

$$\int_0^1 f(t)\,dt = 2.$$

Here is a way of rephrasing integrability.

Proposition 7.1.10. *Let $f : [a,b] \to \mathbb{R}$ be a bounded function. Then f is integrable if and only if, for every $\epsilon > 0$ there exists a partition P such that*

$$U(P,f) - L(P,f) < \epsilon.$$

Proof. $\Rightarrow$: Assume that f is integrable and let $\epsilon > 0$. Set

$$I := \sup_P L(P,f) = \inf_P U(P,f).$$

It follows that there are partitions P_L and P_U such that

$$L(P_L,f) > I - \frac{\epsilon}{2}, \quad U(P_U,f) < I + \frac{\epsilon}{2}.$$

Let $P = P_L \cup P_U$, so by Lemma 7.1.4,

$$L(P,f) \ge L(P_L,f) \quad \text{and} \quad U(P,f) \le U(P_U,f).$$

Therefore,

$$U(P,f) - L(P,f) \le U(P_U,f) - L(P_L,f) < I - \frac{\epsilon}{2} - \left(I + \frac{\epsilon}{2}\right) = \epsilon.$$

$\Leftarrow$: For any $\epsilon > 0$ there is a partition P such that $U(P,f) - L(P,f) < \epsilon$.

$$0 \le \inf_Q U(Q,f) - \sup_Q L(Q,f) \le U(P,f) - L(P,f) < \epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that $\inf_Q U(Q,f) = \sup_Q L(Q,f)$, so that f is integrable. $\square$

Theorem 7.1.11. *If $f : [a,b] \to \mathbb{R}$ is continuous then f is integrable.*

Proof. Let $\epsilon > 0$ be given. Since f is continuous and $[a,b]$ is compact, by Theorem 5.5.5, there exists $\delta > 0$ such that if $x, y \in [a,b]$ and $|x-y| < \delta$ then $|f(x) - f(y)| < \frac{\epsilon}{b-a}$.

Let $P = \{t_0, \dots, t_n\}$ be a partition such that $|t_i - t_{i-1}| < \delta$ for all i. By Proposition 5.3.6, $f([t_{i-1}, t_i])$ is a closed interval, and then by the uniform continuity condition, we see that its maximum and minimum cannot be separated by more than $\epsilon/(b-a)$.[2] In other words,

$$M_i(P,f) - m_i(P,f) < \frac{\epsilon}{b-a}.$$

[2]In fact, we don't really need to use Proposition 5.3.6 here:

$$M_i(P,f) - m_i(P,f) = \sup f([t_{i-1},t_i]) - \inf f([t_{i-1},t_i]) = \sup\{f(x) - f(y) : x, y \in [t_{i-1},t_i]\} \le \frac{\epsilon}{b-a}.$$

Consequently,

$$\begin{aligned} U(P,f) - L(P,f) &= \sum_{i=1}^{n} (M_i(P,f) - m_i(P,f))(t_i - t_{i-1}) \\ &< \sum_{i=1}^{n} \frac{\epsilon}{b-a}(t_i - t_{i-1}) \\ &= \frac{\epsilon}{b-a}(t_n - t_0) = \epsilon. \end{aligned}$$

This verifies the condition in the previous proposition, so we conclude that f is integrable. □

Exercise 7.1.1. Define $f : [-1,1] \to \mathbb{R}$ by $f(x) := x^2$

(i) Let $P := \left\{-1, -\frac{1}{2}, \frac{1}{3}, 1\right\}$. Compute $L(P,f)$ and $U(P,f)$.

(ii) Find a partition P such that $U(P,f) - L(P,f) < \frac{1}{2}$.

Exercise 7.1.2. [Leb16, Exercise 5.1.5] Define $f : [-1,1] \to \mathbb{R}$ by

$$f(x) := \begin{cases} 1, & x > 0; \\ 0, & x \leq 1. \end{cases}$$

Prove that f is integrable and compute $\int_{-1}^{1} f(t)\,dt$.

Exercise 7.1.3. Let $f : [0,1] \to \mathbb{R}$ be an increasing function.

(a) Prove that for any partition P and any i, $M_i(P,f) = m_{i+1}(P,f)$.

(b) For $n \in \mathbb{N}_{\geq 1}$, set $P_n := \{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}\}$. Find a short expression for $U(P_n,f) - L(P_n,f)$.

(c) Prove that f is Riemann integrable.

Exercise 7.1.4. Recall the Dirichlet function from Exercise 5.2.4:

$$f(x) := \begin{cases} 0, & \text{if } x \notin \mathbb{Q}; \\ \frac{1}{q}, & \text{if } x = \frac{p}{q} \text{ in lowest terms (with } p \in \mathbb{Z} \text{ and } q \in \mathbb{N}_{\geq 1}). \end{cases}$$

Let $\epsilon > 0$.

(a) Prove that for any partition P of $[0,1]$,

$$L(P,f) = 0.$$

(b) Prove that there is a number N such that for any partition P of $[0,1]$, the set

$$\{i : M_i(P,f) \geq \epsilon\}$$

has size at most N.

(c) Let N be as in part (b). Suppose that $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[0,1]$, and let

$$\delta := \max\{x_1 - x_0, x_2 - x_1, \ldots, x_n - x_{n-1}\}.$$

Prove that

$$U(f,P) \leq \epsilon + N\delta.$$

(d) Prove that there is a partition P such that

$$U(f, P) < 2\epsilon.$$

(e) Prove that f is integrable and compute $\int_0^1 f(t)\,dt$.

Exercise 7.1.5. [Leb16, Exercise 5.1.7] Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable. Prove that given $\epsilon > 0$ there is a partition $P = \{t_0, \ldots, t_n\}$ of $[a, b]$ such that, for any $\xi_1 \in [t_0, t_1], \ldots, \xi_n \in [t_{n-1}, t_n]$,

$$\left| \int_a^b f(t)\,dt - \sum_{i=1}^n f(\xi_i)(t_i - t_{i-1}) \right| < \epsilon.$$

[*The sum* $\sum_{i=1}^n f(\xi_i)(t_i - t_{i-1})$ *is called a* Riemann sum *for* f.]

Exercise 7.1.6. [Leb16, Exercise 5.1.8] Let $\alpha > 0, \beta \in \mathbb{R}$. Suppose that $f : [\beta, \alpha + \beta] \to \mathbb{R}$ is Riemann integrable. Define $g : [0, 1] \to \mathbb{R}$ by

$$g(x) := f(\alpha x + \beta).$$

Prove that g is Riemann integrable and compute $\int_0^1 g$ in terms of $\int_\beta^{\alpha+\beta} f$.

Exercise 7.1.7. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let P be a partition of $[a, b]$. Show that for $c > 0$,

$$U(P, cf) = cU(P, f) \quad \text{and} \quad L(P, cf) = cL(P, f).$$

What happens if $c < 0$?

7.2 Properties of the integral

Proposition 7.2.1 (Additive Property)**.** *Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let $c \in (a, b)$. Then f is integrable if and only if $f|_{[a,c]}$ and $f|_{[c,b]}$ are both integrable. In this case,*

$$\int_a^b f(t)\,dt = \int_a^c f(t)\,dt + \int_c^b f(t)\,dt.$$

Proof. $\Rightarrow$: Let $\epsilon > 0$ be given. There exists a partition P of $[a, b]$ such that $U(P, f) - L(P, f) < \epsilon$. By possibly adding a point (which leads to a refinement of the original P), we may assume that $c \in P$. Break up the partition P into two partitions corresponding to $[a, c]$ and $[c, b]$: define

$$P_1 := P \cap [a, c] \quad \text{and} \quad P_2 := P \cap [c, b].$$

Then one sees that

$$U(P, f) = U(P_1, f|_{[a,c]}) + U(P_2, f|_{[c,b]})$$

and likewise

$$L(P, f) = L(P_1, f|_{[a,c]}) + L(P_2, f|_{[c,b]}).$$

Since $U(P_2, f|_{[c,b]}) - L(P_2, f|_{[c,b]}) \geq 0$, it follows that

$$U(P_1, f|_{[a,c]}) - L(P_1, f|_{[a,c]}) \leq U(P, f) - L(P, f) < \epsilon.$$

Likewise,

$$U(P_2, f|_{[c,b]}) - L(P_2, f|_{[c,b]}) \leq U(P, f) - L(P, f) < \epsilon.$$

Since ϵ is arbitrary, we see that both $f|_{[a,c]}$ and $f|_{[c,b]}$ are integrable.

$\Leftarrow$ and showing $\int_a^b f = \int_a^c f + \int_c^b f$ are Exercise 7.2.1 $\square$

Proposition 7.2.2 (Linearity of the integral)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be bounded integrable functions and let $c \in \mathbb{R}$. Then $cf + g$ is integrable, and*

$$\int_a^b cf(t) + g(t)\,dt = c\int_a^b f(t)\,dt + \int_a^b g(t)\,dt.$$

Proof. The case $c = 0$ is trivial. For $c > 0$, it is an exercise (Exercise 7.2.2) to show that

$$m_i(P, cf + g) \geq cm_i(P, f) + m_i(P, g) \quad \text{and} \quad M_i(P, cf + g) \leq cM_i(P, f) + M_i(P, g).$$

It follows that

$$L(P, cf + g) \geq cL(P, f) + L(P, g) \quad \text{and} \quad U(P, cf + g) \leq cU(P, f) + U(P, g).$$

Consequently,

$$\begin{aligned}\sup_P L(P, cf + g) &\geq \sup_P cL(P, f) + L(P, g)\\ &\geq c\sup_P L(P, f) + \sup_P L(P, g)\\ &= c\int_a^b f + \int_a^b g.\end{aligned}$$

and likewise

$$\inf_P U(P, cf + g) \leq c\int_a^b f + \int_a^b g.$$

It follows that $cf + g$ is integrable with $\int_a^b cf + g = c\int_a^b f + \int_a^b g$.

For the case $c < 0$, it suffices (by combining with the case we just did) to show that $-f$ is integrable and $\int_a^b -f = -\int_a^b f$. This is Exercise 7.2.3. □

Proposition 7.2.3. *Let $f, g : [a, b] \to \mathbb{R}$ be integrable. If $f(t) \leq g(t)$ for all $t \in [a, b]$ then*

$$\int_a^b f(t)\,dt \leq \int_a^b g(t)\,dt.$$

Proof. Let P be a partition. Then it is straightforward to see that $m_i(P, f) \leq m_i(P, g)$ for all i, and thus

$$L(P, f) \leq L(P, g).$$

Therefore,

$$\int_a^b f = \sup_P L(P, f) \leq \sup_P L(P, g) = \int_a^b g.$$

□

Corollary 7.2.4. *Let $f : [a, b] \to \mathbb{R}$ be integrable. If $m, M \in \mathbb{R}$ are such that*

$$m \leq f(t) \leq M$$

for all $x \in [a, b]$ then

$$m(b - a) \leq \int_a^b f(t)\,dt \leq M(b - a).$$

Exercise 7.2.1. Prove the remaining part of Proposition 7.2.1: that if $f : [a, b] \to \mathbb{R}$ is a bounded function such that $f|_{[a,c]}$ and $f|_{[c,b]}$ is integrable then f is integrable and

$$\int_a^b f(t)\,dt = \int_a^c f(t)\,dt + \int_c^b f(t)\,dt.$$

Exercise 7.2.2. Let $f, g : [a, b] \to \mathbb{R}$ be bounded functions, let P be a partition of $[a, b]$, and let $c > 0$.

(a) Show that $m_i(P, f + g) \geq m_i(P, f) + m_i(P, g)$ and $M_i(P, f + g) \leq M_i(P, f) + M_i(P, g)$ for all i.

(b) Show that $m_i(P, cf) = cm_i(P, f)$ and $M_i(P, cf) = cM_i(P, f)$.

(c) Show that $m_i(P, cf + g) \geq cm_i(P, f) + m_i(P, g)$ and $M_i(P, cf + g) \leq cM_i(P, f) + M_i(P, g)$.

(d) Show that $m_i(P, -f) = -M_i(P, f)$.

Exercise 7.2.3. Let $f : [a, b] \to \mathbb{R}$ be an integrable function. Prove that $-f$ is integrable and

$$\int_a^b -f(t)\,dt = -\int_a^b f(t)\,dt.$$

Exercise 7.2.4. Let $f : [a, b] \to \mathbb{R}$ be a continuous function such that $f(t) \geq 0$ for all $t \in [a, b]$. Suppose that $\int_a^b f(t)\,dt = 0$. Prove that $f(t) = 0$ for all $t \in [a, b]$.
[*Hint. If $f(x) > 0$ then use continuity to obtain γ and $c < d$ such that $f(t) \geq \gamma$ for all $t \in [c, d]$.*]

Exercise 7.2.5. Let $f : [a, b] \to \mathbb{R}$ be continuous. Prove that if $\int_a^b f(t)g(t)\,dt = 0$ for every continuous $g : [a, b] \to \mathbb{R}$, then $f(t) = 0$ for all $t \in [a, b]$.
[*Hint. Find a way to use the previous exercise.*]

Exercise 7.2.6. Let $f : [a, b] \to \mathbb{R}$ be an integrable function (hence bounded). Define $F : [a, b] \to \mathbb{R}$ by

$$F(x) := \int_a^x f(t)\,dt.$$

Prove that F is Lipschitz (see Exercise 5.5.7) and therefore (uniformly) continuous.

Exercise 7.2.7. Let $f : [a, b] \to \mathbb{R}$ be an integrable function. Prove that $|f|$ is integrable and that

$$\left|\int_a^b f(t)\,dt\right| \leq \int_a^b |f(t)|\,dt.$$

7.3 The Fundamental Theorem of Calculus

Theorem 7.3.1 (Fundamental Theorem of Calculus)**.** *Let $f : [a, b] \to \mathbb{R}$ be an integrable function. Define $F : [a, b] \to \mathbb{R}$ by*

$$F(x) := \int_a^x f(t)\,dt.$$

For any $x \in [a, b]$, if f is continuous at x then F is differentiable at x and

$$F'(x) = f(x).$$

Proof. Fix $x \in [a,b]$ such that f is continuous at x. We will show that (if $x < b$) $\lim_{y \to x^+} \frac{F(y)-F(x)}{y-x} = f(x)$ using the δ-ϵ definition of limit (with $\leq$ in place of $<$). A similar argument will show that (if $x > a$) $\lim_{y \to x^-} \frac{F(y)-F(x)}{y-x} = f(x)$, and thus by Exercise 5.1.8,

$$F'(x) = \lim_{y \to x} \frac{F(y) - F(x)}{y - x} = f(x).$$

Note that by the Additive Property, $F(y) - F(x) = \int_x^y f$.

Given $\epsilon > 0$, choose $\delta > 0$ such that if $t \in [a,b]$ and $|t - x| \leq \delta$ then $|f(t) - f(x)| \leq \epsilon$. Let $y \in (x, x+\delta]$. Then for $t \in [x,y]$, we have

$$f(x) - \epsilon \leq f(t) \leq f(x) + \epsilon.$$

Hence by Corollary 7.2.4 (applied to $f|_{[x,y]}$),

$$(f(x) - \epsilon)(y - x) \leq \int_x^y f \leq (f(x) + \epsilon)(y - x),$$

so that

$$f(x) - \epsilon \leq \frac{F(y) - F(x)}{y - x} \leq f(x) + \epsilon.$$

This proves that $\lim_{y \to x^+} \frac{F(y)-F(x)}{y-x} = f(x)$, as required. □

The following is the form in which the Fundamental Theorem of Calculus is typically used: it enables computing integrals via antidifferentiation.

Theorem 7.3.2. *Let $F : [a,b] \to \mathbb{R}$ be a differentiable function, such that $F' : [a,b] \to \mathbb{R}$ is continuous. Then*

$$\int_a^b F'(t)\, dt = F(b) - F(a).$$

Proof. Define $G : [a,b] \to \mathbb{R}$ by

$$G(x) := \int_a^x F'(t)\, dt.$$

Then by the Fundamental Theorem of Calculus, $G' = F'$. It follows from Exercise 6.4.3 that $G - F$ is a constant function, i.e.,

$$F(x) = G(x) + C$$

for some $C \in \mathbb{R}$. Using $x = a$, we find

$$F(a) = G(a) + C = 0 + C,$$

so that $C = F(a)$. Hence,

$$F(b) - F(a) = G(b) = \int_a^b F',$$

as required. □

Interesting fact 7.3.3. The above theorem can be generalized somewhat: instead of assuming that F' is continuous, it suffices to assume that F' exists (i.e., F is differentiable). The proof for this case is a bit longer (and can't be reduced to Theorem 7.3.1).

Exercise 7.3.1. [Sav17, Exercise 7.3.1] Let $f : [a, b] \to \mathbb{R}$ be continuous and define $F : [a, b] \to \mathbb{R}$ by

$$F(x) := \int_x^b f(t)\, dt.$$

Prove that F is differentiable and $F'(x) = -f(x)$ for all $x \in [a, b]$.

Exercise 7.3.2. [Leb16, Exercise 5.3.2] Define $G : \mathbb{R} \to \mathbb{R}$ by $G(x) := \int_0^{x^2} \sin(t^2)\, dt$. Compute $G'(x)$.

Exercise 7.3.3. [Sav17, Exercise 7.3.6] Let $f : [a, b] \to \mathbb{R}$ be continuous, where $a < b$. Prove that if $\int_a^x f(t)\, dt = \int_x^b f(t)\, dt$ for all $x \in [a, b]$, then $f(t) = 0$ for all $t \in [a, b]$.

Exercise 7.3.4. Prove the *Integration by Parts* formula: if $f, g : [a, b] \to \mathbb{R}$ are differentiable functions for which $f', g' : [a, b] \to \mathbb{R}$ are continuous, then

$$\int_a^b f'(t)g(t)\, dt = f(b)g(b) - f(a)g(a) - \int_a^b f(t)g'(t)\, dt.$$

Exercise 7.3.5. Prove the *Change of Variables* formula: if $f : [c, d] \to \mathbb{R}$ is a continuous function and $g : [a, b] \to \mathbb{R}$ is a differentiable function such that $g([a, b]) \subseteq [c, d]$ and $g' : [a, b] \to \mathbb{R}$ is continuous, then

$$\int_a^b f(g(t))g'(t)\, dt = \int_{g(a)}^{g(b)} f(t)\, dt.$$

[*Hint. Define* $F(t) := \int_a^x f(g(t))g'(t)\, dt$ *and* $G(t) := \int_{g(a)}^{g(x)} f(t)\, dt$. *Use FTC and the Chain Rule to compute* F' *and* G'.]

7.4 Improper integrals

So far we have only defined integration for bounded functions on bounded intervals. It is desirable to extend the theory to unbounded functions and/or unbounded intervals. The approach of approximating by Darboux sums, however, will not work: if f is not bounded above then for every partition P, $M_i(P, f)$ will be infinite for some i, and thus

$$U(P, f) = \infty.$$

Instead, to deal with this case, we use limits.

Definition 7.4.1. *Let* $f : (a, b] \to \mathbb{R}$ *be a function such that, for every* $x \in (a, b]$, $f|_{[x,b]}$ *is Riemann integrable. Then we define*

$$\int_a^b f(t)\, dt := \lim_{x \to a^+} \int_x^b f(t)\, dt,$$

provided that this limit exists. Likewise, if $f : [a, b) \to \mathbb{R}$ *is such that* $f|_{[a,x]}$ *is integrable for all* $x \in [a, b)$, *then*

$$\int_a^b f(t)\, dt := \lim_{x \to b^-} \int_a^x f(t)\, dt,$$

provided that this limit exists.

Let $f : [a, \infty) \to \mathbb{R}$ *be a function such that, for every* $x \in [a, \infty)$, $f|_{[a,x]}$ *is integrable. Then we define*

$$\int_a^\infty f(t)\, dt := \lim_{x \to \infty} \int_a^x f(t)\, dt.$$

Likewise, if $f : (-\infty, b] \to \mathbb{R}$ is a function such that $f|_{[x,b]}$ is integrable for all $x \in (-\infty, b]$ then

$$\int_{-\infty}^{b} f(t)\,dt := \lim_{x\to-\infty} \int_{x}^{b} f(t)\,dt.$$

Whenever one of the above limits exist, we say that the improper integral converges.

Note that if $f : [a, b] \to \mathbb{R}$ is integrable then by Exercise 7.2.6,

$$\int_{a}^{b} f(t)\,dt = \lim_{x\to b^-} \int_{a}^{x} f(t)\,dt$$

(and similarly, one can prove $\int_a^b f(t)\,dt = \lim\limits_{x\to a^+} \int_x^b f(t)\,dt$), so there is no ambiguity in the above notation.

The above definition allows integrals to be defined where there is unboundedness at one endpoint. To allow unboundedness at both endpoints, we cut in the middle and combine. For example, suppose $f : (a, \infty) \to \mathbb{R}$ is a function for which $f|_{[x,y]}$ is Riemann integrable for all $a < x < y$. Then we choose some $c \in (a, \infty)$ and define

$$\int_{a}^{\infty} f(t)\,dt := \int_{a}^{c} f(t)\,dt + \int_{c}^{\infty} f(t)\,dt,$$

provided both of these exist (using the above definition).

Exercise 7.4.1. [TBB08, Exercise 8.5.3] For what values of p is the integral $\int_1^\infty \frac{1}{t^p}\,dt$ convergent? (You may use the Fundamental Theorem of Calculus.)

Exercise 7.4.2. [Sav17, Exercise 7.4.5] Suppose that $f : [0, \infty) \to \mathbb{R}$ is a function such that $\int_0^\infty f(t)\,dt$ converges. Prove that for every $\epsilon > 0$ there exists M such that if $a > b > M$ then

$$\left| \int_{a}^{b} f(t)\,dt \right| < \epsilon.$$

Exercise 7.4.3. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(t) := t$. Show that the limit

$$\lim_{x\to\infty} \int_{-x}^{x} f(t)\,dt$$

exists, but explain why $\int_{-\infty}^{\infty} f(t)\,dt$ doesn't converge.

Exercise 7.4.4. In this question, we prove the *Integral Test* (Proposition 3.2.10). Let $f : [1, \infty) \to \mathbb{R}$ be an decreasing function.

(i) Prove that for all $n \in \mathbb{N}_{\geq 1}$,

$$\sum_{n=2}^{N} f(n) \leq \int_{1}^{N+1} f(t)\,dt \leq \sum_{n=1}^{N} f(n).$$

(ii) Prove that if $\int_1^\infty f(t)\,dt$ converges, then $\sum\limits_{n=1}^{\infty} f(n)$ converges.

(iii) Prove that if $F : [1, \infty) \to \mathbb{R}$ is a bounded increasing function then $\lim\limits_{x\to\infty} F(x) = \sup F([0, \infty)$.

(iv) Using (c), prove that if $\sum\limits_{n=1}^{\infty} f(n)$ converges then $\int_1^\infty f(t)\,dt$ converges.

Chapter 8

Sequences and series of functions

Our aim in this chapter is to make sense of and analyze infinite sums of functions: $\sum_{n=1}^{\infty} f_n$, where $f_n : X \to \mathbb{R}$. Inherent in trying to define such a thing is a notion of limit, and therefore it makes sense to start with sequences of functions and develop the notion of a limit in that setting. From there, we define convergence of series of functions in terms of partial sums.

8.1 Pointwise limits

The most obvious way to define a limit of functions is the pointwise limit. We will illustrate with examples that, unfortunately, pointwise limits do not behave well with respect to calculus operations.

Definition 8.1.1. *Let X be a set, let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions, and let $f : X \to \mathbb{R}$. We say that $(f_n)_{n=1}^{\infty}$* converges pointwise *to f if for every $x \in X$,*

$$\lim_{n\to\infty} f_n(x) = f(x).$$

In this case, we may write

$$\text{pw}-\lim_{n\to\infty} f_n = f.$$

Example 8.1.2. Define $f_n : [0,1] \to \mathbb{R}$ by $f_n(x) := x^n$. Then for $x \in [0,1]$,

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^n = \begin{cases} 0, & x \in [0,1); \\ 1, & x = 1. \end{cases}$$

Hence, $(f_n)_{n=1}^{\infty}$ converges pointwise to $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) := \begin{cases} 0, & x \in [0,1); \\ 1, & x = 1. \end{cases}$$

Note that *each f_n is continuous, but the pointwise limit f is not.*

Example 8.1.3. Define $f_n : [0,1] \to \mathbb{R}$ by $f_n(x) := \frac{x^n}{n}$. For $x \in [0,1]$, we can see (using the Squeeze Theorem) that

$$\lim_{n\to\infty} f_n(x) = 0,$$

so that $(f_n)_{n=1}^{\infty}$ converges pointwise to the constant zero function (which we'll write as g here). How do the derivatives behave? f_n' is differentiable and $f_n'(x) = x^{n-1}$. Hence, $(f_n')_{n=1}^{\infty}$ converges pointwise

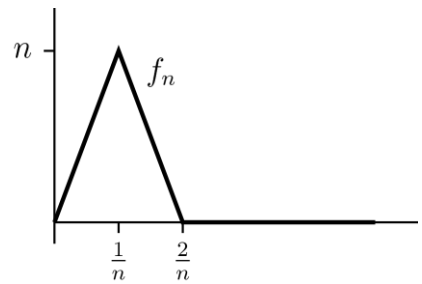
to the function f from the previous example – which is not the same as g' (which is also the zero function). This shows that

$$\left(\text{pw}-\lim_{n\to\infty} f_n\right)' \neq \text{pw}-\lim_{n\to\infty} f_n'.$$

Example 8.1.4. Define $f_n : [0,1] \to \mathbb{R}$ such that

$$f_n(0) = 0, \quad f_n\left(\frac{1}{n}\right) = n, \quad f_n\left(\frac{2}{n}\right) = 0, \quad f_n(1) = 0,$$

and f_n is linear in between.



An explicit formula for f_n is

$$f_n(x) := \begin{cases} n^2 x, & x \in [0, \frac{1}{n}]; \\ 2n - n^2 x, & x \in [\frac{1}{n}, \frac{2}{n}]; \\ 0, & x \in [\frac{2}{n}, 1]. \end{cases}$$

Claim. The sequence $(f_n)_{n=1}^{\infty}$ converges pointwise to the zero function g.

To see this, note that for $x = 0$, we have $f_n(x) = 0$ for all n, and thus $\lim_{n\to\infty} f_n(0) = 0$. On the other hand, for $x > 0$, we have $f_n(x) = 0$ provided $n > \frac{2}{x}$, and thus $(f_n(x))_{n=1}^{\infty}$ is eventually zero, which implies its limit is 0.

However, to compute $\int_0^1 f_n(t)\,dt$, note that this gives the area of a triangle with base length $\frac{2}{n}$ and height n, so

$$\int_0^1 f_n(t)\,dt = 1.$$

The integral of $g = \text{pw}-\lim_{n\to\infty} f_n$, however, is 0. This shows that

$$\int_0^1 \lim_{n\to\infty} f_n(t)\,dt \neq \lim_{n\to\infty} \int_0^1 f_n(t)\,dt.$$

Exercise 8.1.1. Give an example of a sequence of differentiable functions $(f_n : \mathbb{R} \to \mathbb{R})_{n=1}^{\infty}$ which converge pointwise to the function $f(x) := |x|$, which is not differentiable at 0.

Exercise 8.1.2. (cf. [TBB08, Exercise 9.2.1]) Define $f_n : \mathbb{R} \to \mathbb{R}$ by $f_n(x) := \frac{x^{2n}}{1+x^{2n}}$. Determine the pointwise limit of the sequence $(f_n)_{n=1}^{\infty}$.

Exercise 8.1.3. [TBB08, Exercise 9.2.3] There is a sequence $(x_n)_{n=1}^{\infty}$ in $[0,1]$ such that

$$\{x_n : n \in \mathbb{N}_{\geq 1}\} = \mathbb{Q} \cap [0,1].$$

(In other words, $\mathbb{Q} \cap [0,1]$ is *countable.*) Define $f_n : [0,1] \to \mathbb{R}$ by

$$f_n(x) := \begin{cases} x, & x \in \{x_1, \ldots, x_n\}; \\ 0, & \text{otherwise}, \end{cases}$$

and

$$f(x) := \begin{cases} x, & x \in \mathbb{Q}; \\ 0, & \text{otherwise}, \end{cases}$$

Show that f_n converges pointwise to f, and that each f_n is integrable. (From Example 7.1.8, we know that f is *not* integrable.)

Exercise 8.1.4. [TBB08, Exercise 9.2.7] Let $(f_n : [a,b] \to \mathbb{R})_{n=1}^{\infty}$ be a sequence which converges pointwise to $f : [a,b] \to \mathbb{R}$. Which of the following are true:

(a) If each f_n is (weakly) increasing then so is f.

(b) If each f_n is strictly increasing then so is f.

(c) If each f_n is bounded then so is f.

(d) If each f_n is constant then so is f.

(e) If $f_n(x) \geq 0$ for all $x \in [a,b]$ then $f(x) \geq 0$ for all $x \in [a,b]$.

(f) If each f_n is linear (meaning, of the form $f_n(x) = \alpha_n x + \beta_n$) then so is f.

(g) If each f_n is convex (meaning $f_n(tx + (1-t)y) \leq t f_n(x) + (1-t) f_n(y)$ for all $x, y \in [a,b]$ and all $t \in [0,1]$) then so is f.

8.2 Uniform convergence

In this section, we define a stronger notion of convergence of a sequence of functions. Using this notion, we can overcome the bad behaviour with mere pointwise convergence, seen in the last section.

To contextualize the definition of uniform convergence, we unwind the definition of "a sequence $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ converges pointwise to $f : X \to \mathbb{R}$" (using the δ-ϵ definition of convergence in $\mathbb{R}$), we get:

For any $x \in X$ and $\epsilon > 0$ there exists $n_0 \in \mathbb{N}_{\geq 1}$ such that for all $n \geq n_0$ then $\|f_n(x) - f(x)\|_2 < \epsilon$.

Here, n_0 is allowed to depend on both x and ϵ (but not, of course, on n). The idea of "uniformizing" this definition which we will do presently is similar to what we did when we defined uniform continuity (Section 5.5): we modify the definition so that n_0 is not allowed to depend on x.

Definition 8.2.1. *Let X be a set, let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions, and let $f : X \to \mathbb{R}$. We say that $(f_n)_{n=1}^{\infty}$* converges uniformly *to f if for every $\epsilon > 0$ there exists n_0 such that for all $n \geq n_0$ and $x \in X$,*

$$\|f_n(x) - f(x)\|_2 < \epsilon.$$

In this case, we may write

$$\text{u}-\lim_{n \to \infty} f_n = f.$$

By comparing this definition to the characterization of pointwise convergence above, it is clear that

$$\text{u}-\lim_{n\to\infty} f_n = f \quad \text{implies} \quad \text{pw}-\lim_{n\to\infty} f_n = f.$$

We also define uniform convergence of a series of functions.

Definition 8.2.2. *Let X be a set, let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions, and let $f : X \to \mathbb{R}$. We define*

$$\text{u}-\sum_{n=1}^{\infty} f_n := \text{u}-\lim_{N\to\infty} \sum_{n=1}^{N} f_n,$$

provided that this limit exists. When this limit exists, we say that the series $\sum\limits_{n=1}^{\infty} f_n$ converges uniformly *(on X).*

Example 8.2.3. Define $f_n : \mathbb{R} \to \mathbb{R}$ by $f_n(x) := \frac{\sin(nx)}{n}$. Then $\text{u}-\lim\limits_{n\to\infty} f_n = 0$ (the zero function).

To see this, let $\epsilon > 0$ be given. Then pick $n_0 > \frac{1}{\epsilon}$. If $n \geq n_0$ and $x \in \mathbb{R}$ then

$$|f_n(x) - 0| = \frac{|\sin(nx)|}{n} \leq \frac{1}{n} < \epsilon.$$

Example 8.2.4. Let $b \in (0, 1)$ and define $f_n : [0, b] \to \mathbb{R}$ by $f_n(x) := x^n$. Then $\text{u}-\lim\limits_{n\to\infty} f_n = 0$ (the zero function).

To see this, let $\epsilon > 0$ be given. Since $\lim\limits_{n\to\infty} b^n = 0$, there exists n_0 such that if $n \geq n_0$ then $|b^n| < \epsilon$. Now, if $n \geq n_0$ and $x \in [0, b]$ then $x \leq b$ so that $x^n \leq b^n$, and thus

$$|f_n(x) - 0| = x^n < \epsilon.$$

Example 8.2.5. The sequence of functions $(f_n : [0, 1] \to \mathbb{R})_{n=1}^{\infty}$ defined by $f_n(x) := x^n$ does *not* converge uniformly to the function g in Example 8.1.2. This will follow from Theorem 8.3.1 below: if $\text{u}-\lim\limits_{n\to\infty} f_n = g$ then this theorem would tell us that g is continuous, but g is not.

In practice, we will (more) often be interested in whether series of functions converge uniformly, and for this, there is the following useful test.

Theorem 8.2.6 (The Weierstrass M-test)**.** *Let X be a set, let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions, and let $(M_n)_{n=1}^{\infty}$ be a sequence of non-negative real numbers. Suppose that the following hold:*

(i) $|f_n(x)| \leq M_n$ for all $x \in X$, and

(ii) $\sum\limits_{n=1}^{\infty} M_n$ converges.

Then $\sum\limits_{n=1}^{\infty} f_n$ converges uniformly and absolutely.

Proof. For each $x \in X$, by the Comparison Test, the series

$$\sum_{n=1}^{\infty} |f_n(x)|$$

converges, so that $\sum_{n=1}^{\infty} f_n(x)$ converges absolutely. Define

$$g(x) := \sum_{n=1}^{\infty} f_n(x).$$

We will show that $g = \text{u}-\sum_{n=1}^{\infty} f_n$.

Let $\epsilon > 0$. Since $\sum_{n=1}^{\infty} M_n$ converges, we can find N_0 such that

$$\left| \sum_{n=1}^{\infty} M_n - \sum_{n=1}^{N_0} M_n \right| < \epsilon.$$

Then if $N \geq N_0$ and $x \in X$,

$$\begin{aligned}
\left| g(x) - \sum_{n=1}^{N} f_n(x) \right| &= \left| \sum_{n=1}^{\infty} f_n(x) - \sum_{n=1}^{N} f_n(x) \right| \\
&= \left| \sum_{n=N+1}^{\infty} f_n(x) \right| \\
&\leq \sum_{n=N+1}^{\infty} |f_n(x)| \\
&\leq \sum_{n=N+1}^{\infty} M_n \\
&< \epsilon,
\end{aligned}$$

as required. □

Exercise 8.2.1. (cf. [TBB08, Exercise 9.3.1]) For $n \in \mathbb{N}_{\geq 1}$, define $f_n : \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) := \lim_{n\to\infty} \frac{x^{2n}}{1 + x^{2n}}.$$

Prove that $(f_n|_{[b,\infty)})_{n=1}^{\infty}$ converges uniformly, for any (fixed) $b > 1$, and that $(f_n|_{[-c,c]})_{n=1}^{\infty}$ converges uniformly, for any (fixed) $c \in (0, 1)$.

Exercise 8.2.2. [TBB08, Exercise 9.3.3] Let X be a finite set and suppose that $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ is a sequence of functions which converge pointwise to $f : X \to \mathbb{R}$. Prove that $f = \text{u}-\lim_{n\to\infty} f_n$.

Exercise 8.2.3. For a function $f : X \to \mathbb{R}$, define

$$\|f\|_\infty := \sup\{|f(x)| : x \in X\} \in [0, \infty].$$

Prove that a sequence $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ converges uniformly to $f : X \to \mathbb{R}$ if and only if

$$\lim_{n\to\infty} \|f_n - f\|_\infty = 0.$$

Exercise 8.2.4. Let X be a set and let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ and $(g_n : X \to \mathbb{R})_{n=1}^{\infty}$ be sequences which converge uniformly.

(a) [TBB08, Exercise 9.3.9] Prove that $(f_n + g_n)_{n=1}^{\infty}$ converges uniformly.

(b) [TBB08, Exercise 9.3.10] Must the sequence $(f_n g_n)_{n=1}^{\infty}$ converge uniformly? Give a proof or counterexample.

Exercise 8.2.5. [TBB08, Exercise 9.3.4] Let $X = X_1 \cup X_2$, let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions, and let $f : X \to \mathbb{R}$. Prove that if $f|_{X_1} = \text{u}-\lim_{n\to\infty} f_n|_{X_1}$ and $f|_{X_2} = \text{u}-\lim_{n\to\infty} f_n|_{X_1}$ then

$$f = \text{u}-\lim_{n\to\infty} f_n.$$

Exercise 8.2.6. Show that the series

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{2^n}$$

converges uniformly on $\mathbb{R}$.

Exercise 8.2.7. Let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions. Prove that the sequence converges uniformly if and only if it is "uniformly Cauchy", which means: for every $\epsilon > 0$ there exists n_0 such that for all $n, n' \geq n_0$,

$$\|f_n(x) - f_m(x)\|_2 < \epsilon.$$

[*Hint. Use the Cauchy Convergence Criterion for* $\mathbb{R}$ *to first prove that* $g = \text{pw}-\lim_{n\to\infty} f_n$ *exists. Then prove that* $g = \text{u}-\lim_{n\to\infty} f_n$ *by a similar proof to the Cauchy Convergence Criterion.*]

Exercise 8.2.8. (cf. [TBB08, Exercise 9.3.31]) Let X be a set and $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ is a sequence of functions. Suppose that for every sequence $(x_n)_{n=1}^{\infty}$ in X, we have

$$\lim_{n\to\infty} f_n(x_n) = 0.$$

Prove that

$$\text{u}-\lim_{n\to\infty} f_n = 0 \text{ (the zero function)}.$$

Exercise 8.2.9. Let $f : \mathbb{R} \to \mathbb{R}$ be a function and define a sequence of functions $(f_n : \mathbb{R} \to \mathbb{R})_{n=1}^{\infty}$ by

$$f_n(x) := f(x + \frac{1}{n}).$$

(a) Prove that if f is continuous then $f = \text{pw}-\lim_{n\to\infty} f_n$.

(b) Prove that if f is uniformly continuous then $f = \text{u}-\lim_{n\to\infty} f_n$.

(c) Give an example of a function f such that $(f_n)_{n=1}^{\infty}$ doesn't even converge pointwise to f.

8.3 Properties of uniform convergence

Theorem 8.3.1. *Let* $X \subseteq \mathbb{R}$ *be a set and* $a \in X$. *Let* $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ *a sequence of functions which converges uniformly to* $f : X \to \mathbb{R}$. *If each* f_n *is continuous at* a *then so is* f.

Hence, if each f_n *is continuous on* X *then so is* f.

Proof. Let $\epsilon > 0$. Since $f = \text{u}-\lim_{n\to\infty} f_n$, there exists n_0 such that for all $n \geq n_0$ and all $x \in X$,

$$\|f_n(x) - f(x)\|_2 < \frac{\epsilon}{3}.$$

Fix any $n \geq n_0$ Next, since f_n is continuous at a, there exists $\delta > 0$ such that for all $x \in X$, if $\|x - a\|_2 < \delta$ then

$$\|f_n(x) - f_n(a)\|_2 < \frac{\epsilon}{3}.$$

Now let $x \in X$ be a point satisfying $\|x - a\|_2 < \delta$. Then

$$\begin{aligned}\|f(x) - f(a)\|_2 &= \|f(x) - f_n(x) + f_n(x) - f_n(a) + f_n(a) - f(a)\|_2 \\ &\leq \|f(x) - f_n(x)\|_2 + \|f_n(x) - f_n(a)\|_2 + \|f_n(a) - f(a)\|_2 \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon,\end{aligned}$$

as required. □

Applying the above theorem to a sequence of partial sums, we get the following.

Corollary 8.3.2. *Let $X \subseteq \mathbb{R}$ and suppose that $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ is a sequence of continuous functions. If $\sum_{n=1}^{\infty} f_n$ converges uniformly then the function $\sum_{n=1}^{\infty} f_n$ is continuous.*

Example 8.3.3. Consider the series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

If we fix some interval $[-b, b]$ and set $M_n := \frac{b^n}{n!}$, then for every $x \in [-b, b]$, we have

$$\left|\frac{x^n}{n!}\right| \leq M_n.$$

Since $\frac{M_{n+1}}{M_n} \to 0$, it follows by the Ratio Test that $\sum_{n=1}^{\infty} M_n$ converges, so by Weierstrass M-test,

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}$$

converges uniformly on $[-b, b]$. It follows that this defines a continuous function on $[-b, b]$. Since b is arbitrary, it follows that $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is continuous on $\mathbb{R}$.

Theorem 8.3.4. *Let $(f_n : [a, b] \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of continuous functions which converges uniformly to $f : [a, b] \to \mathbb{R}$. Then*

$$\int_a^b f(t)\, dt = \lim_{n\to\infty} \int_a^b f_n(t)\, dt.$$

Remark 8.3.5. In this theorem, we know that each f_n and therefore also f is continuous, and hence they are all Riemann integrable. In Exercise 8.3.4, this theorem is generalized to the case that each f_n is just Riemann integrable (where the main challenge is to show that f is also Riemann integrable).

Proof. Define $M_n := \sup\{|f_n(t) - f(t)| : t \in [a,b]\}$. This is called $\|f_n - f\|_\infty$ in Exercise 8.2.3, and by that same exercise, we know that

$$M_n \to 0 \quad \text{as } n \to \infty.$$

For each n we have

$$\begin{aligned}
0 &\leq \left| \int_a^b f(t)\,dt - \int_a^b f_n(t)\,dt \right| \\
&= \left| \int_a^b f(t) - f_n(t)\,dt \right| \\
&\leq \int_a^b |f(t) - f_n(t)|\,dt && \text{(Exercise 7.2.7)} \\
&\leq M_n(a-b). && \text{(Corollary 7.2.4)}
\end{aligned}$$

By the Squeeze Theorem, it follows that

$$\lim_{n\to\infty} \left| \int_a^b f(t)\,dt - \int_a^b f_n(t)\,dt \right| = 0,$$

as required. □

Applying the above theorem to a sequence of partial sums, we get the following.

Corollary 8.3.6. *Let $(f_n : [a,b] \to \mathbb{R})_{n=1}^\infty$ be a sequence of continuous functions. If the series $\sum_{n=1}^\infty f_n$ converges uniformly then*

$$\sum_{n=1}^\infty \int_a^b f_n(t)\,dt = \int_a^b \sum_{n=1}^\infty f_n(t)\,dt.$$

Example 8.3.7. Let $x \in [0,1)$. By the Weierstrass M-test (with $M_n = x^n$), the series $\sum_{n=0}^\infty t^n$ converges uniformly on $[0,x]$. At each value of t, this is a geometric series, so that we know that

$$\sum_{n=0}^\infty t^n = \frac{1}{1-t}.$$

It follows that

$$\int_0^x \frac{1}{1-t}\,dt = \sum_{n=1}^\infty \int_0^x t^n\,dt = \sum_{n=0}^\infty \frac{x^{n+1}}{n+1} = \sum_{n=1}^\infty \frac{x^n}{n}.$$

We also know, from the Fundamental Theorem of Calculus, that $\int_0^x \frac{1}{1-t} = \ln(1-x)$, so this gives us an interesting formula:

$$\ln(1-x) = \sum_{n=1}^\infty \frac{x^n}{n}, \quad x \in [0,1).$$

Essentially the same argument also proves this formula for $x \in (-1,0]$.

For derivatives, the requirement for uniform convergence is slightly more subtle. In Example 8.1.3, we had a sequence $(f_n : [0,1] \to \mathbb{R})_{n=1}^{\infty}$ converging pointwise to the zero function, although $(f_n')_{n=1}^{\infty}$ converges to a nonzero function. Looking closely at this example, we see that in fact $0 = \text{u}-\lim_{n\to\infty} f_n$, which tells us that uniform convergence of the f_n is not enough to improve the behaviour with respect to differentiation. Instead, what we need is that the sequence $(f_n')_{n=1}^{\infty}$ converges uniformly. We also ask that the sequence consists of continuously differentiable functions; this isn't really necessary (see [TBB08, Theorem 9.37]).

Theorem 8.3.8. *Let $(f_n : [a,b] \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of differentiable functions such that f_n' is continuous for each n. Suppose that the sequence $(f_n')_{n=1}^{\infty}$ converges uniformly to some function $g : [a,b] \to \mathbb{R}$ and that $(f_n)_{n=1}^{\infty}$ converges pointwise to $f : [a,b] \to \mathbb{R}$. Then f is differentiable and $f' = g$.*

Proof. For $x \in [a,b]$, we have

$$\begin{aligned} f(x) &= \lim_{n\to\infty} f_n(x) \\ &= \lim_{n\to\infty} f_n(a) + \int_a^x f_n'(t)\,dt && \text{(Theorem 7.3.2)} \\ &= f(a) + \lim_{n\to\infty} \int_a^x f_n'(t)\,dt \\ &= f(a) + \int_a^x g(t)\,dt && \text{(Theorem 8.3.4).} \end{aligned}$$

Hence by the Fundamental Theorem of Calculus (Theorem 7.3.1), $f' = g$. □

Using partial sums we get the following.

Corollary 8.3.9. *Let $(f_n : [a,b] \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of differentiable functions such that f_n' is continuous for each n, and let $f = \sum_{n=1}^{\infty} f_n$ (converging pointwise). If the series $\sum_{n=1}^{\infty} f_n'$ converges uniformly then*

$$f' = \sum_{n=1}^{\infty} f_n'.$$

Example 8.3.10. We already know that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $x \in (-1,1)$. We would like to differentiate both sides: on the left we get $\frac{1}{(1-x)^2}$. On the right, differentiating term-by-term would give $\sum_{n=0}^{\infty} nx^{n-1}$ – but we don't know yet whether (or where) this converges and whether it agrees with the derivative on the left.

Fix some $b \in (0,1)$. We'll show that $\sum_{n=0}^{\infty} nx^{n-1}$ converges uniformly on $[-b,b]$. Indeed, setting $M_n := nb^{n-1}$, then for all $x \in [-b,b]$ we have

$$|nx^{n-1}| \le M_n.$$

Moreover, $\frac{M_{n+1}}{M_n} \to b < 1$, so by the Ratio Test, $\sum_{n=0}^{\infty} M_n$ converges. Hence by the Weierstrass M-test,

$$\sum_{n=0}^{\infty} nx^{n-1}$$

converges uniformly.

We may now apply the above corollary to conclude that

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} nx^{n-1}, \quad \text{for } x \in [-b, b].$$

Since b is arbitrary, it follows that

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} nx^{n-1}, \quad \text{for } x \in (-1, 1).$$

Exercise 8.3.1. [TBB08, Exercise 9.4.1] Does there exist a sequence of discontinuous functions $(f_n : [a, b] \to \mathbb{R})_{n=1}^{\infty}$ which converges uniformly to a continuous function $f : [a, b] \to \mathbb{R}$?

Exercise 8.3.2. Let $X \subseteq \mathbb{R}$ and let $(f_n : X \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of uniformly continuous functions which converges uniformly to $f : X \to \mathbb{R}$. Prove that f is uniformly continuous.

Exercise 8.3.3. [TBB08, Exercise 9.5.2] Prove that

$$\int_0^{\pi} \sum_{n=1}^{\infty} \frac{\sin(nt)}{n^2}\, dt = \sum_{n=1}^{\infty} \frac{2}{(2n-1)^3}.$$

Exercise 8.3.4. Let $(f_n : [a, b] \to \mathbb{R})_{n=1}^{\infty}$ be a sequence of functions which converges uniformly to $f : [a, b] \to \mathbb{R}$.

(a) Let P be a partition of $[a, b]$, fix $n \in \mathbb{N}_{\geq 1}$, and let $\epsilon > 0$ be such that

$$\epsilon > \|f_n - f\|_{\infty} = \sup\{|f_n(x) - f(x)| : x \in [a, b]\}.$$

Prove that

$$m_i(P, f) > m_i(P, f_n) - \epsilon, \quad M_i(P, f) < M_i(P, f_n) + \epsilon,$$

for all i and

$$L(P, f) > L(P, f_n) - \epsilon, \quad U(P, f) < U(P, f_n) + \epsilon.$$

(b) Prove that if each f_n is Riemann integrable then so is f, and

$$\int_a^b f(t)\, dt = \lim_{n\to\infty} \int_a^b f_n(t)\, dt.$$

[*Exercise 8.2.3 may be handy.*]

Exercise 8.3.5. [TBB08, Exercise 9.5.3] Suppose $(f_n : [a, b] \to \mathbb{R})_{n=1}^{\infty}$ is a sequence of continuous functions which converges uniformly to $f : [a, b] \to \mathbb{R}$. For each n, define $F_n : [a, b] \to \mathbb{R}$ by

$$F_n(x) := \int_a^x f_n(t)\, dt,$$

and define $F : [a, b] \to \mathbb{R}$ by

$$F(x) := \int_a^x f(t)\, dt.$$

Prove that $(F_n)_{n=1}^{\infty}$ converges uniformly to F.

Exercise 8.3.6. Using the previous exercise, prove that if $(f_n : [a,b] \to \mathbb{R})_{n=1}^{\infty}$ is a sequence satisfying the hypotheses of Theorem 8.3.8, then $(f_n)_{n=1}^{\infty}$ converges uniformly.

Exercise 8.3.7. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := 1 + \frac{x^2}{1!} + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots .$$

(a) Verify that the series defining f converges.

(b) [TBB08, Exercise 9.6.2] Prove that

$$f'(x) = 2xf(x), \quad x \in \mathbb{R}.$$

(c) Find a function (defined using a series) $g : \mathbb{R} \to \mathbb{R}$ which satisfies

$$g'(x) = 3x^2 g(x), \quad x \in \mathbb{R}.$$

8.4 Fourier series

Here is an important example of a series of functions.

For a Riemann integrable function $f : [0, 2\pi] \to \mathbb{R}$, define

$$\begin{aligned} a_0(f) &:= \frac{1}{2\pi} \int_0^{2\pi} f(t)\, dt, \\ b_k(f) &:= \frac{1}{\pi} \int_0^{2\pi} f(t) \cos(kt)\, dt, \quad k = 1, 2, \ldots \\ c_k(f) &:= \frac{1}{\pi} \int_0^{2\pi} f(t) \sin(kt)\, dt, \quad k = 1, 2, \ldots . \end{aligned}$$

These are called the *Fourier coefficients* of f. When there is only one function in question, we will write a_0, b_k, c_k instead of $a_0(f), b_k(f), c_k(f)$. The series of functions $[0, 2\pi] \to \mathbb{R}$ given by

$$a_0 + \sum_{k=1}^{\infty} b_k \cos(kt) + \sum_{k=1}^{\infty} c_k \sin(kt)$$

is the *Fourier series* associated to f.

Sometimes (often) this series converges pointwise or uniformly to f. Note that each function in this series is 2π-periodic – that is, it takes the same value at 0 and 2π; therefore, to have any hope of convergence, we focus on functions $f : [0, 2\pi]$ that are 2π periodic:

$$f(0) = f(2\pi).$$

We will explore one general case of convergence of the Fourier series; but first, we note a useful calculation using Integration by Parts.

Proposition 8.4.1. *Let $f : [0, 2\pi] \to \mathbb{R}$ be a function which is differentiable, such that f' is continuous. Suppose that $f(0) = f(2\pi)$. Then*

$$a_0(f') = 0, \quad b_k(f') = kc_k(f), \quad \textit{and} \quad c_k(f') = -kb_k(f),$$

for all $k = 1, 2, \ldots$.

Proof. First, by the Fundamental Theorem of Calculus,

$$a_0(f') = \int_0^{2\pi} f'(t)\,dt = f(2\pi) - f(0) = 0.$$

To calculate $b_k(f')$, we use Integration by Parts (see Exercise 7.3.4) with $g(t) = \cos(kt)$, $g'(t) = -k\sin(kt)$:

$$\begin{aligned}\pi b_k(f') &= \int_0^{2\pi} f'(t)\cos(kt)\,dt \\ &= f(t)\cos(t)\Big|_{t=0}^{2\pi} - (-k)\int_0^{2\pi} f(t)\sin(kt)\,dt \\ &= 0 + k\pi c_k(f).\end{aligned}$$

Similarly for $c_k(f')$, we use Integration by Parts with $g(t) = \sin(kt)$, $g'(t) = k\cos(kt)$:

$$\begin{aligned}\pi c_k(f') &= \int_0^{2\pi} f'(t)\sin(kt)\,dt \\ &= f(t)\sin(t)\Big|_{t=0}^{2\pi} - k\int_0^{2\pi} f(t)\cos(kt)\,dt \\ &= -k\pi b_k(f).\end{aligned}$$

□

The following tells us that two different continuous functions can't have the same Fourier series.

Proposition 8.4.2 (Fourier series uniqueness)**.** *Suppose that $f, g : [0, 2\pi] \to \mathbb{R}$ are continuous 2π-periodic functions. If each Fourier coefficient of f is the same as the corresponding Fourier coefficient of g, then $f = g$.*

Proof. We will not prove this here. It will be seen in MAT3120. □

Proposition 8.4.3. *Suppose that a series*

$$\alpha_0 + \sum_{k=1}^{\infty} \beta_k \cos(kt) + \sum_{k=1}^{\infty} \gamma_k \sin(kt)$$

converges uniformly to a function $g(t)$. Then

$$a_0(g) = \alpha_0, \quad b_k(g) = \beta_k, \quad \textit{and} \quad c_k(g) = \gamma_k.$$

Proof. Exercise 8.4.1. □

We now prove convergence of the Fourier series, when f is periodic and sufficiently smooth.

Proposition 8.4.4. *Let $f : [0, 2\pi] \to \mathbb{R}$ be a function, such that f' and f'' exist and are continuous, and such that f and f' are 2π-periodic. Then the Fourier series converges absolutely and uniformly to f.*

Proof. Since f'' is continuous, it is bounded by the Extreme Value Theorem. So, let M be such that $|f''(t)| \leq M$ for all M. Then we see that $|b_k(f'')| \leq M$ and $|c_k(f'')| \leq 2M$. For $k = 1, 2, \ldots$, we have

$$|b_k(f)| = \frac{|c_k(f')|}{k} = \frac{|b_k(f'')|}{k^2} \leq \frac{2M}{k^2},$$

and similarly $|c_k(f)| \leq \frac{2M}{k^2}$.

Since $\sum\limits_{k=1}^{\infty} \frac{1}{k^2}$ converges, it follows by the Weierstrass M-Test that the Fourier Series converges uniformly and absolutely. Set

$$g(t) := a_0(f) + \sum_{k=1}^{\infty} b_k(f)\cos(kt) + \sum_{k=1}^{\infty} c_k(f)\sin(kt).$$

Then by the last proposition, we see that f and g have the same Fourier coefficients. By Fourier series uniqueness, $f = g$. $\square$

The previous result can be weakened, to only requiring that f' exists and is continuous. In MAT3120 you will also see that asking only for continuity (and not differentiability), the Fourier series converges in a weaker sense.

Exercise 8.4.1. Prove Proposition 8.4.3.

Exercise 8.4.2. Prove that if the Fourier series for f converges uniformly to f, then f is uniformly continuous

Exercise 8.4.3. Suppose that $f : [0, 2\pi] \to \mathbb{R}$ is differentiable, and the Fourier series for f' converges uniformly to f'. Prove that the Fourier series for f converges uniformly, and determine the Fourier coefficients of f' in terms of those of f.

Chapter 9

Power series

In this chapter we look at a particularly fruitful way of writing a function as a series. The series we consider can be considered, in a sense, as polynomials of (possibly) infinite degree. We will make use of the theory developed in previous chapters to help us do calculus with these series.

9.1 Convergence of a power series

Definition 9.1.1. *A* power series *is a series of the form*

$$\sum_{n=0}^{\infty} a_n(x-c)^n,$$

where $(a_n)_{n=1}^{\infty}$ *is a sequence of real numbers,* $c \in \mathbb{R}$*, and* x *is a variable. The numbers* $a_0, a_1, \ldots$ *are the* coefficients *of the power series, and* c *is called the* centre *of the power series.*

If we substitute a real number for x, then either the series converges or it doesn't; if I is a set of real numbers at which the series does converge, then we can view the power series as a function $I \to \mathbb{R}$. We will, additionally, often view a power series as a series of functions (i.e., as $\sum\limits_{n=0}^{\infty} f_n$ where $f_n : \mathbb{R} \to \mathbb{R}$ is given by $f_n(x) = a_n x^n$).

Definition 9.1.2. *Let* $\sum\limits_{n=0}^{\infty} a_n(x-c)^n$ *be a power series. The* interval of convergence *of this power series is the set*

$$\{b \in \mathbb{R} : \sum_{n=0}^{\infty} a_n(b-c)^n \text{ converges}\}.$$

It is clear that the interval of convergence always contains the centre, c.

Example 9.1.3. If we take $c = 0$ and all coefficients equal to 1, we get the power series

$$\sum_{n=0}^{\infty} x^n.$$

When we substitute a real number for x, we get the Geometric Series (Example 3.1.8). We analyzed earlier exactly what real numbers this series converges for, and from that analysis we may conclude that the interval of convergence is $(-1, 1)$. For x in this interval of convergence, we have

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

Example 9.1.4. Consider the power series $\sum_{n=0}^{\infty} n^n x^n$. For an $x \neq 0$, we use the ratio test:

$$\left|\frac{(n+1)^{n+1}x^{n+1}}{n^n x^n}\right| = (n+1)|x|\left(\frac{n+1}{n}\right)^n \geq (n+1)|x|,$$

and therefore,

$$\lim_{n\to\infty}\left|\frac{(n+1)^{n+1}x^{n+1}}{n^n x^n}\right| = \infty.$$

Consequently, the series does not converge. The interval of convergence of this power series is therefore $\{0\}$.

Theorem 9.1.5. *Let* $\sum_{n=0}^{\infty} a_n(x-c)^n$ *be a power series, and define*

$$R := \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}}$$

(interpreted as 0 *if the* lim sup *is* ∞*, and as* ∞ *if the* lim sup *is* 0*). Then for* $b \in \mathbb{R}$*,*

(i) if $|b-c| < R$ *then* $\sum_{n=0}^{\infty} a_n(b-c)^n$ *converges, while*

(ii) if $|b-c| > R$ *then* $\sum_{n=0}^{\infty} a_n(b-c)^n$ *diverges.*

Remark 9.1.6. The number R in the above proposition is called the *radius of convergence* of the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$.

We can rephrase the conclusion of the above proposition in terms of the interval of convergence of the power series. If $R \in (0,\infty)$, it tells us that the interval of convergence is one of

$$(c-R, c+R), \quad (c-R, c+R], \quad [c-R, c+R), \quad \text{or} \quad [c-R, c+R].$$

If $R = 0$ then the interval of convergence is $\{c\}$, whereas if $R = \infty$ then the interval of convergence is $\mathbb{R}$. In particular, in all cases, the interval of convergence *is an interval.*

Proof. Let $b \in \mathbb{R}$, $b \neq c$ (since of course the series converges if $b = c$). We will use the Root Test (Proposition 3.2.7). We compute

$$\limsup_{n\to\infty} \sqrt[n]{|a_n(b-c)^n|} = \limsup_{n\to\infty} |b-c| \sqrt[n]{|a_n|} = \frac{|b-c|}{R}.$$

Hence by the Root Test, the series $\sum_{n=0}^{\infty} a_n(b-c)^n$ converges if $\frac{|b-c|}{R} < 1$, i.e, if $|b-c| < R$, and it diverges if $\frac{|b-c|}{R} > 1$, i.e., if $|b-c| > R$. (Note that these arguments are valid even in the cases $R = 0$ or $R = \infty$, if we allow $\frac{|b-c|}{0}$ to mean ∞ and $\frac{|b-c|}{\infty}$ to mean 0.) $\square$

As noted in the remark before the proof, if the radius of convergence of $\sum_{n=0}^{\infty} a_n(x-c)^n$ is $R \in (0,\infty)$, then we can conclude that the interval of convergence is one of $(c-R, c+R)$, $(c-R, c+R]$, $[c-R, c+R)$, or $[c-R, c+R]$. We already saw in Example 9.1.3 a situation where we get $(c-R, c+R)$. In the following examples, we demonstrate that the other cases are possible.

Example 9.1.7. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$. The radius of convergence is

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{1/n}} = \frac{1}{1} = 1.$$

For the endpoint $b = -1$, we have that

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges by the Alternating Series Test. For the endpoint $b - 1$, we have that

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

is the Harmonic Series, and therefore it diverges. Hence the interval of convergence is $[-1, 1)$.

Example 9.1.8. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}$. The radius of convergence is

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{1/n(n+1)}} = \frac{1}{1} = 1.$$

For the endpoint $b = -1$, once again we can use the Alternating Series Test to conclude that

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)}$$

converges. For the endpoint $b = 1$, we have that

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

converges by Example 3.1.5.

We have seen one example where the radius of convergence is 0 (Example 9.1.4. As a simple example of the opposite case, any polynomial can be viewed as a power series with infinite radius of convergence (it is a power series in which only finitely many coefficients are nonzero). There are also examples with infinitely many nonzero coefficients, such as the following.

Example 9.1.9. Consider the power series $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$. The radius of convergence is

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{1/n^n}} = \frac{1}{\limsup_{n\to\infty} 1/n} = \frac{1}{0} = \infty.$$

Hence the interval of convergence is $\mathbb{R}$.

As we saw in Chapter 8, we cannot conclude much about a function just from knowing that it is a pointwise limit. To advance the theory of power series, we will want to have uniform convergence; this is the purpose of the next result.

Proposition 9.1.10. *Let* $\sum_{n=0}^{\infty} a_n(x-c)^n$ *be a power series and let* R *be its radius of convergence. Let* $[a,b]$ *be any closed bounded interval contained in* $(c-R, c+R)$ *(which is* $\mathbb{R}$ *in the case* $R=\infty$*). Then the series converges uniformly on* $[a,b]$*.*

Proof. We use the Weierstrass M-test (Theorem 8.2.6). Set $M := \max\{|a-c|, |b-c|\}$, so that $M < R$. Let $M_n := |a_n M^n|$. Then using the Root Test as in the previous proof, we know that $\sum_{n=0}^{\infty} M_n$ converges.

For every $x \in [a,b]$, we have $|a_n x^n| \leq M_n$, and therefore it follows by the Weierstrass M-test that the series

$$\sum_{n=0}^{\infty} a_n x^n$$

converges uniformly on $[a,b]$. □

Interesting fact 9.1.11. The above proposition can be strengthened, to say that for any closed bounded interval $[a,b]$ contained in the interval of convergence, the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$ converges uniformly on $[a,b]$ ([TBB08, Theorem 10.10]). This makes a difference when the interval of convergence includes endpoints; for example, in Example 9.1.8, we saw that the power series

$$\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}$$

has interval of convergence $[-1,1]$. The series therefore converges uniformly on $[-1,1]$, although Proposition 9.1.10 is not sufficient to tell us this.

Likewise, the series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges uniformly on $[-1,b)$ for any $b \in (-1,1)$.

Exercise 9.1.1. Determine the interval of convergence of the following power series.

(a) $\sum_{n=0}^{\infty} \frac{n^2}{2^n}(x-3)^n$.

(b) $\sum_{n=0}^{\infty} \frac{2^n}{n^2}(x+3)^n$.

(c) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

(d) $\sum_{n=2}^{\infty} \frac{(x-2)^n}{n\ln(n)}$ (you might use the Integral Test to help sort out convergence at the endpoints).

(e) $\sum_{n=1}^{\infty} \frac{x^n}{n!}$.

(f) $\sum_{n=0}^{\infty} a_n x^n$ where

$$a_n = \begin{cases} 1, & \text{if } n = 4k \text{ or } n = 4k+1 \text{ for some } k \in \mathbb{N}_{\geq 0}; \\ -1, & \text{otherwise.} \end{cases}$$

Exercise 9.1.2. Give an example of a power series whose interval of convergence is $(0, \pi]$.

Exercise 9.1.3. [TBB08, Exercise 10.2.2] Prove that if the limit $\lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$ exists, then it is equal to the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$.

[*Warning: this limit doesn't always exist, whereas the* $\limsup$ *used to define the radius of convergence does always exist.*]

Exercise 9.1.4. Let $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ have radii of convergence R_a and R_b respectively.

(a) Show that the radius of convergence of $\sum_{n=0}^{\infty} (a_n + b_n)x^n$ is at least $\min\{R_a, R_b\}$.

(b) Determine the radius of convergence of $\sum_{n=0}^{\infty} a_n x^{n+1}$.

(c) Determine the radius of convergence of $\sum_{n=0}^{\infty} a_n x^{kn}$ where $k \in \mathbb{N}_{\geq 1}$.

(d) [TBB08, Exercise 10.2.11] If $|a_n| \leq |b_n|$ for all n, determine what the relationship is between R_a and R_b.

Exercise 9.1.5. [TBB08, Exercise 10.2.12] Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence R. Determine the radius of convergence of $\sum_{n=0}^{\infty} a_n x^{2n}$.

Exercise 9.1.6. [TBB08, Exercise 10.2.13] Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R \in (0, \infty)$. Prove that the radius of convergence of $\sum_{n=0}^{\infty} a_n x^{n^2}$ is 1.

9.2 Continuity, integration, and differentiation

Theorem 9.2.1. *Let* $\sum_{n=0}^{\infty} a_n(x-c)^n$ *be a power series with interval of convergence* I*, and define* $f : I \to \mathbb{R}$ *by*

$$f(x) := \sum_{n=0}^{\infty} a_n(x-c)^n.$$

Then:

(i) f *is continuous on* I*, and*

(ii) For any $a, b \in I$*,*

$$\int_a^b f(t)\,dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1}\big((b-c)^{n+1} - (a-c)^{n+1}\big).$$

Proof. We define $f_n : I \to \mathbb{R}$ by $f_n(x) := a_n(x-c)^n$.

(i): Let $x \in I$, and first suppose that $x - c$ is less than the radius of convergence. Then we can find $r > 0$ such that $x \in (c - r, c + r)$. By Proposition 9.1.10,

$$f|_{[c-r,c+r]} = \text{u}\!-\!\sum_{n=0}^{\infty} f_n|_{[c-r,c+r]}.$$

Therefore, by Theorem 8.3.1, $f|_{[c-r,c+r]}$ is continuous, and so (by Exercise 5.2.2) f is continuous at x.

If $x-c$ is not less than the radius of convergence, then x is an endpoint of I, and we use Interesting Fact 9.1.11 to see that the series converges uniformly on a closed interval containing x, and so f is continuous at x.

(ii): Similarly, using either Proposition 9.1.10 or (if one or both of a, b are endpoints of I) Interesting Fact 9.1.11, we get that

$$f|_{[a,b]} = \text{u}\!-\!\sum_{n=0}^{\infty} f_n|_{[a,b]},$$

Therefore, by Theorem 8.3.4,

$$\begin{aligned}\int_a^b f(t)\,dt &= \sum_{n=0}^{\infty} \int_a^b f_n(t)\,dt \\ &= \sum_{n=0}^{\infty} \int_a^b a_n(t-c)^n\,dt \\ &= \sum_{n=0}^{\infty} \frac{a_n}{n+1}\big((b-c)^{n+1} - (a-c)^{n+1}\big).\end{aligned}$$

□

Theorem 9.2.2. *Let* $\sum_{n=0}^{\infty} a_n(x-c)^n$ *be a power series with radius of convergence* $R > 0$*, and define* $f : (c - R, c + R) \to \mathbb{R}$ *by*

$$f(x) := \sum_{n=0}^{\infty} a_n(x-c)^n.$$

Then the power series $\sum_{n=1}^{\infty} na_n(x-c)^{n-1}$ *also has radius of convergence* R*, and for* $x \in (c-R, c+R)$*,*

$$f'(x) = \sum_{n=0}^{\infty} na_n(x-c)^{n-1}.$$

Proof. Let R' be the radius of convergence of the power series $\sum_{n=1}^{\infty} na_n(x-c)^{n-1}$. Then since

$\lim_{n\to\infty} \sqrt[n-1]{n} = 1$[1]

$$\begin{aligned} R' &= \frac{1}{\limsup_{n\to\infty} \sqrt[n-1]{na_n}} \\ &= \frac{1}{\limsup_{n\to\infty} \sqrt[n-1]{n}\,\sqrt[n-1]{a_n}} \\ &= \frac{1}{\limsup_{n\to\infty} \sqrt[n-1]{a_n}} \\ &= R. \end{aligned}$$

Now, let $x \in (c-R, c+R)$. We may find a closed interval $[a,b]$ contained in $(c-R, c+R)$ such that $x \in (a,b)$.

Let $f_n(x) := a_n(x-c)^n$. Using the radius of convergence we just computed, we know that $\sum_{n=0}^{\infty} f_n'$ converges uniformly on $[a,b]$. Hence it follows from Theorem 8.3.8 that $f' = \sum_{n=0}^{\infty} f_n'$ on $[a,b]$, and in particular, at x. □

Corollary 9.2.3. *Let $\sum_{n=0}^{\infty} a_n(x-c)^n$ be a power series with radius of convergence $R > 0$, and define $f : (c-R, c+R) \to \mathbb{R}$ by*

$$f(x) := \sum_{n=0}^{\infty} a_n(x-c)^n.$$

Then for each n,

$$a_n = \frac{f^{(n)}(c)}{n!}.$$

Proof. By induction on n. For $n = 0$, we have

$$\frac{f^{(0)}(c)}{0!} = \frac{f(c)}{1} = a_0.$$

Now assume that it is true for n and let us prove it for $n+1$. Set $g := f'$, so that

$$g(x) = \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n.$$

Since the radius of convergence of this power series is the same as $R > 0$, we may apply the inductive hypothesis to g, which tells us

$$(n+1)a_{n+1} = \frac{g^{(n)}(c)}{n!}.$$

Noting that $f^{(n+1)} = g^{(n)}$, and dividing both sides by $n+1$, we get

$$a_{n+1} = \frac{f^{(n+1)}(c)}{(n+1)n!} = \frac{f^{(n+1)}(c)}{(n+1)!}.$$

□

[1] This follows from the fact that $\frac{r^n}{n} \to \infty$ for any $r > 1$: for if $\sqrt[n-1]{n} > 1$ then there would exist $r > 1$ such that $\sqrt[n-1]{n} > r$ for all n sufficiently large, and from this, we get $r^{n-1}/n < 1$, a contradiction.

Exercise 9.2.1. [TBB08, Exercise 10.4.2] Find power series expansions for

$$\frac{x}{1+x^2} \quad \text{and} \quad \frac{x}{(1+x^2)^2}$$

(centred at 0).

Exercise 9.2.2. (cf. [TBB08, Exercise 10.4.6]) Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$, and define $f : (-R, R) \to \mathbb{R}$ by

$$f(x) := \sum_{n=0}^{\infty} a_n x^n.$$

(a) Prove that if f is *even*, meaning that $f(x) = f(-x)$ for all x, then $a_n = 0$ for all n odd.

(b) Prove that if f is *odd*, meaning that $f(x) = -f(x)$ for all x, then $a_n = 0$ for all n even.

Exercise 9.2.3. In this question, we want to prove that there is a function $f : (-1, 1) \to \mathbb{R}$ which satisfies

$$f(0) = 1, \quad f'(x) = f(x^2).$$

(a) Suppose that there were such a function, and that we could write it as a power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Find recurrence equations for the coefficients $(a_n)_{n=0}^{\infty}$ (i.e., an equation expressing a_n in terms of $a_0, \ldots, a_{n-1}$).

(b) With $(a_n)_{n=0}^{\infty}$ as in (a), prove that $a_n \leq 1$ for all n.

(c) With $(a_n)_{n=0}^{\infty}$ as in (a), prove that the series

$$\sum_{n=0}^{\infty} a_n x^n$$

converges on $(-1, 1)$, and that the function f it defines does satisfy the conditions we began with.

(d) *(Difficult)* Compute exactly the radius of convergence of this power series.

9.3 Taylor series

The final corollary of the previous section tells us that *if* a function f can be "represented" as a power series centred at c (here by "represented" we mean, that the power series converges to f at all points in some open ball around c), then we can recover the coefficients of the power series by taking repeated derivatives of f and evaluating them at c. This suggests the question: when can a function be represented as a power series (centred at a point c)? Here we develop tools to answer this question.

Definition 9.3.1. *Let $I \subseteq \mathbb{R}$ be an open interval and $f : I \to \mathbb{R}$ be a function which is infinitely differentiable (meaning that $f^{(n)}$ exists for all n). For $c \in I$, the* Taylor series *of f (centred at c) is the power series*

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

For $N \in \mathbb{N}_{\geq 0}$, the Nth Taylor polynomial *of f (centred at c) is*

$$P_N(x) := \sum_{n=0}^{N} \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

In other words, the Nth Taylor polynomial is the unique polynomial of degree at most N such that

$$P_N^{(i)}(c) = f^{(i)}(c) \text{ for } i = 0, \ldots, N.$$

Theorem 9.3.2 (Lagrange Remainder Theorem)**.** *Let $f : (a,b) \to \mathbb{R}$ be a function for which $f', f^{(2)}, \ldots, f^{(N+1)}$ all exist on (a,b), let $c \in (a,b)$, and let $P_N(x)$ be the Nth Taylor polynomial of f centred at c. Then for $x \in (a,b)$, there exists z between c and x such that*

$$f(x) - P_N(x) = \frac{f^{(N+1)}(z)}{(N+1)!}(x-c)^{N+1}.$$

Proof. Fix x and assume (without loss of generality) that $x > c$. Define $M := \frac{f(x)-P_N(x)}{(x-c)^{N+1}}$, so that

$$f(x) - P_N(x) = M(x-c)^{N+1}.$$

We need to prove that $M = \frac{f^{(N+1)}(z)}{(N+1)!}$ for some $z \in (x,c)$. Define $g : (a,b) \to \mathbb{R}$ by

$$g(t) := f(t) - P_N(t) - M(t-c)^{N+1}.$$

For $i = 0, \ldots, N$, we have $f^{(i)}(c) = P_N^{(i)}(c)$, while the ith derivative of $M(t-c)^{N+1}$, evaluated at c, is 0. Hence,

$$g^{(i)}(c) = 0.$$

We also have $g(x) = 0$ (by definition of M) so that we may apply Rolle's Theorem to g on $[c,x]$, to obtain z_1 such that

$$g'(z_1) = 0.$$

Now, since $g'(c) = g'(z_1) = 0$, we may apply Rolle's Theorem to the function g' on $[c, z_1]$ to obtain z_2 such that

$$g''(z_2) = 0.$$

We continue in this way, obtaining $z_1, \ldots, z_{N+1}$ with $z_i \in [c, z_{i-1}] \subseteq [c,x]$ such that

$$g^{(i)}(z_i) = 0 \text{ for } i = 1, \ldots, N+1.$$

Now, since P_N is a polynomial of degree at most N, its $(N+1)^{\text{th}}$ derivative is zero, whereas the $(N+1)^{\text{th}}$ derivative of $(t-c)^{N+1}$ is $(N+1)!$. Hence,

$$\begin{aligned} 0 &= g^{(N+1)}(z_{N+1}) \\ &= f^{(N+1)}(z_{N+1}) - M(N+1)!, \end{aligned}$$

and thus $M = \frac{f^{(N+1)}(z_{N+1})}{(N+1)!}$, as required. $\square$

Example 9.3.3. Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) := \sin(x)$. Then we have for $k \in \mathbb{N}_{\geq 0}$

$$f^{(2k)}(x) = (-1)^k \sin(x), \quad f^{(2k+1)}(x) = (-1)^k \cos(x),$$

and from this we see that the Taylor series for f centred at 0 is

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}.$$

Let us use the Lagrange Remainder Theorem to prove that the Taylor series converges to f everywhere. If we let $P_N(x)$ be the Nth Taylor polynomial, our task is to prove that $\lim_{N\to\infty} P_N(x) = f(x)$ for all $x \in \mathbb{R}$.

So, fix $x \in \mathbb{R}$. For each N, by the Lagrange Remainder Theorem, there exists z between 0 and x such that

$$f(x) - P_N(x) = \frac{f^{(N+1)}(z)}{(N+1)!} x^{N+1}.$$

Since $f^{(N+1)}(z)$ is either $\pm \sin(z)$ or $\pm \cos(z)$, we see that $|f^{(N+1)}(z)| \leq 1$, and so

$$|f(x) - P_N(x)| \leq \frac{|x|^{N+1}}{(N+1)!}.$$

Now, for any x, we have $\frac{|x|^{N+1}}{(N+1)!} \to 0$ as $N \to \infty$, and therefore by the Squeeze Theorem, $P_N(x) \to f(x)$, as required.

Exercise 9.3.1. Compute the Taylor series for the function $f(x) := e^x$ centred at 0, and use the Lagrange Remainder Theorem to prove that the Taylor series converges to $f(x)$ for all $x \in \mathbb{R}$.

Exercise 9.3.2. Fix $\alpha \in \mathbb{R}$. For $n \in \mathbb{N}_{\geq 1}$, define

$$\binom{\alpha}{n} := \frac{\alpha(\alpha-1)\cdots(\alpha-n-1)}{n!}.$$

Consider the function $f : [0, \infty) \to \mathbb{R}$ given by $f(x) := x^\alpha$.

(a) Prove that for all $n \in \mathbb{N}_{\geq 1}$ and all $x \in (0, \infty)$,

$$f^{(n)}(x) = n! \binom{\alpha}{n} x^{\frac{1}{2}-n}.$$

(You may use the "Power Rule" for differentiation.)

(b) Determine the Taylor polynomial $P_n(x)$ and the Taylor series for f centred at 1.

(c) Assuming $\alpha > 0$, prove that

$$|f(x) - P_n(x)| \leq \left| \binom{\alpha}{n+1} \right| (x-1)^{n+1} x^\alpha, \quad \text{for } x \geq 1,$$

$$|f(x) - P_n(x)| \leq \left| \binom{\alpha}{n+1} \right| \left(\frac{1-x}{x} \right)^{n+1}, \quad \text{for } x \leq 1.$$

(d) Prove that

$$\left|\binom{\alpha}{n}\right| < 1$$

for all $\alpha \in (0,1)$ and all $n \in \mathbb{N}_{\geq 1}$.

(e) Prove that if $\alpha \in (0,1)$ and $x \in (\frac{1}{2}, 1)$ then the Taylor series for f (centred at 1) is equal to f at x.

Exercise 9.3.3. Define

$$f(x) := \begin{cases} e^{-1/x^2}, & x \neq 0; \\ 0, \quad x = 0. \end{cases}$$

(a) Prove that for each n, there is a polynomial $q_n(x)$ such that

$$f^{(n)}(x) = q_n\left(\frac{1}{x}\right) f(x),$$

for $x \neq 0$.

(b) Prove that f is infinitely differentiable and that

$$f^{(n)}(0) = 0$$

for all n. You may use (without proof) that $\lim_{x \to 0} \frac{f(x)}{x^k} = 0$, for any $k \in \mathbb{N}_{\geq 1}$.

(c) Determine the Taylor series for f centred at 0. What is the radius of convergence of this power series? For what values of x is f equal to its Taylor series?

Review

Chapter 1 – The real numbers

- *Should be comfortable with working with $\mathbb{R}$ (field structure, order structure, $|\cdot|$), but don't need to know the axioms per se,* ***apart from completeness.***
- Important definition: supremum, infimum.

Chapter 2 – Sequences

- Definitions: boundedness, convergence/divergence, increasing/decreasing/monotone, subsequence, Cauchy sequence, $\limsup/\liminf$.
- *Should be well-versed at ϵ-n_0 proofs (proving that a sequence converges, proving that a sequence does not converge, proving some other property assuming that a sequence converges).*
- Uniqueness of limits
- If a sequence converges then it is bounded.
- Algebra of limits.
- If $a_n \leq b_n$ for all n then $\lim\limits_{n\to\infty} a_n \leq \lim\limits_{n\to\infty} b_n$ (assuming the limits exist).
- Squeeze Theorem.
- Every real number is a limit of rationals.
- Monotone convergence criterion.
- If $\lim\limits_{n\to\infty} a_n = L$ then $\lim\limits_{k\to\infty} a_{n_k} = L$.
- Every sequence has a monotone subsequence.
- Bolzano–Weierstrass Theorem.
- Cauchy Convergence Criterion.
- $\liminf\limits_{n\to\infty} a_n \leq \limsup\limits_{n\to\infty} a_n$.
- $\liminf\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} \inf\{a_n, a_{n+1}, a_{n+2}, \dots\}$ (and likewise for $\limsup$).
- $\lim\limits_{n\to\infty} a_n = L$ if and only if $\limsup\limits_{n\to\infty} a_n = \liminf\limits_{n\to\infty} a_n = L$ (Theorem 2.7.6).

- Some good practice exercises: Exercise 2.4.1, 2.6.1, 2.7.6.

Chapter 3 – Series

- Definition: convergence.
- Examples: harmonic series, geometric series.
- Linearity.
- Order-preserving: if $a_n \leq b_n$ then $\sum\limits_{n=1}^{\infty} a_n \leq \sum\limits_{n=1}^{\infty} b_n$ (assuming both converge).
- $\sum\limits_{n=1}^{\infty} a_n$ converges iff $\sum\limits_{n=m}^{\infty} a_n$ converges.
- Divergence Test.
- Boundedness Test.
- Comparison Test.
- Absolute Convergence Test.
- Ratio Test.
- Root Test.
- Alternating Series Test.
- Cauchy Convergence Criterion for Series.
- Some good practice exercises: Exercise 3.2.3, 3.2.6.

Chapter 4 – Topological concepts

- Definitions: open and closed sets, interior, accumulation, boundary, isolated, and limit points, compactness.
- Examples: $\mathbb{R}$, $(0, 1]$ are not compact. Every finite set is compact. (You should be able to prove these **without** using the Heine–Borel Theorem.)
- Permanence properties of open sets.
- Sequential characterization of closedness.
- Compactness.
- Heine–Borel Theorem.
- Permanence properties of compact sets.
- Some good practice exercises: Exercise 4.1.4, 4.1.8, 4.2.4.

Chapter 5 – Continuous functions

- Definitions: limit of a function, one-sided limits, continuity, (strictly/weakly) increasing/decreasing, uniform continuity, infinite limits and limits at ∞.
- Uniqueness of the limit.
- Sequential Characterization of Limits.
- Algebra of Limits, Squeeze Theorem.
- Composition of continuous functions is continuous.
- Algebra of continuous functions (Proposition 5.2.6).
- If K is compact and f is continuous then $f(K)$ is compact.
- Extreme Value Theorem.
- Intermediate Value Theorem.
- If I is an interval and $f : I \to \mathbb{R}$ is continuous and injective, then it is strictly increasing or strictly decreasing, and its inverse is continuous.
- If K is compact and $f : K \to \mathbb{R}^d$ is continuous then it is uniformly continuous.
- Some good practice exercises: Exercise 5.1.7, 5.3.3, 5.5.5.

Chapter 6 – Differentiation

- Definitions: the derivative/differentiability, local minima/maxima.
- If f is differentiable at a then f is continuous at a.
- Linearity, Product Rule, Chain Rule, Inverse Rule.
- If f is differentiable at a and has a local min/max at a then $f'(a) = 0$.
- Rolle's Theorem.
- Mean Value Theorem (you don't need to know Cauchy's Mean Value Theorem).
- Some good practice exercises: Exercise 6.1.5, 6.4.3.

Chapter 7 – Integration

- Definitions: partition, $m_i(P, f)$, $M_i(P, f)$, upper/lower Darboux sums, integrable/integral, improper integrals.
- Examples: a function that is constant except at one point, a function that is not Riemann integrable.
- $L(P, f) \leq U(P', f)$ for any partitions, P, P'.
- f is integrable iff $\forall \epsilon > 0\ \exists$ a partition P such that $U(P, f) - L(P, f) < \epsilon$.
- If f is continuous then f is integrable.

- Additive Property.
- Linearity.
- If $f(x) \leq g(x)$ for all x then $\int_a^b f \leq \int_a^b g$ (assuming both exist).
- Fundamental Theorem of Calculus and its consequence (Theorem 7.3.2).
- Some good practice exercises: Exercise 7.1.6. 7.2.1, 7.2.7

Chapter 8 – Sequences and series of functions

- Definitions: pointwise convergence, uniform convergence, Fourier coefficients and the Fourier series.
- Examples: sequences $(f_n)_{n=1}^{\infty}$ that converge pointwise to f such that:
  - Each f_n is continuous but f is not.
  - $f' \neq \lim_{n\to\infty} f_n'$.
  - $\int_a^b f \neq \lim_{n\to\infty} \int_a^b f_n$.
- The Weierstrass M-test.
- If each f_n is continuous then so is $\text{u}-\lim_{n\to\infty} f_n$.
- If each f_n is integrable then $\int_a^b \text{u}-\lim_{n\to\infty} f_n = \lim_{n\to\infty} \int_a^b f_n$.
- If each f_n is differentiable and (f_n') converges uniformly then $(\lim_{n\to\infty} f_n)' = \text{u}-\lim_{n\to\infty} f_n'$.
- If f'' exists and is continuous, and f, f' are 2π-periodic, then the Fourier series converges absolutely and uniformly.
- Some good practice exercises: Exercise 8.2.9, 8.3.3, 8.3.7.

Chapter 9 – Power series

- Definitions: power series, interval of convergence, radius of convergence, Taylor series/polynomial.
- A power series converges on $(c - R, c + R)$ and diverges on $\mathbb{R} \setminus [c - R, c + R]$ (where c is the centre and R is the radius of convergence).
- A power series converges uniformly on $[a, b]$ for any $[a, b] \subseteq (c - R, c + R)$.
- Continuity, integration, and differentiation of power series.
- Lagrange Remainder Theorem.
- Some good practice exercises: Exercise 9.1.2, 9.2.2, 9.3.1, 9.3.3.

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